

# REPRESENTATION DRIFT IS NOT FORGETTING: DIAGNOSING FINE-TUNING DAMAGE IN TIME-SERIES FOUNDATION MODELS

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## ABSTRACT

Fine-tuning a time-series foundation model reshapes its internal representations, and that reshaping is routinely read as catastrophic forgetting to be prevented. Preventing it presupposes something worth preserving. **We test that presupposition, and on the standard benchmarks it mostly fails.** Screening **32 held-out cells** across three backbones (Moirai-S/B/L, Chronos-T5, TimesFM-2.5) and six datasets against a lookback-96 linear regression *fit on the target data*, only **five** show the pre-trained model beating that baseline, and all five are ETTh2; all three backbones lose to it on ETTh1, ETTm2, Weather and Electricity, as does Moirai on ILI. Against an eight-baseline ladder, **no cell beats every admissible rung.** On the five that pass, fine-tuning *improves* MSE by 31–42% in three, and the two with mean harm carry 4/10 and 5/10 seed agreement. **No cell meets our own definition of damaging a demonstrated pre-trained capability**, so the value-gate we pre-registered has no positive instance to predict: TP 0, FP 8, and precision 0.00 at every threshold from 0.10 to 0.60. Two measurement findings show how easily this is missed. First, the frozen-encoder comparison must be scored on held-out windows—4 of 31 cells otherwise reverse sign, every one of them flattering the frozen encoder. Second, an earlier version of this work drew the opposite conclusion because its linear baseline was an unfitted per-window trend extrapolation that a constant predictor beats; correcting the estimator moves 16 of 21 Moirai cells from gate-pass to gate-fail. Across the 31 intervention cells the outcome still spans −51.8% to +46.1% against zero-shot, and the magnitude of CKA-measured representation change is not sufficient to predict whether allowing encoder adaptation helps or hurts (within-backbone Spearman  $\rho = +0.168$  against the frozen-encoder contrast, clustered CI  $[-0.469, +0.717]$ —too wide to establish a reliable relationship in either direction). We additionally retract a probe-based diagnostic reported in an earlier version.

## 1 INTRODUCTION

Fine-tuning a time-series foundation model reshapes its internal representations, and a large literature treats that reshaping as damage to be prevented—by freezing the encoder, by regularising toward the pre-trained weights, by confining updates to low-rank adapters. Every one of those prescriptions presupposes that the pre-trained representation was worth preserving *on the target task*. **We test that presupposition on the benchmarks the field actually uses, and it mostly fails.**

Our test is deliberately weak. We ask only whether the pre-trained model, zero-shot (ZS), beats a lookback-96 linear regression *fit on the practitioner’s own target data*: not a tuned supervised forecaster, not an ensemble—a single ridge-OLS map estimated once on the training split and applied unchanged. We write the margin as  $R_{\text{task}}^2(\text{PT}) = 1 - \text{MSE}_{\text{ZS}}/\text{MSE}_{\text{Linear}}$  and screen **32 held-out cells** spanning three backbones—Moirai (Small 13.8M, Base 91M, Large 311M), Chronos-T5 Ansari et al. (2024) and TimesFM-2.5 Das et al. (2024), with MOMENT Goswami et al. (2024) screened and excluded—six datasets and two forecast horizons (§4). **Five cells clear it**, and all five are the same dataset on the same backbone family: ETTh2 on Moirai. On the five that do clear it, full fine-tuning *improves* MSE by 31–42% in three; the two whose mean shows harm carry only 4/10 and 5/10 seeds agreeing on the sign.

**Terminology.** *Zero-shot* (ZS) is the frozen pre-trained model evaluated without fine-tuning; condition B is full fine-tuning and condition D a frozen encoder trained on the same data for the same epochs, so  $\Delta_{\text{enc}} \equiv \text{B} - \text{D}$  is the incremental value of allowing encoder-side adaptation, and is the quantity this design manipulates. We measure representational change with *centred kernel alignment* (CKA, geometric similarity between pre-trained and fine-tuned encoders; CKA=1 means no change) and *weight drift* ( $\ell_2$  parameter distance), both defined in §3; neither is manipulated, so both enter only as candidate predictors of  $\Delta_{\text{enc}}$ . EWC Kirkpatrick et al. (2017), L2-SP and LoRA appear as mitigation conditions; LR is learning rate,  $n$  sample size, ILI the influenza-like-illness benchmark.

**The definition no cell meets.** We fixed, in advance, what it would take for a cell to count as damage to a demonstrated pre-trained capability: the cell must clear the gate, full fine-tuning must end worse than not fine-tuning at all in *every* seed, and the frozen encoder must end better in *every* seed. **No cell in the matrix satisfies it** (§5.3). The single candidate, Moirai-Small on ETTh2 at  $h=192$ , splits 5/10 on the sign of the harm. The prospective test we pre-registered therefore has no positive instance to predict: scored on eight held-out cells the gate returns TP 0, FP 8, and precision 0.00 at every threshold from 0.10 to 0.60. Within Moirai it is worse than uninformative—higher pre-trained value predicts encoder adaptation helping *more* (Spearman  $\rho = -0.518$  over 22 cells from six series).

## Contributions.

1. **A negative result on the premise of forgetting mitigation.** Screened against a linear regression fit on the target data, only 5 of 32 held-out cells show a pre-trained advantage to preserve, and all five are ETTh2 on Moirai; three of those five are *improved* 31–42% by full fine-tuning (§4).
2. **Zero cells meet a pre-registered definition of degradation**, so the value-gate we proposed has no positive instance: TP 0, FP 8, precision 0.00 at every threshold in  $[0.10, 0.60]$ , and  $\rho = -0.518$  against the intervention outcome within Moirai—a direction, not an interval claim (§5.3, §7).
3. **Two measurement preconditions for getting this right.** Held-out scoring of the frozen-encoder comparison (4 of 31 cells reverse sign, all one-directional), and a fitted rather than per-window linear baseline (17 of 21 Moirai cells flip). Both corrections move results against the conclusion we originally drew (§7, Appendix A19).
4. **The drift–utility dissociation, on three backbones.** Outcomes from  $-51.8\%$  to  $+46.1\%$  with no reliable drift signature (within-backbone  $\rho = +0.168$ , clustered CI  $[-0.469, +0.717]$ ); four of Chronos-T5’s five cells sit at  $\text{CKA} \in [0.092, 0.169]$ —indistinguishable drift—and the outcome across all five spans 62 points, from  $-41.4\%$  to  $+20.6\%$  (§6). A reproducibility bundle (configs, seed lists, metric scripts, uni2ts patch, both linear baselines) accompanies this paper.

**What is a claim here, and what is a control.** Those four are the paper’s claims, argued in §§4–7; we do not claim that foundation models are useless for forecasting, nor that fine-tuning is safe. Everything else is a control or a boundary case, labelled as such where it appears: ILI and an IoT anomaly detector are external cells that gate-fail, reported as case studies rather than evidence (§6); the mitigation spectrum, a strict-freeze condition and the LoRA sweeps ask only whether the frozen-encoder comparison depends on how it is implemented (Appendices A5, A6, A8); and a linear-probe  $\Delta R^2$  axis from an earlier version is retracted outright (Appendix A29). No claim above rests on any of them.

## 2 BACKGROUND AND RELATED WORK

**Related work: three lines, and the gap between them.** (i) *Forgetting* is studied in continual learning, NLP and fine-tuning dynamics Kirkpatrick et al. (2017); Li & Hoiem (2018); Luo et al. (2023); Kumar et al. (2022); Andreassen et al. (2023), and supplies the prescriptions we test; we use “forgetting” throughout in the narrower task-level sense of degradation relative to zero-shot, not the continual-learning sense of loss on a prior task. (ii) *TSM adaptation* recipes span probabilistic fine-tuning, generative pre-training, LLM reprogramming and cross-task transfer Rasul et al. (2024); Liu et al. (2024); Jin et al. (2024); Zhou et al. (2023), and are evaluated on target-task accuracy. (iii) *Representation analysis* supplies the measurement—representation similarity

and probing Kornblith et al. (2019); Hewitt & Manning (2019); Pimentel et al. (2020)—including work probing frozen TSFM internals directly Wiliński et al. (2025), where we instead track how representations *change under fine-tuning*. **None of the three establishes that the pre-trained representation was worth preserving on the target task**, which is the premise (i)’s prescriptions rest on and what this paper measures.

**Pre-training value gate.** The gate is deliberately cheap and *downstream-only*: the practitioner runs the frozen model zero-shot on the target set and compares against a lookback-96 linear regression *fit on that same dataset*—one ridge-OLS map  $\mathbb{R}^{96 \times F} \rightarrow \mathbb{R}^{h \times F}$  estimated on the training split and applied unchanged to the evaluation windows. Neither pre-training data nor pre-training compute is required. We report  $R_{\text{task}}^2(\text{PT}) = 1 - \text{MSE}_{\text{ZS}}/\text{MSE}_{\text{Linear}}$  as a continuous quantity for every cell (Table 3) and mark 0.20 only as an operating point, so readers can apply alternative thresholds; §7 shows the choice makes no difference to the outcome here. The threshold is calibrated to ETT short-horizon benchmarks, where linear baselines are famously competitive with deep forecasters Zeng et al. (2023). That competitiveness is the whole reason the gate has teeth: **5 of our 32 screened cells clear it**, all of them ETTh2 on Moirai, and the remaining 27 are cells in which the pre-trained model has no measured advantage for fine-tuning to destroy. Substituting a per-window trend extrapolation—a linear fit to each evaluation window’s own last 96 steps, using no training data—inflates the denominator enough that a constant predictor beats it on ETTh1 and Weather, and moves 17 cells across the threshold. An earlier version of this work used that estimator; Appendix A19 documents the correction and its consequences in full.

**Task-level consequence: forgetting.** We define forgetting as signed relative change in zero-shot-canonical task metric on held-out test:  $\text{forg.\%} = 100 \cdot (M_{\text{FT}} - M_{\text{ZS}})/|M_{\text{ZS}}|$  (MSE for forecasting, so positive = worse).  $\text{Forg.} > 0$  indicates *task-level forgetting*: degraded performance against the zero-shot baseline.

**Decoupling harmful forgetting from benign specialisation.** A reviewer-critical distinction: representational change that redirects features toward the target (*benign specialisation*) must be separated from change that discards generally useful structure (*harmful forgetting*). An earlier version of this work operationalised the distinction with a *probe-sign asymmetry* criterion: trained-objective decodability up, task-orthogonal decodability flat or down. That criterion is retracted (Appendix A29). What remains is *value-gate conditioning*. Representational change can be harmful only where the structure it discards was valuable, i.e. in a gate-pass cell; in a gate-fail cell the same change is benign by construction, because no demonstrated advantage existed to lose. Gate-passing plus  $\text{forgetting} > 0$  is therefore necessary for harm but not sufficient: the question of whether the encoder’s *change* carried the loss is answered by intervention—comparing full fine-tuning against a frozen encoder on the same data for the same epochs—and not by observing the representation. §3 states the resulting three-clause definition, and §5.3 reports that no cell we screened satisfies it. We stress that this is a statement about what these benchmarks can evidence, not a proof that harmful forgetting never occurs; §8 sets out what would be needed to exhibit it.

### 3 PROTOCOL AND DEFINITIONS

**Moirai for forecasting and anomaly scoring.** Moirai Woo et al. (2024) predicts a probabilistic distribution over a target horizon given context. For ETT forecasting we report MSE on the median of 20 forecast samples at  $h=96$  and  $h=192$  (Appendix A3). For IoT the anomaly score is the fraction of timesteps outside the  $1-\alpha$  CI ( $\alpha=0.05$ ) over a 32-timestep horizon.

**Fine-tuning objective.** Condition B uses NLL only. Condition C adds supervised contrastive loss Khosla et al. (2020):  $\mathcal{L}(\theta) = \mathcal{L}_{\text{NLL}}(\theta) + \lambda \mathcal{L}_{\text{SupCon}}(\mathbf{z}; \mathbf{y}; \tau)$ , with  $\lambda = 0.5$  and  $\tau = 0.07$ .

**Training details.** AdamW ( $\eta = 10^{-4}$ , weight decay  $10^{-2}$ ), batch size 32, early stopping with patience 3, gradient clipping at 1.0. ETT fine-tuning runs 20 epochs at  $n=500$  (10 seeds) and is swept across  $n \in \{500, 1k, 2k, 5k, 10k\}$ .

**uni2ts PackedStdScaler patch.<sup>1</sup>** Any Moirai fine-tuning reproduction of this paper requires the patch.

<sup>1</sup>We patched an in-place tensor mutation in uni2ts’s PackedStdScaler that silently detaches autograd; a differentiable `torch.where` replaces the in-place write (Appendix A2).

**Representation diagnostics.** We track representation change with two metrics: (1) CKA Kornblith et al. (2019) between pre-trained and fine-tuned encoder representations (geometric similarity); and (2) weight drift ( $\ell_2$  distance from pre-trained parameters).

**The value score and its baseline.** We write  $R_{\text{task}}^2(\text{PT}) = 1 - \text{MSE}_{\text{ZS}}/\text{MSE}_{\text{Linear}}$  for the continuous pre-training value score; it is positive when the pre-trained model beats the linear baseline, and 0.20 marks the operating point used as a screen rather than a boundary.  $\text{MSE}_{\text{Linear}}$  is one ridge map  $\mathbb{R}^{96 \times F} \rightarrow \mathbb{R}^{h \times F}$  ( $\lambda=10^{-4}$ ) fit by least squares on the *training* windows and applied unchanged to the evaluation windows, on the train-normalised scale the runs report, over the same window set the model is scored on (up to 300 windows). Both terms therefore see the same targets; the pre-trained model sees  $96+h$  steps of context against the baseline’s 96, which if anything makes the gate a lower bound on the linear model’s strength.

**Two splits, and the estimand, named once.** The *selection* split is the set of windows used for early stopping and checkpoint choice; the *held-out* split is disjoint from it and is where every number in this paper is scored. We write  $\Delta_{\text{enc}} \equiv \text{B} - \text{D}$  for the difference in forgetting between full fine-tuning and the frozen encoder, positive when freezing does better; it estimates the incremental downstream value of allowing encoder-side adaptation *under this protocol*.  $\Delta_{\text{enc}}$  is the estimand because it is the contrast this design *manipulates*. Representation drift is not manipulated: CKA is measured after the fact, so it enters as a candidate *predictor* of  $\Delta_{\text{enc}}$  and never as the treatment.

**Definition (degradation of a demonstrated capability).** A cell counts as degradation when all three clauses hold: (i)  $R_{\text{task}}^2(\text{PT}) \geq 0.20$  on held-out windows, so there is a measured pre-trained advantage; (ii) full fine-tuning ends worse than not fine-tuning at all,  $\text{forg}_{\text{B}} > 0$ , in *every* seed; and (iii) the frozen encoder ends better,  $\text{forg}_{\text{D}} < 0$ , in *every* seed, which localises the loss to encoder weight updates rather than to the data or the epoch budget. Clauses (ii) and (iii) require unanimity because at three to ten seeds an exact sign-flip test cannot reach  $p < 0.05$  on 28 of our 31 cells; every cell is additionally reported with a paired interval, an effect size and two  $p$ -values in Appendix A22. This definition is fixed for the whole paper; §5.3 reports how many cells meet it.

**Conditions.** **B** NLL-only fine-tune (baseline: how much does unconstrained fine-tuning drift and forget?); **C** NLL+SupCon; **D** frozen encoder (control: separates encoder updates from head training and overfitting); **E** LoRA ( $r=8$ ); **F** L2-SP and **G** EWC. Conditions D and B are the comparison every claim in this paper turns on.

**What condition D does and does not hold fixed.** D freezes the encoder’s own weights but leaves the input projection and mask encoding trainable, so it is a control on *encoder weight updates*, not a guarantee of unchanged representations: its measured CKA is close to but never exactly 1 on Moirai (0.969–0.9999 across our cells) and falls as low as 0.38 on TimesFM, where the encoder stack’s weight drift is exactly zero. **H** (strict freeze) additionally freezes those two modules and is the only condition with CKA exactly 1.000; it checks whether  $\Delta_{\text{enc}}$  depends on the projections being trainable, and on 4 of the 16 cells we ran it on—every Moirai/ETTm2 cell— $\Delta_{\text{enc}}$  reverses sign (Appendix A6).

**Dispersion.** Every  $\pm$  in the body and in every generated table is the standard error of the mean over seeds; where an appendix table or passage uses a standard deviation instead it says so, and says which, where it does.

## 4 EXPERIMENT 1: IS THERE A CAPABILITY WORTH PRESERVING?

To ask whether fine-tuning damages valuable pre-trained features we first have to find a cell where such features are demonstrably present. This section reports that search, and it comes out negative: across **32 screened cells**—21 Moirai (three capacities, five datasets, two horizons), 5 Chronos-T5, 5 TimesFM-2.5 and 1 Moirai/ILI—**five** clear the value gate, and all five are ETTh2 on Moirai. Thirty-one of the 32 also carry a paired frozen-encoder intervention; TimesFM/Electricity is screened but has no B/D pair, so the intervention matrix has 31 rows throughout. Figure 1 shows both: the screen emptying, and the gate running backwards on the cells that clear it.

**Setup** follows §3:  $h \in \{96, 192\}$  for Moirai and  $h=24$  elsewhere,  $n=500$  or  $n=1,000$  samples, 3–10 seeds per cell. Every gate value is computed on held-out test windows against a lookback-96 ridge map fit on the cell’s own training split.

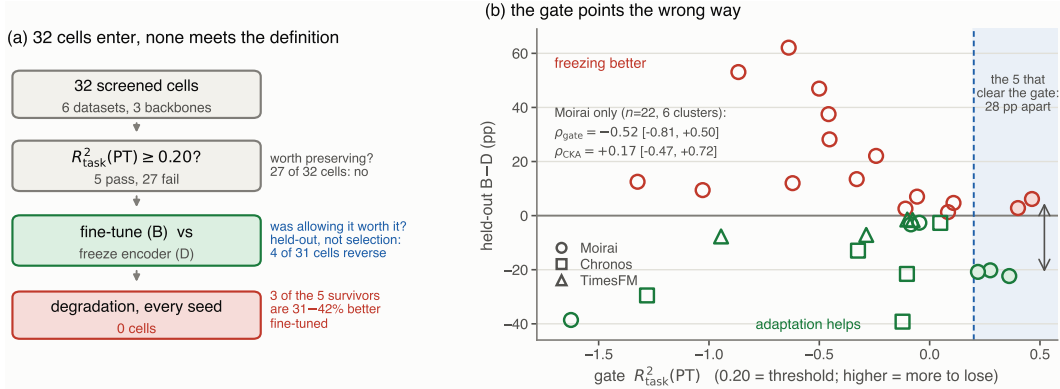


Figure 1: **(a) The screen empties.** Of 32 screened cells, five clear  $R_{\text{task}}^2(\text{PT}) \geq 0.20$  and *none* satisfies the three-clause degradation definition of §3; three of the five survivors are *improved* by full fine-tuning. **(b) The gate points the wrong way.** The 31 cells with a paired frozen-encoder run, plotted as gate score against held-out  $\Delta_{\text{enc}}$  (positive = freezing better, so the upper-right quadrant beyond the dashed threshold is the only region a degradation cell could occupy). Within Moirai—the only backbone with enough cells to ask, and pooling across backbones is what §6 forbids—higher measured pre-trained value predicts encoder adaptation helping *more* ( $\rho = -0.52$ ), while CKA drift does not reliably order them ( $\rho = +0.17$ , clustered CI  $[-0.47, +0.72]$ ). Both correlations are directions rather than interval claims (Appendix A27). Filled markers clear the gate. Every value is read from `scripts/cell_matrix.py`; the figure asserts each count quoted here at draw time and fails loudly if the run records move, so it cannot silently disagree with the text.

**Moirai: 5 of 21** (Appendix Table 3). The five are ETTh2 at Small  $h=96$  (+0.400) and  $h=192$  (+0.464), Base  $h=96$  (+0.361) and  $h=192$  (+0.275), and Large  $h=96$  (+0.219). Every other dataset fails at every capacity: ETTh1 spans  $-0.243$  to  $-0.501$ , Weather  $-0.621$  to  $-1.324$ , Electricity7  $-0.085$  and  $-0.047$ , and ETTm2  $-0.110$  to  $+0.108$ —never reaching the threshold even where the sign turns. Capacity does not help: Moirai-Base is  $6.3\times$  larger than Moirai-Small and *further* behind the baseline on both ETTh1 and Weather.

**Chronos-T5-Small: 0 of 5** and **TimesFM-2.5: 0 of 5** (§6, Appendix Table 7). Across the ten cells the scores run from  $-1.281$  (ETTh2) to  $+0.132$  (Electricity), per cell in Appendix Table 7. The two backbones agree on the ordering of the five datasets despite unrelated pre-training corpora and architectures, which is what one expects if the ranking is a property of the *benchmarks* rather than of either model. Note in particular that ETTh2—the only dataset on which Moirai clears the gate—is where Chronos posts its second-worst score. The advantage is not a property of the dataset alone.

**Why the estimator matters more than the threshold.** Moving the threshold anywhere in  $[0.10, 0.60]$  changes which cells pass but changes nothing below (§7). Changing the *baseline* changes everything. Swapping the fitted regression for the per-window trend extrapolation used earlier—a denominator a constant predictor beats—moves 17 of Moirai’s 21 cells across the threshold (Appendix A19). So we report a **ladder**: eight estimators on the same windows and the same numerator, with the training mean as an admissibility floor. The count clearing 0.20 against a denominator no worse than that floor is 5, 2, 4, 2, 0, 4, 6 and 7 of 32, so the screen’s emptiness is not one estimator’s artefact. Measured against the *strongest* admissible rung each cell faces—a minimum, since a weaker rival inflates the score—**no cell of the 32 clears** 0.20, and the five ETTh2 survivors land between  $-0.74$  and  $-1.46$ , every one of them beaten by DLinear. That count can only fall as rungs are added. When this paper says five cells clear the gate, it means five cells clear *this* gate (Appendix A20).

## 5 EXPERIMENT 2: DOES ENCODER ADAPTATION HELP OR HARM?

Experiment 1 leaves five cells with a measured pre-trained advantage. This section applies the intervention to them—and, for the drift comparisons of §6, to all 31 cells that carry a paired frozen-encoder run—and asks the question the design actually manipulates: does allowing the encoder to adapt help or hurt?

## 5.1 WHAT FINE-TUNING DOES TO THE FIVE SURVIVORS

The five gate-passing cells are the entire evidence base for a claim about damage to valuable features, and they do not support one (Appendix Table 21).

**On three of the five, full fine-tuning is a large improvement.** On Moirai-Base at  $h=96$  and  $h=192$  and Moirai-Large at  $h=96$ , forgetting is  $-31.4\%$ ,  $-36.3\%$  and  $-42.4\%$  against zero-shot, negative in 3/3 seeds each, at CKA 0.460, 0.396 and 0.626. These are the cells with a demonstrated pre-trained advantage *and* the largest representational change among the five, and fine-tuning them cuts MSE by roughly a third. Freezing the encoder captures only part of that gain ( $-9.1\%$ ,  $-16.1\%$ ,  $-21.6\%$ ), so encoder adaptation is where most of the improvement comes from.

**On the remaining two, the mean harm does not survive inspection.** Moirai-Small/ETTh2 at  $h=192$  reads  $\text{forg}_B = +1.99 \pm 3.32$  with 5 of 10 seeds positive—a coin flip on the sign, and the closest the matrix comes to degradation. At  $h=96$  the positive mean ( $+9.71 \pm 10.58$ , 4 of 10 seeds) is produced by one seed that diverges under the *frozen* encoder too, so it is an optimisation failure and not an encoder-drift effect; we report means including it throughout and decompose it in Appendix A19.

**Consequence.** There is no cell in this matrix where we can say that fine-tuning reliably destroyed a demonstrated pre-trained capability. §5.3 states that verdict against the three-clause definition and reports what it costs the diagnostic we proposed.

## 5.2 SAMPLE SIZE, AND WHAT MITIGATION IS FOR

Sweeping  $n \in \{500, \dots, 10k\}$  on Moirai-Small/ETTh2 at  $h=96$ —one of the five survivors, so a sweep where a pre-trained advantage does exist—separates the two readings directly: drift grows monotonically with  $n$  (CKA 0.949 at 500 to 0.518 at 10k) while forgetting is non-monotonic and reverses sign,  $+14\%$  at 500,  $-2.9\%$  by 2k,  $-5.3 \pm 2.1\%$  at 10k (7/10 negative). **If drift magnitude predicted harm these columns would track; they do not.** Whatever harm exists is confined to Small at  $n \leq 1k$ , while the largest spread across seeds sits at the  $n=5k$  crossover (Table 4, Figure 2 and Appendix A10.1 for that bimodal cell and the protocol comparison). This sweep predates held-out scoring and is scored on one split throughout, which is why its  $n=500$  value ( $+14.0 \pm 3.9$ ) differs from the held-out mean above ( $+9.71 \pm 10.58$ ); §7 is about exactly that gap.

**What mitigation is for.** Five strategies spanning unconstrained fine-tuning to full freezing—LoRA, EWC, L2-SP, frozen encoder, strict freeze—are reported in Appendix A5, on Moirai-Small/ETTh2, the one cell where we ran the whole spectrum: weight anchoring does not help there and LoRA does, but extending LoRA to eight further value-cells leaves full fine-tuning ahead on four of them (Appendix A7). The screen bounds what any of it can be for: on 27 of the 32 cells we screened there is no demonstrated advantage to protect, and on three of the remaining five, unconstrained fine-tuning beats zero-shot by 31–42% and the frozen encoder by 20–22 points.

## 5.3 NO CELL MEETS THE DEGRADATION DEFINITION

Applying the three-clause definition of §3 to all 31 intervention cells returns **zero cells**. The count comes from `scripts/cell_matrix.py`, not from hand assembly.

Clause (i) removes 26 of the 31 outright: they have no measured pre-trained advantage, so a worse post-fine-tuning score is benign replacement rather than damage. Of the five that survive it, three fail clause (ii) in the strongest possible way—full fine-tuning *improves* MSE by 31–42% with 3/3 seeds agreeing. That leaves two, and both fail the unanimity requirement: Moirai-Small/ETTh2 at  $h=192$  has  $\text{forg}_B = +1.99 \pm 3.32$  with 5 of 10 seeds positive, and at  $h=96$  the positive mean is a single diverged seed (§5.1). The  $h=192$  cell satisfies clauses (i) and (iii) and is the closest the matrix comes. Relaxing unanimity—to a majority, 70%, 80%, or a  $t$ -test, at either gate threshold—admits at most that one cell (Appendix A21). Nor does replacing the count with inference: aggregated over the 15 value-cells with clustered resampling,  $\Delta_{\text{enc}} = -4.3$  with CI  $[-13.1, +4.5]$  and 7 of 15 positive, so the contrast straddles zero rather than favouring either regime (Appendix A22). Changing the gate’s *estimator* instead admits none under seven of the ladder’s eight rungs and one under a seasonal-naïve denominator weaker than ours on that cell, so loosening the baseline is what manufactures degradation cells (Appendix A20).

**What we are and are not saying.** Not that fine-tuning never damages a pre-trained capability, but that across 32 screened cells on six standard benchmarks and three backbones we could not construct one instance that survives its own definition—fixed before the corrected gate was computed. These benchmarks are the wrong place to look: on no cell does the advantage survive every admissible rung of an eight-baseline ladder, and on the five that clear our ridge gate, fine-tuning on 500–1,000 samples usually improves on it. §8 states what a cell would have to look like to settle the question.

**This is a change of conclusion, not of emphasis.** An earlier version of this work reported eight degradation cells; all eight fail clause (i) once the gate is computed against the regression it is *defined* as. Every reading this paper has withdrawn, and why, is in Appendix A18.

## 6 EXPERIMENT 3: DOES MEASURED DRIFT PREDICT THE OUTCOME?

Experiment 2 established what allowing encoder adaptation does. Neither CKA nor weight drift is manipulated by that design—both are measured after the fact—so this section asks whether either can *predict*  $\Delta_{\text{enc}}$  without running the frozen control.

**Drift and outcome come apart inside a single backbone**—within our Moirai matrix, the strongest form of the finding. Beyond the sweep of §5.2: within Moirai-Base the two *least* drifted cells are the only two that degrade (ETTh1 at CKA 0.755 and 0.669, +22.8% and +39.5%) while its two most drifted improve the most within the arm (ETTh2 at CKA 0.460 and 0.396, −31.4% and −36.4%); and inside Moirai-Large’s LoRA arm the learning rate, not the rank, decides the sign at near-identical CKA (0.973 against 0.989; Appendix A8).

Against the fitted baseline none of the ten non-Moirai cells gate-passes, so what the two extra backbones supply is the dissociation, which does not depend on the gate.

**Near-identical drift, opposite consequences.** On Chronos-T5-Small four of the five datasets restructure to a degree that would, on the usual reading, be called catastrophic—CKA  $\in [0.092, 0.169]$ , at or below the  $0.215 \pm 0.054$  an *untrained* encoder of this architecture scores (Appendix A9)—yet outcomes across the five span −41.4% on ETTh2 to +20.6% on ETTh1, a 62-point spread, with ETTh1 the *least* drifted of the four and the only one that degrades. TimesFM-2.5 adds four cells at milder drift (CKA 0.246–0.561), all improving (−51.8% to −3.0%). Pooled across all 31 cells CKA does correlate with the intervention ( $\rho = +0.567$ , clustered CI [+0.239, +0.828]), but that pooled figure is a backbone artefact by our own rule: within Moirai, the only arm with enough cells to ask,  $\rho = +0.168$  with clustered CI [−0.469, +0.717], which admits a moderate positive relationship but does not order our cells. Nor does the second diagnostic §3 defines rescue it: pooled  $\ell_2$  weight drift gives  $\rho = +0.118$  with a CI including zero, on the 25 cells that retain a drift record, and neither do five further geometric measures on the 59 encoders that retained checkpoints (Appendices A24, A25). The same dissociation holds *inside* one encoder: unfreezing the top 3 of 6 layers and unfreezing all 6 are indistinguishable on the task (−5.9±2.0% against −7.9±2.1%, 10 seeds each) at CKA 0.716 against 0.465, a gap of 3.5 standard errors (Appendix A16). And where a cell does degrade, freezing does not save it: on Chronos/ETTh1 the frozen encoder is worse still (+59.8% against +20.6%), so encoder adaptation is not the locus of the damage.

**Not an artefact of cells with nothing to lose.** Drift might fail to order  $\Delta_{\text{enc}}$  only because 16 of the 31 cells beat no rung of the ladder on held-out MSE. Grading the screen answers that: on the 15 clearing 0.20 against some admissible rung the relation is *weaker* ( $\rho = +0.454$ , CI [−0.490, +0.890]) than on the 16 not (+0.826, CI [−0.321, +0.913]), and as a single coefficient the dependence is  $b_3 = -45.3$  pp per unit CKA per unit value, CI [−225.6, +55.0]—the wrong sign for the objection and far too wide to settle it (Appendix A23). Measured value does not order  $\Delta_{\text{enc}}$  by itself either ( $\rho = +0.023$ , CI [−0.516, +0.599]).

**ILI: a small-data external case study.** Moirai-Small on ILI (weekly influenza,  $h=24$ , 10 seeds) restructures heavily (CKA  $0.456 \pm 0.008$ ) and improves (−50.8±0.5% held-out, 10/10 negative), but **gate-fails decisively** at  $R_{\text{task}}^2(\text{PT}) = -1.625$ . Two further corrections attach to it: the pre-registered CKA-intermediate prediction (0.90–0.95) is **falsified**, and an earlier report of *zero* frozen-encoder improvement in all 10 seeds was a script defect, now  $\Delta_{\text{enc}} = -38.6 \pm 1.3$  (Appendix A28). At  $\approx 450$  windows this is a case study, not a pillar; an IoT anomaly-detection negative control behaves the same way (Appendix A12).

Table 1: **The pre-registered prospective test.** Predictions frozen in git before any condition-B or -D run for these cells existed. “Degradation” applies the paper’s own definition unchanged:  $\text{gate\_passing} \wedge \text{forg}_B > 0$  in every seed  $\wedge \text{forg}_D < 0$  in every seed.

| Cell (new)                     | $\text{gate}_{\text{val}}$ | $\text{forg}_B$ | B–D held-out | Degradation? | Predicted      |
|--------------------------------|----------------------------|-----------------|--------------|--------------|----------------|
| base Weather ( $h=96$ )        | +0.41                      | −11.7 (0/3)     | +9.4±2.0     | no           | gate ×, data × |
| base Weather ( $h=192$ )       | +0.59                      | −18.1 (0/3)     | +12.5±8.7    | no           | gate ×, data × |
| base ETTm2 ( $h=96$ )          | +0.30                      | −13.1 (0/3)     | +7.0±2.4     | no           | gate ×, data ✓ |
| base ETTm2 ( $h=192$ )         | +0.26                      | −8.4 (0/3)      | +1.3±3.4     | no           | gate ×, data ✓ |
| small Electricity7 ( $h=96$ )  | +0.71                      | −16.0 (0/3)     | −3.2±1.5     | no           | gate ×, data ✓ |
| small Electricity7 ( $h=192$ ) | +0.76                      | −18.6 (0/3)     | −2.6±2.1     | no           | gate ×, data ✓ |
| large ETTh1 ( $h=96$ )         | +0.35                      | +14.1 (3/3)     | +28.2±4.9    | no           | gate ×, data × |
| large Weather ( $h=96$ )       | +0.47                      | +7.9 (3/3)      | +12.0±7.5    | no           | gate ×, data × |

## 7 EXPERIMENT 4: A PROSPECTIVE TEST, AND THE MEASUREMENT CHOICES IT RESTS ON

**Held-out scoring is a precondition, not a refinement.** Every number in this paper is scored on windows disjoint from the ones used for early stopping. That is not a stylistic preference. Scoring  $\Delta_{\text{enc}}$  on the selection windows instead **reverses its sign in 4 of the 31 intervention cells**, and *every one of the four reverses in the same direction*: from “the frozen encoder was better” to “full fine-tuning was better” (Appendix Table 6). The largest is Chronos/ETTh1, which moves from +6.8 to −39.2, a 46-point swing. Two of the four are among the five gate-passing cells of §5.1, so this is not confined to cells nobody would have looked at.

The mechanism is visible in the decomposition. The reversals are driven by the frozen arm, whose held-out penalty is much larger than its selection-window penalty ( $\Delta_{\text{frozen}} +53.9$  against  $\Delta_{\text{adapted}} +8.0$  on Chronos/ETTh1), and they concentrate where the two splits are least exchangeable: the untrained checkpoint’s held-out-to-selection error ratio spans 0.98 to 5.07 across these four cells, so on Moirai-Large/ETTh2 the test split is five times harder than the split used to choose the checkpoint. A frozen encoder cannot adapt to that shift and a fine-tuned one partly can, which is exactly the asymmetry that makes same-split scoring flatter freezing.

**A pre-registered prospective test with no positive instance.** Everything above is retrospective: the cells were run, then read. A diagnostic that only ever explains cells it has already seen is worth little, so we tested ours forward. **Predictions were written to results/v47\_prospective/preregistration.json and committed to git at a named revision before any outcome run for those cells existed;** the scoring script reads that file and the finished runs and decides nothing. Two rules were registered against each other:

- **gate rule** —  $\text{gate}_{\text{val}} \geq 0.20 \Rightarrow$  at risk of degradation. This is the paper’s own criterion.
- **dataset rule** —  $\text{dataset} \in \{\text{ETTh1, Weather}\} \Rightarrow$  degradation. A *post hoc* pattern read off the original Moirai cells, registered as a competitor so that the gate’s performance has a reference point rather than being graded against nothing.

Eight held-out cells were then run (Weather, ETTm2, Electricity7, ETTh1 × Small/Base/Large, 3 seeds each) and scored by the pre-registered outcome definition, unchanged (Table 1).

**Result: the outcome the gate was registered to predict never occurs.** None of the eight cells degraded. The gate flagged all eight, giving **TP 0, FP 8, FN 0**: precision 0.00, and recall undefined because the denominator is zero. The competing dataset rule scores no better, for the same reason. This is a stronger negative than the one we expected to report, and it is a different kind: with the corrected baseline there are no degradation cells to miss, so the screen has nothing to be right about (Appendix A18).

**The 0.20 threshold is not what fails.** Re-scoring the same pre-registered predictor against the same pre-registered outcome at thresholds from 0.10 to 0.60—moving nothing else—gives precision 0.00 at every one, with 8, 8, 6, 5, 3 and 2 cells flagged respectively (Appendix Table 8). **The problem is not the cutoff and it is not even the predictor: it is that the predicted class is empty.** So the 0.00



is arithmetic rather than a measured error rate—with no positives to find, any rule scores it—and what carries information is the negative correlation below.

**The gate is worse than uninformative about the intervention.** Within Moirai ( $n=22$  cells from six series, the only arm with enough cells to ask), the gate’s continuous score correlates *negatively* with held-out  $\Delta_{\text{enc}}$ : Spearman  $\rho=-0.518$ . Resampling series rather than cells widens the interval across zero, so we report the direction and not the interval (Appendix A27). Higher measured pre-trained value predicts encoder adaptation helping *more*, not less—the opposite of what a preservation heuristic needs. No other quantity we measured orders  $\Delta_{\text{enc}}$  either (Appendix A26).

**What survives.** Not the gate as a predictor of harm, and not as a screen for when preservation is worth the effort—on these cells it would have fired eight times out of eight and been wrong eight times. What survives is the gate as a *disqualifier*: a cell that fails it cannot supply evidence about damage to pre-trained features, and 27 of our 32 fail it—a much weaker instrument than we proposed.

## 8 LIMITATIONS

Full detail is in Appendix A1; broader impact is in the Ethics Statement. (1) *A null result is not a proof of absence, and the screen is a low bar.* We report that no cell in our matrix meets the degradation definition, not that degradation does not occur. What would settle it is a cell with a *large* pre-trained margin over a *strong* baseline, and that is the combination our benchmarks do not supply: 27 of 32 cells lose to a fitted lookback-96 ridge map, the five that beat it are *all Moirai on ETTh2*, and no cell in the matrix beats every admissible rung of the eight-baseline ladder (Appendix A20). So the part of our result that rests on a demonstrated pre-trained advantage is a claim about one series measured against one weak baseline, and the five-survivor agreement in §5.3 could reflect something about ETTh2 rather than about fine-tuning. (2) *Condition D is a control on encoder weight updates, not on representations.* It leaves the input projection and mask encoding trainable, so its CKA is 0.969–0.9999 on Moirai and as low as 0.38 on TimesFM. Condition H (strict freeze, CKA exactly 1.0000) closes that gap on 16 of the 31 cells; it confirms all five survivors but *reverses*  $\Delta_{\text{enc}}$ ’s sign on all four ETTm2 cells, and 15 rest on D alone (Appendix A6). (3) *Seed counts are small.* Cells carry 3 to 10 seeds, and the unanimity requirement in clauses (ii) and (iii) is doing real work at those counts: the closest candidate is excluded on a 5/10 split. (4) *Horizon confounds the cross-backbone comparison.* The non-Moirai arm runs at  $h=24$  and the Moirai arm at  $h \in \{96, 192\}$ , on different normalisations; we report them separately and never pool them. (5) *A retracted axis.* The linear-probe  $\Delta R^2$  diagnostic is withdrawn (Appendix A29), so we offer no account of *what* the encoder restructured—only whether the restructuring was necessary, which the frozen-encoder comparison answers. (6) *Similarity is not mechanism.* A representation can drift far and still carry the task information in a transformed basis, and a small drift can destroy one critical direction; our negative claim is about *identifying* damage from similarity alone, not about CKA’s validity as a measure of similarity. (7) *Also unsettled:* the forgetting sign at  $n=10k$  is protocol-dependent; the ILI cell is small ( $\approx 450$  windows); and two pre-registered predictions failed (§7, §6).

## 9 CONCLUSION

We set out to measure how much fine-tuning damages a time-series foundation model’s pre-trained capability, and found that on these benchmarks it is mostly not *measurably* there to damage: 5 of 32 cells beat a lookback-96 regression fit on the target data, all five on one dataset, and none beats every admissible rung of an eight-baseline ladder. On three of those five, unconstrained fine-tuning improves MSE by 31–42%, and no cell meets a definition of degradation we fixed in advance. So check the features beat a fitted baseline before spending effort on LoRA, EWC or L2-SP; if they do, settle the question against a frozen encoder scored on windows you did not select on. What survives as a positive finding is the dissociation: the outcome spans  $-51.8\%$  to  $+46.1\%$  against zero-shot across the 31 cells, and the magnitude of CKA-measured change is not sufficient to predict  $\Delta_{\text{enc}}$ ’s sign (within-backbone  $\rho=+0.168$ , clustered CI  $[-0.469, +0.717]$ ). The cheap rule we had hoped to report rests on a premise these benchmarks do not supply.

## ETHICS STATEMENT

**Data.** Every experiment uses public benchmark data and involves no human subjects and no personally identifying information. The forecasting datasets (ETTh1, ETTh2, ETTm2, Weather and Electricity) are standard public releases of instrument and utility measurements. The ILI cell uses public aggregate influenza-surveillance counts, not individual health records. The IoT negative control uses N-BaIoT, network traffic captured from a laboratory testbed of consumer devices rather than from a deployed network. We introduce no new data collection.

**The main risk in this paper is that its negative result is read too broadly.** We report that on these benchmarks we could not construct a cell in which fine-tuning demonstrably damages a pre-trained capability, and that a diagnostic we proposed for predicting such damage has no positive instance to predict. That is a statement about six standard forecasting benchmarks and three backbones, and §8 states what it does not license. It is *not* evidence that fine-tuning is safe in general, and a practitioner who reads it as permission to skip the frozen-encoder control would be drawing the opposite of our recommendation: our prescription is to *run* that control, because we found the observational shortcuts unreliable. Domains where a pre-trained checkpoint holds a large, well-established advantage — which these benchmarks do not, on 27 of the 32 cells we screened — are exactly where the question remains open.

**Broader impact.** This is a diagnostic study on public forecasting benchmarks, and the effect we intend is to raise the bar for claims about forgetting in time-series foundation models: a paper reporting that fine-tuning damaged pre-trained features should say what baseline established that the features were valuable, and how that baseline was fit. Our own screen is the worked example of why — the earlier version of this work reported eight degradation cells on a baseline a constant predictor beats. The controlled comparison remains the way to answer the question for any particular cell, and §5.3 claims only that the harm cannot be demonstrated on these benchmarks.

**Research integrity.** This version corrects an earlier version of the same work in two ways that changed its conclusions rather than its presentation, and we flag both rather than absorbing them silently. An earlier version’s value gate used an unfitted per-window trend extrapolation as its linear baseline; a constant predictor beats that baseline, and correcting it changes the gate status of 17 of the 21 Moirai cells and removes every degradation instance the paper had — that is, the entire class of cell the diagnostic was built to predict. Separately, we retract a linear-probe diagnostic reported earlier: the statistic it rests on is non-zero only where the probe fails to generalise, and Appendix A13 reports the measurement that establishes this on the same encoders. Neither correction required re-running a model; both are documented with their audit trails in Appendix A29 and Appendix A19. Where a number in an earlier version could not be traced to a run record, we say so at the site rather than reprinting it.

**Compute.** The experiments are small by contemporary standards: no run exceeds 10,000 training windows, and the sweep reported here runs on a single GPU or on Apple silicon. We report the protocol deviations forced by the hardware available (Appendix A16) rather than presenting the runs as environment-independent.

## REPRODUCIBILITY STATEMENT

The paper is organised so that every quantity can be traced to a stored run record, and we have tried to make the failure modes we hit ourselves cheap for others to avoid.

**Where the numbers come from.** Every table the source `\inputs` is written by a named emitter that reads the per-run JSON records — none is assembled by hand — and `scripts/rederive_all.sh` runs all of them in one command and then reports which generated files changed. A clean run means the PDF agrees with the run records; any diff is a number that was stale. There is one exception and it is marked where it appears: the two inline tabulars in §A29 transcribe a *withdrawn* computation whose probe records are too sparse to re-derive, and they support no claim in this paper. The per-run records carry the fields the tables use — `final_cka`, `forgetting_pct`, `final_val_mse`, `final_weight_drift` and the early-stopping block — together with the seed, epoch budget and sample count, so the dispersion conventions can be recomputed rather than taken on trust.

**What “reproducible” means here, exactly, and what it does not.** Auditability is not independent reproduction, and we distinguish three tiers rather than letting the stronger word cover the weaker work. (i) Every table and figure is re-derivable from a clean clone with no data and no checkpoints, because the run records are in the release; that is what `rederive_all.sh` checks. (ii) Re-running the value gate or any fine-tuning cell additionally needs the benchmark CSVs, which are public but not ours to redistribute. The obstacle there is not the download but knowing you fetched the same file: ETT and Weather circulate in variants that differ in column order and date parsing, and a variant with the same name changes every window and so every MSE. `results/data_manifest.json` therefore records the SHA-256, shape and date range of each CSV we used, and `scripts/data_manifest.py -check` verifies a local copy against it. (iii) The drift-metric battery’s geometry pass needs the fine-tuned encoders themselves, which are excluded for size; its stored output is released, so the table is re-derivable even though the measurement behind it is not. **No third party has run any of this.** What we offer is re-derivation from committed run records and a bounded two-cell from-scratch check, not independent replication.

**What the two-cell check found, and why we report it rather than a tolerance.** `scripts/rerun_two_cells.sh` retrains one Moirai cell (Small/ETTh2  $h=96$ ,  $n=500$ , seed 42, condition B) and one non-Moirai cell (TimesFM/ETTh1  $h=24$ ,  $n=1,000$ , seed 42, condition B) from the raw CSVs and diffs every stored field. **TimesFM reproduces exactly:** all nine fields, including CKA and weight drift, are bit-identical. **Moirai reproduces its MSEs to within a few percent but not its forgetting percentage:** zero-shot MSE moves  $-0.8\%$ , held-out MSE  $+0.8\%$ , the selection-window MSE  $+3.2\%$ , CKA  $-0.6\%$  (0.957 to 0.952), and weight drift  $+25.2\%$ , while `forgB` moves from  $+6.81$  to  $+11.14$ , a 63.5% relative change. That last figure is the informative one and it is not a failure of the smaller ones, it is those smaller ones compounded: `forgB` is  $(\text{MSE}_{\text{val}} - \text{MSE}_{\text{ZS}})/\text{MSE}_{\text{ZS}}$  on two nearly equal quantities, so a  $-0.8\%$  move in the denominator against a  $+3.2\%$  move in the numerator’s larger term is amplified into  $+63.5\%$  in the ratio. Moirai scores the median of 20 sampled forecasts and TimesFM emits a point forecast, which is why only one of the two is reproducible to the bit. Every sign-bearing field kept its sign, which is what the paper’s claims are stated in terms of — but a reader should treat any single-seed Moirai forgetting figure as carrying at least this much run-to-run variation, and that is why the paper reports cells at 3–10 seeds with a dispersion and never a single run.

**Two things are required to reproduce the Moirai runs at all.** First, the `uni2ts PackedStdScaler` performs an in-place write that silently detaches the gradient tape for losses depending on scaled statistics; without the patch of Appendix A2 training can appear to run while not converging. Second, the frozen-encoder condition is not a strict freeze — the input projections and mask encoding remain trainable, which is why its CKA is not 1.000 — and the strictly-frozen variant is reported separately (Appendix A6). Comparisons against a “frozen” baseline are not comparable across papers unless which parameters were frozen is stated.

**Scoring protocol.** All headline comparisons are scored on windows disjoint from those used for early stopping. This is not a refinement: §7 shows that scoring on the selection windows instead reverses the sign of the frozen-encoder comparison in 4 of the 31 intervention cells, always in the same direction. Reproducing our numbers therefore requires reproducing the split discipline, and Appendix Table 6 gives the per-arm decomposition needed to check which split a given number was scored on.

**Pre-registration.** The prospective test of §7 was registered before any outcome run for those cells existed. The predictions were written to `preregistration.json` under `results/`, and committed before the first condition-B or condition-D run of that arm was launched; the scorer, `score_prospective.py`, reads that file and the finished runs and makes no decisions of its own. Both are in the release, and the commit that added the registration precedes the runs it scores, so the ordering is checkable from the history rather than asserted by us.

**What we release.** The analysis and experiment scripts, the per-run JSON records behind every table, the emitter for every table and the one-command re-derivation script, the benchmark-data manifest, the pre-registration file, and the L<sup>A</sup>T<sub>E</sub>X source and style files needed to build this paper. Model checkpoints are excluded for size; the appendices state which runs retained a checkpoint, because two re-measurements we would otherwise have made were blocked by runs whose checkpoints were not kept (Appendix A14).

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*Notation.* The appendices write the paper’s estimand in its condition form, B–D, because they compare conditions directly (for instance B–D against B–H). It is the same quantity the body calls  $\Delta_{\text{enc}}$ ; the two are interchangeable throughout.

## A1 EXTENDED LIMITATIONS AND FUTURE WORK

**(1) Single-backbone scope of the dissociation claim.** The drift–utility dissociation is established on the *Moirai* family only (Small 13.8M, Base 91M, Large 311M across ETTh1, ETTh2, ETTm2). We attempted three further backbones as second-model controls across three ETT datasets (Table 2).

Table 2: The initial non-Moirai screen, at  $h=96$  with a lookback of 192, one seed (42) and 300 test windows. **This is a different protocol from the paper’s non-Moirai screen** (Table 7), which runs  $h=24$  at lookback 96 with three seeds and a paired frozen-encoder condition; the two are therefore not comparable cell by cell, and only the latter feeds the 32-cell count. Chronos-T5-Base/Large and MOMENT appear here and nowhere else. Advantage is  $R^2_{\text{task}}(\text{PT})$  as a percentage. Rows trace to `results/v10_chronos_etth2.json`, `v14_*`, `v18_backbone_gate/*` and `v12_moment_etth2_gate.json`.

| Backbone         | Dataset | ZS MSE | Linear MSE | Advantage |
|------------------|---------|--------|------------|-----------|
| Chronos-T5-Small | ETTh2   | 0.304  | 0.213      | −42.6%    |
| Chronos-T5-Small | ETTh1   | 0.118  | 0.092      | −28.8%    |
| Chronos-T5-Small | ETTh2   | 0.188  | 0.110      | −70.2%    |
| Chronos-T5-Base  | ETTh2   | 0.286  | 0.213      | −34.3%    |
| Chronos-T5-Large | ETTh2   | 0.288  | 0.213      | −35.2%    |
| TimesFM-2.5-200M | ETTh2   | 0.242  | 0.213      | −13.8%    |
| TimesFM-2.5-200M | ETTh1   | 0.105  | 0.092      | −13.9%    |
| TimesFM-2.5-200M | ETTh2   | 0.190  | 0.110      | −72.6%    |
| MOMENT-1-base    | ETTh2   | 0.528  | 0.213*     | −148%     |

\*MOMENT was screened at lookback 96, whose baseline is 0.235; its margin was nonetheless computed against the lookback-192 baseline 0.213, and we print the comparator actually used. Against 0.235 the margin is −125%, so the verdict is unchanged either way.

All nine backbone×dataset cells gate-fail (ZS under-performs the Linear baseline by 14% to 148%), consistent with ETT-family distributions being outside the pre-training coverage of these backbones at these sizes. The Chronos capacity sweep (Chronos-T5-Small/Base/Large on ETTh2) confirms the gap is size-invariant: scaling from Small (210M) to Large (710M) yields no improvement (−42.6% → −34.3% → −35.2%). TimesFM-2.5-200M and MOMENT-1-base gate-fail across all tested datasets. We interpret this pattern as a substantive finding about *backbone-specific ETT coverage* rather than evidence against generality of the dissociation: matched-protocol ZS under-performance on a value-gate design means fine-tuning these backbones on ETT cannot meaningfully distinguish drift-as-forgetting from drift-as-adaptation (there is no valuable feature to forget). A gate-passing second backbone on a different dataset (Traffic, ILI, Exchange, or a model with explicit ETT pre-training coverage) remains the most direct path to a multi-backbone replication and is the highest-priority future-work direction.

**(2) Sample-sweep depth and seed count.** The central sample-size sweep (Table 4) covers ETTh2 at  $n \in \{500, 1k, 2k, 5k, 10k\}$ . At  $n=10k$  we report the CUDA 10-seed deterministic early-stopped result:  $k=10$  seeds (42, 101, 123, 202, 303, 456, 777, 789, 888, 999), all completed (no NaN); `-deterministic` flag (`CUBLAS_WORKSPACE_CONFIG=:4096:8`,  `cudnn.deterministic=True`, `seeded DataLoader`, `PYTHONHASHSEED`). Forg.  $-5.3 \pm 6.6\%$ , 7/10 negative; mean best epoch  $4.2 \pm 1.3$ . *Dispersions in this item are standard deviations over seeds (ddof=1), matching Table 4 rather than the body’s SEM convention (§3), because the four protocols compared below differ in seed count and the comparison is about their spread.* The Ridge  $\Delta R^2 = +0.67 \pm 0.24$  (**10/10** positive) that an earlier version reported alongside these is **retracted** (§A29) and is quoted here only as a record of the withdrawn computation; the forgetting and CKA columns are unaffected. The three near-zero positive-forgetting seeds (123: +1.8%, 456: +4.4%, 789: +0.4%) follow the  $n=5k$  overshoot pattern (Fig. 2, right panel). For comparison, MPS early-stopped  $k=10$  gave  $-5.7 \pm 10.4\%$  (8/10 neg, `v18_mps_deterministic_n10k`); a second early-stopped run ( $k=5$ ) gave  $-11.7 \pm 5.3\%$  (all

5 negative, `v17_etth2_n10k_es`); the final-epoch runs, the only ones without early stopping, gave  $+10.6 \pm 9.8\%$  on  $k=3$  (`v16_etth2_n10k`), which is the protocol dependence §5.2 discusses. Moirai-Base is now evaluated at both  $n=500$  and  $n=10k$  (10 seeds each); full  $n$ -sweep on Moirai-Large is future work.

**(3) Probe-head capacity.** Ridge and MLP absolute  $R^2(\text{FT})$  remains negative on the 96-step-ahead head at all sampled  $n$ , which is exactly why the  $\Delta R^2$  axis is **retracted** (§A29): relative movement between two fits that never decode the target is not evidence about the encoder. The probe-design sweep of Appendix A14 shows only that the direction is stable across  $\alpha$ , pooling and probe seed *within* that failed-fit regime; it does not rescue the statistic, and no claim in this paper rests on it. Probe capacity is therefore no longer a limitation of a live result but a diagnosis of a withdrawn one.

**(4) Sample-sweep on other sizes.** Moirai-Large is evaluated at  $n=500$  only; full  $n$ -sweep is future work. **(5) IoT scope.** IoT is a negative control (ZS AUC 0.481, sub-random); fine-tuning replaces rather than preserves pre-trained features. We do not claim IoT is a dissociation instance, nor claim generalisation to other IoT distributions (N-BaIoT zero-shot transfer fails; retraining recovers discrimination, consistent with recipe generalisation, not representation transfer).

## A2 UNI2TS PACKEDSTDSCALER PATCH DETAILS

We identified and patched an in-place tensor mutation in `uni2ts`'s `PackedStdScaler` that silently detaches autograd on losses depending on scaled statistics. The issue arises when `PackedStdScaler.transform` performs an in-place `.copy_()` that breaks the gradient tape. Our patch replaces the in-place write with a differentiable `torch.where` operation. All Moirai fine-tuning results in this paper use the patched scaler (released with our code); unpatched `uni2ts` runs may converge differently and we have observed silent non-convergence on `uni2ts@main`. We recommend upstreaming this patch to `uni2ts`.

## A3 MOIRAI MEDIAN FORECAST SELECTION

Moirai generates probabilistic forecasts as a mixture of distributions (StudentT, NormalFixedScale, NegativeBinomial, LogNormal). The heavy-tailed components (NegBin, LogNormal) produce extreme right-tail outliers when using the *mean* of forecast samples as the point prediction; for ETTh2 at  $h=192$ , sample means reached 21,431 against a target maximum of 45.5, inflating MSE from 0.30 to 8.58.

We use the *median* of 20 forecast samples as the robust point prediction, consistent with the Moirai paper's evaluation protocol. This eliminates sensitivity to the heavy-tailed mixture components while preserving the distributional forecast's central tendency.

## A4 ETTh2 FORECASTING DETAILS

**Dataset.** ETTh2 (Electricity Transformer Temperature, Hourly, dataset 2) contains 17,420 timesteps of 7 features (HUFL, HULL, MUFL, MULL, LUFL, LULL, OT) recorded hourly from electrical transformers. We use the standard chronological split: 12 months training, 4 months validation, 4 months test.

**Training protocol.** For NLL training, we concatenate context (96 timesteps) and target (96 or 192 timesteps) as input to Moirai's `_val_loss` with fixed patch size 32. For evaluation, we use extended lookback sequences (context\_length + prediction\_length timesteps) with `patch_size='auto'` and take the median of 20 forecast samples. All data is fed to Moirai in raw scale (Moirai's internal `PackedStdScaler` handles instance normalization); predictions and targets are then normalized using training-set mean and standard deviation for MSE/MAE computation, matching the standard time-series forecasting evaluation protocol.

**Temporal contrastive labels.** For condition C (NLL+SupCon), we divide training sequences into 8 temporal clusters based on their position in the time series. Sequences within the same cluster are positive pairs; sequences from different clusters are negatives. This encourages the encoder to maintain temporal-regime awareness.

Table 3: Task-native gate score  $R_{\text{task}}^2(\text{PT})=1-\text{MSE}_{\text{ZS}}/\text{MSE}_{\text{Linear}}$  for every Moirai cell, on held-out (test) windows. **Not a linear probe:** this is a ratio of forecast MSEs computed through Moirai’s own forecasting head, and is unaffected by the retraction of the linear-probe  $\Delta R^2$  axis. **Estimator, exactly:**  $\text{MSE}_{\text{ZS}}$  is the median of 20 Moirai forecast samples at extended lookback (context + horizon);  $\text{MSE}_{\text{Linear}}$  is a single lookback-96 ridge-OLS linear map  $\mathbb{R}^{96 \times D} \rightarrow \mathbb{R}^{h \times D}$  fit on the cell’s training windows and applied unchanged out of sample; both are computed on the same 300 matched test windows, train-normalised. The gate involves no fine-tuning, so a cell has one value at every  $n$ : the  $n=500$  and  $n=10\text{k}$  (ES) operating points reported in §4 share the entry shown here. Bold = gate-pass at the 0.20 threshold. “—” = not run.

| Backbone            | Dataset      | $h=96$        | $h=192$       |
|---------------------|--------------|---------------|---------------|
| <i>Moirai-Small</i> |              |               |               |
|                     | ETTh1        | −0.243        | −0.331        |
|                     | ETTh2        | <b>+0.400</b> | <b>+0.464</b> |
|                     | ETTm2        | −0.110        | +0.108        |
|                     | Weather      | −0.867        | −0.639        |
|                     | Electricity7 | −0.085        | −0.047        |
| <i>Moirai-Base</i>  |              |               |               |
|                     | ETTh1        | −0.458        | −0.501        |
|                     | ETTh2        | <b>+0.361</b> | <b>+0.275</b> |
|                     | ETTm2        | −0.057        | +0.083        |
|                     | Weather      | −1.029        | −1.324        |
| <i>Moirai-Large</i> |              |               |               |
|                     | ETTh1        | −0.454        | —             |
|                     | ETTh2        | <b>+0.219</b> | —             |
|                     | Weather      | −0.621        | —             |

**Reading:** five of these 21 cells clear the 0.20 threshold and all five are ETTh2; on every other dataset the released checkpoint does not beat a supervised lookback-96 linear regression out of sample, at any capacity.  $R_{\text{task}}^2(\text{PT}) < 0$  means the linear baseline wins outright. These values supersede the ones printed in earlier versions of this table, which were computed against a per-window trend extrapolation rather than the fitted regression the gate is defined with; the substitution and its consequences are documented in Appendix A19. *Electricity7* is Electricity restricted to 7 of its 370 series (six MT series plus OT), a deliberate protocol deviation that keeps the cell dimensionality-matched to ETT/Weather.

**Horizon selection.** We evaluate  $h=96$  and  $h=192$ , the two horizons at which Moirai-Small on ETTh2 clears the value gate: measured against the fitted lookback-96 baseline the gate is defined with (Appendix A19),  $R_{\text{task}}^2(\text{PT})=+0.400$  at  $h=96$  and  $+0.464$  at  $h=192$  on held-out windows. Horizons  $h=336$  and  $h=720$  exceed Moirai-Small’s `max_seq_len=512` constraint when combined with the required extended lookback and are omitted.

## A5 ETTh2 DRIFT DIAGNOSIS AND MITIGATION (FULL TABLES)

This section contains the full tables referenced in §4. Table 9 shows the mitigation spectrum, conditions B through H on Moirai-Small/ETTh2 at both horizons. An earlier version also printed a three-row B/C/D subset of that table separately, with a caption asserting that Moirai’s 28% and 45% zero-shot margins over “Linear” confirmed the pre-trained features were valuable; those margins were measured against the superseded trend baseline (Appendix A19), and the subset is dropped rather than recaptioned since every row it carried appears here.

Table 4 gives the sample-size sweep on ETTh2 ( $h=96$ , condition B) referenced in §4. The three linear-probe columns it used to carry are retracted, and are transcribed in §A29 rather than reprinted here.

**The full 31-cell intervention evidence base.** Table 5 lists every cell carrying a paired B/D intervention behind the headline claim of §1—31 of the 32 screened cells, the screen itself being a gate value and needing no intervention run: three backbones, six datasets, three Moirai capacities and two horizons, each with its held-out gate and the intervention contrast B–D (positive = the frozen encoder was better). It is generated by `scripts/cell_matrix.py -latex` directly from the



Table 4: **Sample-size sweep on Moirai-Small/ETTh2** ( $h=96$ , condition B). Drift grows monotonically (CKA $\downarrow$  as  $n\uparrow$ ) while forgetting is *non-monotonic* and reverses sign: positive at  $n\leq 1,000$ , negative at  $n=2,000$ , positive again at  $n=5,000$  and negative at  $n=10,000$  (7/10 seeds negative, Wilson CI [0.40, 0.89]). **If drift magnitude predicted harm the CKA and Forg. columns would track; they do not**—the dissociation of §5.2 measured inside a single cell. **Dispersion here is the sample standard deviation over seeds** (ddof=1), not the standard error the rest of the paper reports (§3): what this table is about is how the spread across seeds moves with  $n$ —the  $n=5,000$  crossover is a rise in seed variance—and SEM would hide that behind  $k$ , which rises at the same place.

| Samples                | $k$ | MSE             | Forg.%          | CKA             |
|------------------------|-----|-----------------|-----------------|-----------------|
| 500                    | 3   | .148 $\pm$ .008 | +14.0 $\pm$ 6.7 | .949 $\pm$ .010 |
| 1,000                  | 3   | .147 $\pm$ .010 | +13.4 $\pm$ 9.1 | .936 $\pm$ .025 |
| 2,000                  | 3   | .126 $\pm$ .008 | −2.9 $\pm$ 5.9  | .814 $\pm$ .042 |
| 5,000                  | 7   | .137 $\pm$ .012 | +6.6 $\pm$ 10.0 | .546 $\pm$ .082 |
| 10,000 <sup>cuda</sup> | 10  | .120 $\pm$ .007 | −5.3 $\pm$ 6.6  | .518 $\pm$ .199 |

<sup>cuda</sup>The  $n=10,000$  row is the CUDA arm (A10G): 10 deterministic seeds, 10 epochs *with* early stopping (mean best epoch 4.2 $\pm$ 1.3). Every smaller  $n$  is MPS, 20 epochs *without* early stopping. That is the protocol dependence §5.2 discusses; Appendix A1 (2) gives the  $n=10,000$  seeds and the matched MPS comparisons, and Appendix A10.1 the trajectories of the 7  $n=5,000$  seeds. Rows read `final_val_mse`, `forgetting_pct` and `final_cka` from the per-run records in `results/v8_final/n{500,1000,2000,5000}`, `results/v9_n5k_seeds` and `results/v19_cuda_etth2_n10k`. The three withdrawn linear-probe columns this table used to carry are retained as a record in §A29.

committed run JSON; no value in it is transcribed by hand. The two protocol arms (§4 at  $h=96/192$ , §6 at  $h=24$ ) are listed together for completeness but are never pooled into a single statistic.

**The mitigation spectrum, cell by cell.** Table 9 reports five strategies spanning unconstrained fine-tuning to full freezing, all on Moirai/ETTh2. The body (§5.2) states only the conclusion; the per-method readings are here. **LoRA** ( $r=8$ ,  $\alpha=16$ ) improves over zero-shot at  $h=96$  at CKA 0.979, but is not uniformly safe: on Moirai-Large it fails at the default learning rate (+25.7%) and only a  $10\times$  LR reduction rescues it (−8.5%, CKA 0.989), so tune the learning rate before the rank (Appendix A8). **EWC and L2-SP** reduce drift without buying utility: at  $n=1,000$ ,  $h=96$ , each scored against its own zero-shot, they land at +13.4 to +16.0%, against the +11.1% full fine-tuning reaches at  $n=500$ . Those two figures are not matched on training-set size—no full-fine-tuning arm at  $n=1,000$  on this cell has more than a single seed—so what we read off them is that anchoring buys no utility, not an ordering between anchoring and full fine-tuning. **Freezing** eliminates drift but cannot adapt, and the strict freeze of Appendix A6 shows that even the residual adaptation condition D permits is doing work. The screen is what makes this spectrum hard to act on: none of these methods has a demonstrated advantage to protect on 27 of our 32 cells, and on three of the remaining five the unconstrained condition beats both of the others run there—zero-shot by 31–42% and the frozen encoder by 20–22 points.

**Condition coverage differs by cell, and we state it rather than let the widest row imply it.** The full six-condition comparison exists only for Moirai-Small/ETTh2, so “best of six” is not a claim available for any other cell. Every intervention cell has B and D. Condition H (strict freeze) adds 16 cells (Appendix A6) and condition E (LoRA) adds 8 (Appendix A7), both on the seeds given in their own tables; L2-SP and EWC exist on Moirai-Small/ETTh2 alone. A claim about a cell is therefore a claim about the conditions that cell ran, and the per-cell seed counts are printed in every table rather than summarised into one figure.

The joint trajectory is plotted in Fig. 2; panels (a) and (d) carry the claim: a monotone drift curve beside a non-monotone outcome curve.

## A6 THE STRICT-FREEZE CONTROL

Condition D freezes the encoder’s own weights but leaves the input projection (`in_proj`, 247,680 parameters) and `mask_encoding` (384) trainable alongside the distribution head (`param_proj`, 2,956,800). What the encoder *receives* therefore keeps changing, so its output is not a fixed function

Table 5: All 31 intervention cells (the 31 of 32 screened cells with a paired B/D run): held-out gate  $R^2_{\text{task}}$ , B–D on validation and on held-out windows (positive = the frozen encoder was better), and whether the validation and held-out readings disagree in sign. Every row names its own  $n$  because  $n$  is *not* constant down the column: the ETTh2 spectrum cells ran at 500 while their same-dataset neighbours ran at 1,000, and a row that stayed silent would read as sharing its neighbour’s. ILI is the one cell with no cap — it trains on its whole 452-window split, since it is the only series small enough to — so its  $n$  is a size rather than a subsample. Generated by `scripts/cell_matrix.py -latex`.

| Cell                                        | Seeds | Gate $R^2_{\text{task}}$ | B–D validation   | B–D held-out                       | Reverses? |
|---------------------------------------------|-------|--------------------------|------------------|------------------------------------|-----------|
| Chronos / ETTh1 ( $h=24, n=8000$ )          | 3     | $-0.123^f$               | $+6.8 \pm 2.9$   | <b><math>-39.2 \pm 13.9</math></b> | yes       |
| Moirai-S / ILI ( $h=24, n=452$ )            | 10    | $-1.625^f$               | $-62.6 \pm 1.4$  | <b><math>-38.6 \pm 1.3</math></b>  | no        |
| Chronos / ETTm2 ( $h=24, n=8000$ )          | 3     | $-1.281^f$               | $-24.1 \pm 9.8$  | <b><math>-29.5 \pm 6.4</math></b>  | no        |
| Moirai-B / ETTh2 ( $h=96, n=1000$ )         | 3     | $+0.361$                 | $+3.8 \pm 4.1$   | <b><math>-22.3 \pm 2.9</math></b>  | yes       |
| Chronos / Weather ( $h=24, n=8000$ )        | 3     | $-0.103^f$               | $-22.4 \pm 4.5$  | <b><math>-21.6 \pm 3.2</math></b>  | no        |
| Moirai-L / ETTh2 ( $h=96, n=500$ )          | 3     | $+0.219$                 | $+7.2 \pm 5.8$   | <b><math>-20.8 \pm 1.4</math></b>  | yes       |
| Moirai-B / ETTh2 ( $h=192, n=1000$ )        | 3     | $+0.275$                 | $-3.9 \pm 6.2$   | <b><math>-20.2 \pm 1.1</math></b>  | no        |
| Chronos / ETTh2 ( $h=24, n=8000$ )          | 3     | $-0.325^f$               | $-19.9 \pm 3.2$  | <b><math>-12.9 \pm 0.6</math></b>  | no        |
| TimesFM / ETTm2 ( $h=24, n=1000$ )          | 3     | $-0.945^f$               | $-12.2 \pm 12.4$ | <b><math>-7.7 \pm 2.6</math></b>   | no        |
| TimesFM / ETTh2 ( $h=24, n=1000$ )          | 3     | $-0.287^f$               | $+0.2 \pm 2.9$   | <b><math>-7.1 \pm 3.1</math></b>   | yes       |
| Moirai-S / Electricity7 ( $h=96, n=1000$ )  | 3     | $-0.085^f$               | $-10.4 \pm 1.1$  | <b><math>-3.2 \pm 1.5</math></b>   | no        |
| Chronos / Electricity ( $h=24, n=8000$ )    | 3     | $+0.048^f$               | $-10.3 \pm 3.2$  | <b><math>-2.7 \pm 1.9</math></b>   | no        |
| Moirai-S / Electricity7 ( $h=192, n=1000$ ) | 3     | $-0.047^f$               | $-8.2 \pm 1.9$   | <b><math>-2.6 \pm 2.1</math></b>   | no        |
| TimesFM / Weather ( $h=24, n=1000$ )        | 3     | $-0.077^f$               | $-4.8 \pm 0.6$   | <b><math>-1.6 \pm 1.5</math></b>   | no        |
| TimesFM / ETTh1 ( $h=24, n=1000$ )          | 3     | $-0.101^f$               | $-5.4 \pm 0.9$   | <b><math>-1.4 \pm 2.8</math></b>   | no        |
| Moirai-B / ETTm2 ( $h=192, n=1000$ )        | 3     | $+0.083^f$               | $+3.5 \pm 1.5$   | <b><math>+1.3 \pm 3.4</math></b>   | no        |
| Moirai-S / ETTm2 ( $h=96, n=1000$ )         | 5     | $-0.110^f$               | $+6.5 \pm 2.9$   | <b><math>+2.6 \pm 3.4</math></b>   | no        |
| Moirai-S / ETTh2 ( $h=96, n=500$ )          | 10    | $+0.400$                 | $+10.6 \pm 2.8$  | <b><math>+2.8 \pm 3.7</math></b>   | no        |
| Moirai-S / ETTm2 ( $h=192, n=1000$ )        | 5     | $+0.108^f$               | $+8.6 \pm 3.3$   | <b><math>+4.7 \pm 3.2</math></b>   | no        |
| Moirai-S / ETTh2 ( $h=192, n=500$ )         | 10    | $+0.464$                 | $+17.4 \pm 4.3$  | <b><math>+6.1 \pm 3.4</math></b>   | no        |
| Moirai-B / ETTm2 ( $h=96, n=1000$ )         | 3     | $-0.057^f$               | $+3.9 \pm 3.7$   | <b><math>+7.0 \pm 2.4</math></b>   | no        |
| Moirai-B / Weather ( $h=96, n=1000$ )       | 3     | $-1.029^f$               | $+19.3 \pm 0.8$  | <b><math>+9.4 \pm 2.0</math></b>   | no        |
| Moirai-L / Weather ( $h=96, n=1000$ )       | 3     | $-0.621^f$               | $+4.3 \pm 6.5$   | <b><math>+12.0 \pm 7.5</math></b>  | no        |
| Moirai-B / Weather ( $h=192, n=1000$ )      | 3     | $-1.324^f$               | $+30.6 \pm 12.0$ | <b><math>+12.5 \pm 8.7</math></b>  | no        |
| Moirai-S / ETTh1 ( $h=192, n=1000$ )        | 5     | $-0.331^f$               | $+14.8 \pm 1.9$  | <b><math>+13.5 \pm 1.7</math></b>  | no        |
| Moirai-S / ETTh1 ( $h=96, n=1000$ )         | 5     | $-0.243^f$               | $+16.1 \pm 1.5$  | <b><math>+22.1 \pm 2.3</math></b>  | no        |
| Moirai-L / ETTh1 ( $h=96, n=1000$ )         | 3     | $-0.454^f$               | $+13.4 \pm 2.1$  | <b><math>+28.2 \pm 4.9</math></b>  | no        |
| Moirai-B / ETTh1 ( $h=96, n=1000$ )         | 3     | $-0.458^f$               | $+15.8 \pm 4.6$  | <b><math>+37.5 \pm 8.2</math></b>  | no        |
| Moirai-B / ETTh1 ( $h=192, n=1000$ )        | 3     | $-0.501^f$               | $+15.6 \pm 2.3$  | <b><math>+47.0 \pm 13.8</math></b> | no        |
| Moirai-S / Weather ( $h=96, n=1000$ )       | 3     | $-0.867^f$               | $+55.4 \pm 4.3$  | <b><math>+53.1 \pm 10.2</math></b> | no        |
| Moirai-S / Weather ( $h=192, n=1000$ )      | 3     | $-0.639^f$               | $+70.2 \pm 12.3$ | <b><math>+62.1 \pm 13.6</math></b> | no        |

of the input, which is why D’s measured CKA is 0.76–0.90 on ILI and 0.97–0.99 on Moirai-Base rather than 1.000. A reader can reasonably object that wherever “freezing wins”, it wins partly by re-fitting the input projection rather than by holding the encoder still.

Condition **H** removes the objection: `in_proj` and `mask_encoding` are frozen too, so only `param_proj` trains and the encoder’s output is pinned by construction (measured CKA 1.0000 exactly, as it must be). B–H is then a clean read of what allowing *any* encoder-side change was worth, and B–D versus B–H separates encoder adaptation from input re-fitting. We fixed the reading rule before running: the claim is that the sign and reading of B–D survive strict freezing, so B–H is reported beside B–D on every cell, and divergence on any cell is reported as that cell’s gap being partly input re-fitting rather than being buried.

**Result** (Table 10). The sixteen cells fall into three groups. Seven were chosen before the gate was corrected: they are the six that met the degradation definition under the superseded trend baseline (Appendix A19), plus ILI. Nothing was dropped; two cells the prospective arm later added (Moirai-Large on ETTh1 and on Weather) postdate this control and carry no H run. That selection no longer

Table 6: **The four sign reversals between selection and held-out windows** (§7).  $\Delta$  adapted and  $\Delta$  frozen are the change in each arm’s forgetting when scoring moves from the selection split to the held-out split;  $z_{\text{test}}/z_{\text{val}}$  is the untrained checkpoint’s error ratio between the two splits, an exchangeability check that involves no fine-tuning. All four reversals run from “frozen better” to “adapted better”.  $n$  is per row and not constant (Table 5), and is taken from the published row rather than re-read here. Generated by `scripts/heldout_decomposition.py`.

| Cell                                | B–D val | B–D held-out | $\Delta$ adapted (B) | $\Delta$ frozen (D) | $z_{\text{test}}/z_{\text{val}}$ |
|-------------------------------------|---------|--------------|----------------------|---------------------|----------------------------------|
| Chronos / ETTh1 ( $h=24, n=8000$ )  | +6.8    | −39.2        | +8.0                 | +53.9               | 0.98                             |
| Moirai-B / ETTh2 ( $h=96, n=1000$ ) | +3.8    | −22.3        | +12.2                | +38.3               | 2.31                             |
| Moirai-L / ETTh2 ( $h=96, n=500$ )  | +7.2    | −20.8        | −52.9                | −24.8               | 5.07                             |
| TimesFM / ETTh2 ( $h=24, n=1000$ )  | +0.2    | −7.1         | −11.7                | −4.3                | 1.09                             |

Table 7: **Cross-backbone results** ( $h=24$ ) (§6). Every non-Moirai cell with a paired frozen-encoder run: held-out gate  $R_{\text{task}}^2$  ( $^f$ =gate-fail at the 0.20 operating point — all nine fail, as does the screened-only TimesFM/Electricity cell at +0.132), CKA after full fine-tuning, and the two outcome readings as % change against zero-shot (*negative is better than zero-shot*). **Protocol note:** this arm is univariate OT, lookback 96,  $h=24$ , 200 held-out windows, per-window  $z$ -scored; the Moirai arm of §4 is lookback 192,  $h=96$ , 300 windows, train-normalised. Gates for both arms come from one source (`results/gate_test_side.json`); the two arms are never pooled.

| Backbone     | Dataset     | Gate $R_{\text{task}}^2$ | CKA   | Full FT | Frozen |
|--------------|-------------|--------------------------|-------|---------|--------|
| Chronos-T5-S | ETTh2       | −1.281 <sup>f</sup>      | 0.112 | −41.4%  | −11.9% |
| Chronos-T5-S | Weather     | −0.103 <sup>f</sup>      | 0.092 | −5.7%   | +15.9% |
| Chronos-T5-S | Electricity | +0.048 <sup>f</sup>      | 0.232 | −2.3%   | +0.4%  |
| Chronos-T5-S | ETTh2       | −0.325 <sup>f</sup>      | 0.094 | −0.4%   | +12.5% |
| Chronos-T5-S | ETTh1       | −0.123 <sup>f</sup>      | 0.169 | +20.6%  | +59.8% |
| TimesFM-2.5  | ETTh2       | −0.945 <sup>f</sup>      | 0.360 | −51.8%  | −44.1% |
| TimesFM-2.5  | ETTh2       | −0.287 <sup>f</sup>      | 0.246 | −20.4%  | −13.3% |
| TimesFM-2.5  | Weather     | −0.077 <sup>f</sup>      | 0.335 | −7.8%   | −6.1%  |
| TimesFM-2.5  | ETTh1       | −0.101 <sup>f</sup>      | 0.561 | −3.0%   | −1.6%  |

picks out a privileged class—against the fitted baseline no cell meets the definition—but it remains the set chosen to be *most* favourable to the degradation story, which is the useful property here. Five are the gate-passing survivors of §5.1, run afterwards precisely because the first seven did not reach them: until they existed, the survivors’ readings rested on condition D alone. The last four are the Moirai/ETTh2 cells that clear the screen against a rung other than the fitted line (Appendix A23), added so that the control covers value-cells outside ETTh2 — all four of them, at three seeds each, so ETTh2 is covered completely rather than partially. Measured CKA is 1.0000 everywhere, confirming the construction.

**On twelve of the sixteen the reading survives; on four it reverses, and the four are the ones we added.** B–H keeps B–D’s sign on the original seven and on the five survivors. There the magnitude changes by −13.1 to +28.9 pp, and on the original seven by −5.8 to +0.5 pp, so at most 5.8 pp of a gap as large as 62 pp is input-projection re-fitting. On ILI, where adaptation helps, strict freezing makes adaptation look *more* valuable (−38.6 → −43.6), which is the same direction: holding more of the model still costs more where the encoder was earning its keep. B–D is averaged over the seeds condition H ran, so Moirai-Small/ETTh1’s entries here are 3-seed values against the 5-seed values quoted in Table 5.

**All four ETTh2 cells diverge, and we said in advance how that would be read.** On every one of them  $\Delta_{\text{enc}}$  changes sign between the two controls: Moirai-Small  $h=96$  goes from B–D +7.3±2.9 to B–H −6.8±2.0, Moirai-Base  $h=96$  from +7.0±2.4 to −9.2±3.1, Moirai-Small  $h=192$  from +4.2±5.8 to −2.7±6.1 and Moirai-Base  $h=192$  from +1.3±3.4 to −2.4±2.9—shifts of −14.2, −16.2, −6.9 and −3.7 pp. The last of those is the weakest of the four and we do not lean on it: both

Table 8: Gate threshold sensitivity on the eight pre-registered cells (§7). The predictor and the outcome definition are both as pre-registered; only the threshold moves. Bold is the operating point used in the paper.

| Threshold | Flagged | TP | FP | FN | Precision | Recall     |
|-----------|---------|----|----|----|-----------|------------|
| 0.10      | 8       | 0  | 8  | 0  | 0.00      | <i>nan</i> |
| 0.20      | 8       | 0  | 8  | 0  | 0.00      | <i>nan</i> |
| 0.30      | 6       | 0  | 6  | 0  | 0.00      | <i>nan</i> |
| 0.40      | 5       | 0  | 5  | 0  | 0.00      | <i>nan</i> |
| 0.50      | 3       | 0  | 3  | 0  | 0.00      | <i>nan</i> |
| 0.60      | 2       | 0  | 2  | 0  | 0.00      | <i>nan</i> |

Table 9: **Mitigation spectrum on Moirai-Small/ETTh2**. Rows ordered by representation preservation (CKA at  $h=96$ ). On *this* cell LoRA preserves representations (CKA $\approx$ 0.98) while *improving* on zero-shot, and weight anchoring reduces drift without buying utility. **Neither reading generalises, and we checked rather than assumed:** extending condition E to 8 further value-cells (Appendix A7) puts CKA<sub>E</sub> anywhere from 0.55 to 0.97 and leaves full fine-tuning ahead of LoRA on 4 of the 8. **Every row is scored against its own zero-shot.** Forgetting% is the mean over seeds of each run’s (fine-tuned – zero-shot)/zero-shot, so no row borrows another’s denominator; earlier versions of this table divided the anchoring and LoRA rows by a single shared value and read up to 3.4 pp away as a result. MSE, CKA and  $\ell_2$  drift are means over seeds with population std (ddof=0); the body’s SEM convention (§3) does not apply here. **The rows are not matched on training-set size.** Conditions B, C and D use  $n=500$  training samples (zero-shot MSE .1256 at  $h=96$ , .1580 at  $h=192$ ); E, F and G use  $n=1,000$  (zero-shot MSE .1274 and .1596); the two rightmost columns give the seed count and the training-set size separately. We therefore claim no ordering *between* the two groups: there is no full-fine-tuning arm at  $n=1,000$  on this cell with more than a single seed, so “anchoring ties full fine-tuning” is not a comparison these runs support. The dissociation the table is cited for—CKA not predicting utility—is within-group and unaffected. Condition D reads CKA 0.99996 rather than 1: it leaves `in_proj` and `mask_encoding` trainable, so it is a frozen-*encoder* control and not a strict freeze. Only condition H (Appendix A6) is 1.0000 by construction. Earlier versions of this table printed D as 1.000, which rounded that distinction away.

| Method                         | $h = 96$        |             |                     |                | $h = 192$       |             |                     |                | seeds | $n$   |
|--------------------------------|-----------------|-------------|---------------------|----------------|-----------------|-------------|---------------------|----------------|-------|-------|
|                                | MSE             | Forg.%      | CKA                 | Drift          | MSE             | Forg.%      | CKA                 | Drift          |       |       |
| B: Full fine-tune              | .140 $\pm$ .014 | +11.1       | .937 $\pm$ .028     | 3.96 $\pm$ .56 | .183 $\pm$ .020 | +15.7       | .970 $\pm$ .009     | 4.12 $\pm$ .32 | 10    | 500   |
| F: L2-SP ( $\lambda=0.01$ )    | .150 $\pm$ .006 | +16.0       | .939 $\pm$ .023     | 4.98 $\pm$ .34 | .179 $\pm$ .009 | +12.7       | .945 $\pm$ .007     | 4.84 $\pm$ .22 | 3     | 1,000 |
| C: NLL+SupCon                  | .138 $\pm$ .010 | +10.1       | .949 $\pm$ .014     | 3.72 $\pm$ .43 | .178 $\pm$ .011 | +12.9       | .964 $\pm$ .015     | 3.97 $\pm$ .34 | 10    | 500   |
| F: L2-SP ( $\lambda=0.1$ )     | .147 $\pm$ .003 | +13.4       | .965 $\pm$ .006     | 2.94 $\pm$ .08 | .181 $\pm$ .003 | +13.6       | .980 $\pm$ .002     | 2.84 $\pm$ .02 | 3     | 1,000 |
| G: EWC ( $\lambda=1000$ )      | .149 $\pm$ .015 | +14.9       | .972 $\pm$ .004     | 3.29 $\pm$ .79 | .188 $\pm$ .004 | +18.3       | .978 $\pm$ .006     | 4.46 $\pm$ .38 | 3     | 1,000 |
| E: LoRA ( $r=8$ ) <sup>†</sup> | .116 $\pm$ .001 | <b>−9.1</b> | .979 $\pm$ .006     | 0.00 $\pm$ .00 | .157 $\pm$ .002 | <b>−1.8</b> | .986 $\pm$ .005     | 0.00 $\pm$ .00 | 5     | 1,000 |
| D: Frozen encoder              | .126 $\pm$ .006 | +0.5        | .99996 $\pm$ .00001 | 1.11 $\pm$ .15 | .155 $\pm$ .002 | −1.7        | .99995 $\pm$ .00002 | 1.30 $\pm$ .18 | 10    | 500   |

<sup>†</sup> LoRA ( $\alpha=16$ ) adds low-rank updates that do not modify base encoder parameters;  $\ell_2$  drift is exactly zero by construction.

of its intervals straddle zero, so what that cell shows is a shift in the mean and not a resolved reversal. The pattern is carried by the two  $h=96$  cells, where the shift is four to five times the SEM. The consistency of the *direction* across all four is what the reading rests on: four cells, one dataset, four shifts, all negative. The rule fixed before the runs was that divergence means part of that cell’s gap is input re-fitting, and that is what these say: on ETTh2, condition D’s apparent “freezing wins” was carried *entirely* by the input projection adapting to the target, and once it is held fixed too, allowing the encoder to adapt is the better choice. Two readings follow, and they point the same way as the rest of the paper rather than against it. First, the direction of the correction is towards adaptation on every diverging cell, so nothing here recovers a degradation case; ETTh2 joins ETTh2 in the column where full fine-tuning wins under the strict control. Second, and more important for §5, it is direct evidence that **condition D does not measure encoder adaptation alone**: on four of the sixteen cells where we can separate the two, the loose control’s sign is set by the projections rather than by the encoder. That is a limit on B–D as an estimand, and it is why §3 defines  $\Delta_{\text{enc}}$  against the control actually used and reports H beside it rather than substituting one for the other.

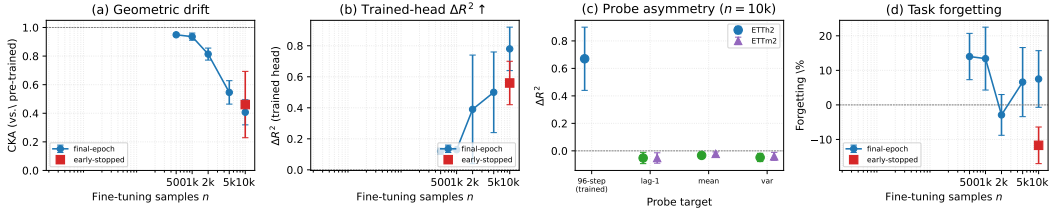


Figure 2: Cost-benefit dynamics on Moirai-Small/ETTh2 (condition B,  $h=96$ ). (a) CKA decays monotonically with  $n$  (final-epoch vs. early-stopped markers); (d) forgetting is non-monotonic and protocol-dependent. Panels (b) and (c) plot the linear-probe  $\Delta R^2$  statistic and its sign asymmetry, both **retracted** (§A29); they are reproduced unaltered as a record of the withdrawn analysis and support no claim here. The dissociation this figure is cited for is the contrast between (a) and (d): drift monotone in  $n$ , outcome not.

**On the five survivors the control cuts against freezing, not for it.** These are the cells the objection mattered most for, and on the three where full fine-tuning already won, strict freezing *widens* its margin rather than narrowing it: B–D of  $-22.3$ ,  $-20.2$  and  $-20.8$  becomes B–H of  $-28.0$ ,  $-31.8$  and  $-33.9$  on Moirai-Base  $h=96$ , Base  $h=192$  and Large  $h=96$ , a move of  $-5.6$  to  $-13.1$  pp. Part of what condition D was scoring on these cells was therefore the input projection re-fitting to the target data, and removing it makes encoder adaptation look *more* valuable, not less. On Moirai-Small  $h=192$ —the cell that comes closest to degradation—B–H is  $+1.2 \pm 8.5$  against B–D’s  $+4.1 \pm 8.6$ : freezing still reads as better, by less, with a standard error several times the mean either way. Condition H also improves on the un-tuned model in every seed of all five ( $\text{forg}_H$  from  $-0.2\%$  to  $-8.5\%$ ), so on the survivors both parts of the degradation definition behave the same way under the strict control as under the loose one.

**The one large change is the diverged seed, and it corroborates the earlier reading.** Moirai-Small/ETTh2  $h=96$  moves from B–D  $+13.7$  to B–H  $+42.6 \pm 29.5$ , the only cell whose magnitude shifts by more than 13 pp. It is seed 42, which §5.1 reports as a training failure rather than an encoder-drift effect: that seed diverges under full fine-tuning ( $+98.9\%$ ) *and* under the frozen encoder ( $+91.1\%$ ), but it cannot diverge under condition H, where only `param_proj` trains and the run therefore sits at zero-shot ( $-0.1\%$ ). Over the other two seeds B–H is  $+14.4$  against B–D’s  $+16.7$ . The strict control thus supplies independent support for the seed-42 diagnosis: what fails on that seed is optimisation, and holding the encoder’s *inputs* fixed as well as its weights is what prevents it.

**One clause does not survive, and it is not the headline one.** “Degradation” has two parts: fine-tuning ends up worse than the un-tuned model, *and* freezing improves on it. The first part is a property of condition B and is untouched by this control. The second is condition D’s, and on two cells the strictly-frozen model does *not* beat the un-tuned model: Moirai-Small/ETTh1  $h=192$  at  $\text{forg}_H = +1.2 \pm 0.4\%$ , positive in all 3 seeds where  $\text{forg}_D$  is  $-4.8\%$ , and Moirai-Small/ETTh2  $h=192$  at  $+1.3 \pm 0.3\%$ , positive in all 3 seeds where  $\text{forg}_D$  is  $-5.8\%$ . On those cells the head alone cannot recover the zero-shot forecast; D’s improvement came partly from re-fitting the input projection. The other fourteen cells keep both parts ( $\text{forg}_H$  from  $-0.2\%$  to  $-12.8\%$ ), so “freezing also beats not fine-tuning” holds strictly on 14 of 16. The first part is unaffected in either direction: where freezing wins it keeps winning under the strict control (B–H positive on all six candidate cells), and where full fine-tuning wins it wins by more (the five survivors above). We report the cells rather than re-defining the term around them.

## A7 LoRA ON THE VALUE-CELLS: THE CONSTRAINED MIDDLE OF THE ADAPTATION AXIS

Conditions B and D bracket the adaptation axis—everything trains, or the encoder body does not. Neither tells a practitioner whether a *cheap constrained* adaptation would have beaten the extreme their cell happened to run. Condition E (LoRA,  $r=8$ ,  $\alpha=16$ ) is that middle point, and until this revision it existed on one cell, which is why every “LoRA is the cheap win” sentence in earlier

Table 10: The strict-freeze control (condition H) beside condition D on sixteen cells: the six that met the degradation definition under the superseded trend baseline, plus ILI, plus the five gate-passing survivors of §5.1, plus all four Moirai/ETTh2 value-cells. **B–H keeps B–D’s sign on twelve and reverses it on all four ETTh2 cells**, where condition D’s margin was the input projection re-fitting rather than the encoder being held still. B–D is restricted to the seeds condition H ran, and a cell with fewer than three H seeds is excluded and named by the emitter rather than printed thin.  $n$  is shown on every row and is not constant down the column; ILI’s 452 is its whole training split rather than a cap (Table 5). Generated by `scripts/cell_matrix.py -latex`.

| Cell                                   | Seeds | B–D        | B–H               | for <sub>H</sub> | CKA <sub>H</sub> |
|----------------------------------------|-------|------------|-------------------|------------------|------------------|
| Moirai-S / Weather ( $h=192, n=1000$ ) | 3     | +62.1±13.6 | <b>+56.7±13.5</b> | −10.6±0.3        | 1.0000           |
| Moirai-S / Weather ( $h=96, n=1000$ )  | 3     | +53.1±10.2 | <b>+50.1±6.6</b>  | −12.8±0.6        | 1.0000           |
| Moirai-B / ETTh1 ( $h=192, n=1000$ )   | 3     | +47.0±13.8 | <b>+47.5±13.7</b> | −8.0±0.4         | 1.0000           |
| Moirai-B / ETTh1 ( $h=96, n=1000$ )    | 3     | +37.5±8.2  | <b>+32.5±8.2</b>  | −9.7±0.4         | 1.0000           |
| Moirai-S / ETTh1 ( $h=96, n=1000$ )    | 3     | +23.2±3.6  | <b>+20.4±3.6</b>  | −0.6±0.2         | 1.0000           |
| Moirai-S / ETTh1 ( $h=192, n=1000$ )   | 3     | +14.5±1.4  | <b>+8.7±1.5</b>   | +1.2±0.4         | 1.0000           |
| Moirai-S / ETTh2 ( $h=96, n=500$ )     | 3     | +13.7±9.0  | <b>+42.6±29.5</b> | −0.2±0.1         | 1.0000           |
| Moirai-S / ETTh2 ( $h=96, n=1000$ )    | 3     | +7.3±2.9   | <b>−6.8±2.0</b>   | −2.9±0.4         | 1.0000           |
| Moirai-B / ETTh2 ( $h=96, n=1000$ )    | 3     | +7.0±2.4   | <b>−9.2±3.1</b>   | −3.9±1.8         | 1.0000           |
| Moirai-S / ETTh2 ( $h=192, n=1000$ )   | 3     | +4.2±5.8   | <b>−2.7±6.1</b>   | +1.3±0.3         | 1.0000           |
| Moirai-S / ETTh2 ( $h=192, n=500$ )    | 3     | +4.1±8.6   | <b>+1.2±8.5</b>   | −1.4±0.2         | 1.0000           |
| Moirai-B / ETTh2 ( $h=192, n=1000$ )   | 3     | +1.3±3.4   | <b>−2.4±2.9</b>   | −6.0±0.4         | 1.0000           |
| Moirai-B / ETTh2 ( $h=192, n=1000$ )   | 3     | −20.2±1.1  | <b>−31.8±1.0</b>  | −4.5±2.0         | 1.0000           |
| Moirai-L / ETTh2 ( $h=96, n=500$ )     | 3     | −20.8±1.4  | <b>−33.9±0.1</b>  | −8.5±0.5         | 1.0000           |
| Moirai-B / ETTh2 ( $h=96, n=1000$ )    | 3     | −22.3±2.9  | <b>−28.0±3.3</b>  | −3.5±0.4         | 1.0000           |
| Moirai-S / ILI ( $h=24, n=452$ )       | 10    | −38.6±1.3  | <b>−43.6±0.7</b>  | −7.1±0.5         | 1.0000           |

versions generalised from Moirai-Small/ETTh2. We ran E on the eight Moirai value-cells that lacked it, seeds 42/123/456, each at its own arm’s  $n$  and epoch budget (Table 11).

**What E can and cannot measure, fixed before the runs.** LoRA adds trainable low-rank updates to the encoder’s attention projections, so the encoder’s *function* changes while its base weights do not:  $\ell_2$  weight drift is exactly 0.0 on all 24 records by construction, and it is omitted from the table rather than printed as a column of zeros that would invite the reading that the encoder was held still. CKA is the measure that still reads on E.  $\Delta_{\text{enc}}$  remains B–D; E is not a second estimate of it.

**The one-cell reading does not survive.** CKA<sub>E</sub> spans 0.5482 to 0.9674 across these eight cells, against the  $\approx 0.98$  the single-cell spectrum reports, so LoRA is not a representation-preserving method in general—it is one on some cells. Full fine-tuning finishes ahead of LoRA ( $B-E < 0$ ) on **4 of the 8**. LoRA is itself worse than not fine-tuning at all on two cells ( $\text{for}_E = +11.9\%$ , 2 of 3 seeds, on Moirai-Base/ETTh2  $h=96$ ;  $+11.2\%$ , 3 of 3, on Moirai-Base/ETTh2  $h=192$ ).

**Against the frozen encoder, read as a paired contrast.** D–E is paired per seed rather than taken as the difference of two means, which matters: on Moirai-Small/ETTh1 the B–E and B–D means sit 0.2 pp apart with a SEM of 4.9, and a difference of means there would read as “LoRA beat freezing” on a cell that cannot separate them. Paired and requiring a unanimous sign, **LoRA beats freezing on 3 of 8 cells, freezing beats LoRA on 2, and 3 are inseparable**. Moirai-Base/ETTh2  $h=192$  is the one cell that orders all three points cleanly—full fine-tuning, then LoRA, then frozen—and it is a gate-passing cell, so the ordering is on a cell with something to preserve.

**Why this belongs in a paper about drift as a diagnostic.** It is a third condition on which the paper’s central claim holds. Across these eight cells CKA<sub>E</sub> does not order the outcome of constraining adaptation ( $\rho = -0.167$  against B–E;  $-0.333$  against B–D on the same cells): the highest-CKA cell here, Moirai-Large/ETTh2 at 0.9674, is one where the constrained option *loses* to full fine-tuning by 27.8 pp, and the lowest, Moirai-Base/ETTh2  $h=96$  at 0.5482, is one where it wins. As with B–D, how much the representation moved does not tell you whether constraining the move helped.

**Two limits of this arm.** The two Moirai-Small/ETTh2 cells already carrying E are excluded here: their E arm ran at  $n=1,000$  against B/D at  $n=500$ , a mismatch disclosed in Table 9’s caption and not improved by adding a third  $n$ . And  $\alpha$  is not written into the run records—only  $r$  is—so  $\alpha=16$

Table 11: **Condition E (LoRA,  $r=8$ ,  $\alpha=16$ ) on the eight Moirai value-cells that lacked it.** All contrasts are held-out test MSE, normalised by the cell’s zero-shot test MSE, in percentage points;  $\pm$  is SEM over the seeds in the Seeds column. Sign convention matches B–D and B–H throughout: **positive means full fine-tuning did worse than the comparison arm.** B–D is recomputed on E’s seeds, so it is comparable to B–E within a row and may differ from the same cell’s row in Table 5, which averages that cell’s full seed set. Parenthesised counts are the seeds sharing the mean’s sign.  $\ell_2$  weight drift is exactly 0.0 for every run and is not shown: LoRA leaves the base parameters untouched by construction, which is why CKA, not drift, is the representation measure here. Rows ordered by D–E.

| Cell                                    | Seeds | B–D             | B–E              | D–E                    | forge                  | CKA <sub>E</sub> |
|-----------------------------------------|-------|-----------------|------------------|------------------------|------------------------|------------------|
| Moirai-S / ETTh2 ( $h=96$ , $n=1000$ )  | 3     | +7.3 $\pm$ 2.9  | +13.9 $\pm$ 2.1  | +6.5 $\pm$ 0.8 (3/3)   | −23.6 $\pm$ 0.3 (0/3)  | 0.8438           |
| Moirai-S / ETTh2 ( $h=192$ , $n=1000$ ) | 3     | +4.2 $\pm$ 5.8  | +7.3 $\pm$ 6.8   | +3.1 $\pm$ 1.0 (3/3)   | −8.6 $\pm$ 0.9 (0/3)   | 0.9280           |
| Moirai-B / ETTh2 ( $h=192$ , $n=1000$ ) | 3     | −20.2 $\pm$ 1.1 | −17.7 $\pm$ 1.1  | +2.5 $\pm$ 0.9 (3/3)   | −18.6 $\pm$ 3.0 (0/3)  | 0.6944           |
| Moirai-S / ETTh1 ( $h=96$ , $n=1000$ )  | 3     | +23.2 $\pm$ 3.6 | +23.4 $\pm$ 4.9  | +0.2 $\pm$ 1.3 (2/3)   | −3.6 $\pm$ 1.1 (0/3)   | 0.9093           |
| Moirai-B / ETTh2 ( $h=96$ , $n=1000$ )  | 3     | +7.0 $\pm$ 2.4  | +4.9 $\pm$ 2.9   | −2.2 $\pm$ 1.8 (1/3)   | −18.0 $\pm$ 2.7 (0/3)  | 0.5482           |
| Moirai-L / ETTh2 ( $h=96$ , $n=500$ )   | 3     | −20.8 $\pm$ 1.4 | −27.8 $\pm$ 0.7  | −7.0 $\pm$ 1.2 (0/3)   | −14.6 $\pm$ 0.7 (0/3)  | 0.9674           |
| Moirai-B / ETTh2 ( $h=192$ , $n=1000$ ) | 3     | +1.3 $\pm$ 3.4  | −19.6 $\pm$ 7.0  | −20.9 $\pm$ 5.3 (0/3)  | +11.2 $\pm$ 5.3 (3/3)  | 0.6495           |
| Moirai-B / ETTh2 ( $h=96$ , $n=1000$ )  | 3     | −22.3 $\pm$ 2.9 | −43.3 $\pm$ 14.1 | −21.0 $\pm$ 11.4 (1/3) | +11.9 $\pm$ 10.5 (2/3) | 0.9173           |

is traceable to `scripts/run_lora_valuecells.sh` and its committed command line rather than to the per-run JSONs, the same limitation the learning rate carries.

## A8 LORA RANK AND HYPERPARAMETER SENSITIVITY

Table 12 shows LoRA on **Moirai-Small** is robust to rank ( $r \in \{4, 8, 16, 32\}$ ): all ranks achieve  $\text{CKA} \geq 0.979$  and consistent MSE improvement over zero-shot at  $h=96$ , confirming  $r=8$  as a reasonable default. On **Moirai-Large** (seed 42,  $h=96$ ,  $n=500$ ), rank escalation alone does *not* rescue the failure mode: forgetting stays +19 to +30% across  $r \in \{8, 16, 32, 64\}$ . Table 13 extends the sweep to  $\alpha$ , target-module scope, and LR (all at  $r=8$ , seed 42): varying  $\alpha \in \{8, 16, 32\}$  and module scope provides no rescue, but reducing LR to  $1 \times 10^{-5}$  does (forge =  $-8.5 \pm 0.7\%$ ,  $\text{CKA} = 0.989 \pm 0.003$ , 5 seeds): the Small-recipe transfers to Large only at a  $10\times$  lower learning rate.

Table 12: LoRA rank sensitivity on ETTh2 (condition E, 500 training samples). Moirai-Small: all ranks achieve near-identical MSE and  $\text{CKA} \geq 0.979$ . Moirai-Large: all ranks forget +19 to +30%, with no rank recovering Moirai-Small’s negative forgetting. Mean  $\pm$  std across 3 seeds (5 for  $r=8$  Small); Moirai-Large single-seed ( $s=42$ ) for  $r \in \{16, 32, 64\}$  (std not reported; point estimates only), 3 seeds for  $r=8$ .

| Backbone | Rank | $h$ | MSE             | Forg.%          | CKA             | Seeds |
|----------|------|-----|-----------------|-----------------|-----------------|-------|
| Small    | 4    | 96  | .116 $\pm$ .001 | −9.5 $\pm$ 0.9  | .992 $\pm$ .001 | 3     |
| Small    | 4    | 192 | .146 $\pm$ .003 | −8.8 $\pm$ 1.9  | .995 $\pm$ .002 | 2     |
| Small    | 8    | 96  | .116 $\pm$ .001 | −9.1 $\pm$ 1.1  | .979 $\pm$ .006 | 5     |
| Small    | 8    | 192 | .157 $\pm$ .002 | −1.8 $\pm$ 3.1  | .986 $\pm$ .005 | 5     |
| Small    | 16   | 96  | .117 $\pm$ .001 | −10.2 $\pm$ 1.2 | .987 $\pm$ .002 | 3     |
| Small    | 16   | 192 | .155 $\pm$ .002 | −1.7 $\pm$ 1.2  | .988 $\pm$ .001 | 3     |
| Small    | 32   | 96  | .119 $\pm$ .001 | −8.3 $\pm$ 0.9  | .986 $\pm$ .001 | 3     |
| Small    | 32   | 192 | .159 $\pm$ .003 | +1.1 $\pm$ 1.4  | .989 $\pm$ .002 | 3     |
| Large    | 8    | 96  | .159 $\pm$ .004 | +25.7 $\pm$ 3.2 | .973 $\pm$ .012 | 3     |
| Large    | 16   | 96  | .154            | +22.5           | .955            | 1     |
| Large    | 32   | 96  | .150            | +19.0           | .986            | 1     |
| Large    | 64   | 96  | .165            | +29.9           | .984            | 1     |

## A9 CKA CALIBRATION: RANDOM-INITIALISATION BASELINE

A CKA of 0.09 is only interpretable against a floor: the value two *unrelated* encoders of this architecture score against each other. We measure it by re-initialising the Moirai-Small

Table 13: LoRA-Large HP ablation (ETTh2,  $r=8$ ,  $n=500$ ,  $h=96$ ). Baseline (default LR= $1e-4$ ,  $\alpha=16$ , modules = q,v,out) forg.= +23.7%, seed 42. Only LR= $1e-5$  rescues LoRA-Large (5 seeds); all other axes fail.

| Axis     | Setting                  | Val MSE                         | Forg.%                         | CKA                             | Seeds    |
|----------|--------------------------|---------------------------------|--------------------------------|---------------------------------|----------|
| LR       | $1e-4$ (default)         | .157                            | +23.7                          | .980                            | 1        |
|          | $5e-5$                   | .146                            | +15.3                          | .982                            | 1        |
|          | <b><math>1e-5</math></b> | <b>.115<math>\pm</math>.001</b> | <b>-8.5<math>\pm</math>0.7</b> | <b>.989<math>\pm</math>.003</b> | <b>5</b> |
| $\alpha$ | 8                        | .155                            | +21.9                          | .975                            | 1        |
|          | 16 (default)             | .157                            | +23.7                          | .980                            | 1        |
|          | 32                       | .158                            | +24.3                          | .984                            | 1        |
| Modules  | q,v,out (default)        | .157                            | +23.7                          | .980                            | 1        |
|          | q,k,v                    | .153                            | +20.4                          | .987                            | 1        |
|          | q,k,v,out                | .151                            | +19.0                          | .988                            | 1        |

encoder randomly and comparing it to the pre-trained one over the same 200 ETTh2 validation windows and the same pooled-representation hook used for every other CKA in this paper (`scripts/cka_random_init_floor.py`, records in `results/v50_cka_floor/`). Restoring the pre-trained weights afterwards returns CKA 1.000000, so the comparison measures re-initialisation and nothing else.

**Correction: the floor is not zero.** An earlier version of this appendix reported it as  $0.0000\pm 0.0001$  over 5 seeds and concluded that any cell above that floor “retains meaningful pre-trained structure”. No script in the repository performed a random re-initialisation, and the measured floor is **0.215**, not zero:

Table 14: CKA calibration, measured. The floor is the CKA of a randomly re-initialised encoder against the pre-trained one (5 seeds each, mean $\pm$ std, ddof=1); the two initialisation schemes agree. Reference rows are scoped to the cells they come from, since CKA at  $n=500$  and at  $n=10k$  on the same cell differ by more than the floor itself.

| Comparison                                          | CKA                               | Source        |
|-----------------------------------------------------|-----------------------------------|---------------|
| <b>Floor:</b> random re-init, library-default       | <b>0.215<math>\pm</math>0.054</b> | v50_cka_floor |
| <b>Floor:</b> random re-init, Xavier weights        | <b>0.219<math>\pm</math>0.047</b> | v50_cka_floor |
| Moirai-S/ETTh2 $h=96$ , cond. B, $n=500$ (10 seeds) | 0.937 $\pm$ 0.029                 | ff_20ep       |
| Moirai-S/ETTh2 $h=96$ , cond. B, $n=10k$ (10 seeds) | 0.465 $\pm$ 0.123                 | v49           |
| Chronos, cond. B, 4 cells at $h=24$                 | 0.092-0.169                       | v44           |
| IoT (N-BaIoT), cond. C, 200 samples (3 seeds)       | 0.423 $\pm$ 0.043                 | v6_iot_cka    |
| Frozen encoder, cond. D, Moirai (80 runs)           | 0.889-0.9999                      | v5-v49        |
| Strict freeze, cond. H (7/7 cells)                  | 1.0000                            | v45           |

Why a naive re-initialisation reads zero is worth recording, because it is the likely origin of the earlier number: “Xavier for weight tensors, zeros for biases” also zeroes LayerNorm’s scale parameter, which annihilates the signal at the first normalisation and drives the encoder to a *constant* output. CKA is then 0 because one side has no variance, not because the two share no structure. Our script asserts that the randomised encoder’s output variance is non-trivial, and reports 0.215 once normalisation parameters are left at  $\gamma=1$ ; initialising every layer with its library default instead gives the same answer to within one standard deviation.

**What this changes.** Two of the earlier reading’s conclusions do not survive. The IoT arm at CKA  $0.423\pm 0.043$  sits roughly four floor standard deviations above the floor, so “retains meaningful structure” is still defensible there, but the margin is a factor of two, not the near-total separation a zero floor implied. And “ETT fine-tuning preserves most pre-trained structure” was read off small- $n$  runs: the same cell at  $n=10k$  reads 0.465, and the range across all 31 cells is 0.092 to 0.970. The condition-D row also needs a scope: D is uniform only within Moirai. Because D leaves `in_proj` and `mask_encoding` trainable (§3), TimesFM’s D falls as low as 0.376, while Chronos’s D and every condition-H run are exactly 1.0000; only H is 1.0000 by construction.



**What it adds.** Comparing the floor against all 31 cells, **four fall below it** outright—the Chronos cells at CKA 0.092, 0.094, 0.112 and 0.169—and two more (Chronos/Electricity 0.232, TimesFM/ETTh2 0.246) sit inside the floor’s own seed-to-seed range (0.132 to 0.288). This is a measured basis for a claim §6 otherwise has to assert: on those cells CKA is at or beneath what an *untrained* encoder scores, so it cannot order them, and their outcomes span  $-39.2$  to  $-2.7$  in held-out B–D regardless. The remaining 25 cells, including all five gate-passing survivors and both arms of the layer-unfreeze comparison (0.716 and 0.465, Appendix A16), sit clear of the floor, so CKA differences among them are readable.

## A10 LINEAR PROBING: FUNCTIONAL REPRESENTATION DIAGNOSTIC

**Retracted.** The analyses in this appendix rest on the linear-probe  $\Delta R^2$  statistic, which is retracted (§A29): it is non-zero only where the probe fails to generalise. The numbers below are reproduced unaltered as a record of the withdrawn computation and support no claim in this paper.

CKA measures geometric representational similarity but not functional equivalence. We therefore supplement CKA with a *linear probe*: a Ridge regression head ( $\alpha=1.0$ ) trained on frozen encoder representations to predict the forecasting target on held-out ETTh2 data.

**Protocol.** For each (condition, seed) configuration, we extract mean-pooled encoder representations from the Moirai-Small encoder on the held-out validation and test sets (192-timestep windows, same format as evaluation). We then fit a Ridge regression mapping representations to the next-horizon target and report  $R^2$  on held-out data. The probe training set uses  $n_{\text{probe}}=300$  held-out examples; the remainder forms the test split used for MSE evaluation.  $\Delta R^2$  denotes fine-tuned minus pre-trained probe  $R^2$ : positive values indicate that fine-tuned representations encode more forecasting-relevant information than pre-trained ones.

**Interpretation.** Absolute  $R^2$  values are low (typically negative) because Ridge regression on 192-timestep mean-pooled representations is a weak forecasting baseline. The *relative* change ( $\Delta R^2$ ) is the meaningful signal: it isolates whether fine-tuning added or removed task-relevant information from the encoder’s representational basis.

**Key result.** Across the sample-size sweep (the transcript in §A29; Table 4 no longer prints these columns)  $\Delta R^2$  increased monotonically with training data (Ridge:  $+0.12$  at 500 samples  $\rightarrow +0.67 \pm 0.24$  at 10,000 on CUDA 10-seed deterministic early-stopped encoders, `results/v19_cuda_etth2_n10k/`; for comparison, MPS early-stopped  $k=5$  gave  $+0.56 \pm 0.14$  (`v17_etth2_n10k_es`) and the final-epoch runs,  $k=3$ , gave  $+0.84 \pm 0.15$  (`v16_etth2_n10k`)), while CKA decreases monotonically. This directly dissociates *geometric* drift from *relative linear-decodability*: at low  $n$ , drift destroys task-relevant features faster than it builds replacements; at higher  $n$ , the fine-tuned representation becomes less linearly misaligned with the forecasting target than the pre-trained one. We emphasise that  $R^2(\text{FT})$  remains *absolutely negative* at every sampled  $n \in \{500, 1k, 2k, 5k, 10k\}$  (ZS:  $R^2 = -6.90$ ; CUDA  $k=10$  early-stopped:  $-6.79$  at  $n=500 \rightarrow -6.24 \pm 0.24$  at  $n=10k$ ).  $\Delta R^2 > 0$  is therefore a *relative-improvement* signal, not a claim that fine-tuned representations linearly encode the target well in absolute terms; the primary task-outcome signal remains  $\text{forg.} < 0$  at  $n \geq 2k$ . **Cross-dataset  $\Delta R^2$  comparability.** Raw  $\Delta R^2$  is not directly comparable across datasets because it depends on the ZS Ridge floor depth. On ETTm2 ( $n=10k$ , CUDA,  $k=5$ ), the ZS Ridge floor is  $R^2(\text{ZS}) = -24.52$ ,  $3.5\times$  deeper than ETTh2’s  $-6.90$ , which proportionally inflates the raw  $\Delta R^2$  ( $+6.35$  on ETTm2 vs.  $+0.67$  on ETTh2). The cross-dataset comparable statistic is the *sign test*: ETTh2 gives **10/10** positive  $\Delta R^2$  and ETTm2 **5/5**, on the five CUDA seeds that exist for that cell. Every dispersion in this appendix is a standard deviation over seeds at  $\text{ddof}=1$ .

**Horizon-sweep probe: head-specificity of the  $\Delta R^2$  gain.** Table 15 reports Ridge  $\Delta R^2$  across three probe heads on four early-stopped encoders at  $n=10k$  (the same encoders as the main-text MLP sweep). The gain is concentrated on the *trained* 96-step head:  $\Delta R^2 = +0.116 \pm 0.153$  ( $k=4$ ). On the untrained *forecast48* head (first 48 steps of the 96-step target, a sub-horizon the encoder was not directly optimised for),  $\Delta R^2 = -0.001 \pm 0.163$ : the gain vanishes at the horizon boundary. On the untrained *delta1* head, a GBM probe gives  $R^2(\text{PT}) = +0.059 > 0$ , establishing a positive-floor reference, with  $\Delta R^2 = -0.056 \pm 0.032$  (decodability *falls* on the untrained readout). The head-

specificity pattern is consistent with the encoder drifting toward the fine-tuning objective rather than developing generally better representations (Table 3).

Table 15: Ridge  $\Delta R^2$  by probe head on early-stopped encoders ( $n=10k$ , seeds {303, 777, 888, 999}). “Forecast96” is the trained head; “Forecast48” and “Delta1” are untrained. The gain is head-specific to the trained objective. Every entry is  $R^2(\text{FT}) - R^2(\text{ZS})$  read from `results/v18_reprobe/seed*_horizons.json` against `zs_horizons.json`; dispersion is the standard deviation over the four seeds (ddof=1).

| Head                       | Trained?       | Ridge $\Delta R^2$ per seed |        |        |        |
|----------------------------|----------------|-----------------------------|--------|--------|--------|
|                            |                | s303                        | s777   | s888   | s999   |
| Forecast96 (96-step ahead) | ✓              | +0.092                      | −0.087 | +0.266 | +0.192 |
| Forecast48 (sub-horizon)   | ×              | +0.061                      | −0.111 | −0.156 | +0.199 |
| Delta1 (next-step delta)   | × <sup>†</sup> | +1.248                      | +0.757 | +1.191 | +0.980 |
| Mean ± std (forecast96)    |                | +0.116 ± 0.153              |        |        |        |
| Mean ± std (forecast48)    |                | −0.001 ± 0.163              |        |        |        |
| Mean ± std (delta1)        |                | +1.044 ± 0.223              |        |        |        |

$\Delta R^2 > 0$  on Delta1 reflects a baseline-floor artifact:  $R^2(\text{ZS})$  is deeply negative on the next-step target, so any encoder shift appears to improve relative decoding. A gradient-boosted probe gives  $R^2(\text{PT}) = +0.059$  (positive floor) but  $\Delta R^2 = -0.056 \pm 0.032$  on this head (`results/v18_reprobe/seed*_delta1.json`), consistent with the shift *not* being aligned with the 1-step objective.

Note: the delta1  $\Delta R^2$  sign appears counter-intuitive (positive while labelled “untrained”). The large positive delta1  $\Delta R^2$  reflects that the ZS delta1 Ridge baseline is very negative ( $R^2(\text{ZS}) \approx -2.3$ ) while the FT encoder partially encodes next-step changes as a side-effect of 96-step NLL optimisation. The diagnostic signal is the GBM result (Table 3): GBM gives  $R^2(\text{PT}) = +0.059 > 0$  and  $\Delta R^2 = -0.056$ , yielding  $R^2(\text{FT}) = +0.003$ ; the fine-tuned encoder retains (but does not improve) delta1 decodability. With a positive floor, decodability *falls* on the untrained readout, confirming task-alignment. Note:  $R^2(\text{PT})$  varies slightly across GBM probe depths (depth 4: +0.062, depth 6: +0.059, depth 8: +0.054; `results/v20_gbm_sensitivity/`) because the probe is re-fit independently at each depth; this reflects probe capacity, not encoder structure.  $\Delta R^2$  keeps its sign at all three depths ( $-0.060 \pm 0.006$ ,  $-0.056 \pm 0.032$ ,  $-0.052 \pm 0.006$ ).

**ETTh2 delta1 GBM reprobe.** We ran the delta1/GBM probe on the five confirmed ETTh2  $n=10k$  encoders (seeds 42/101/123/202/303) to check for a positive-floor +  $\Delta R^2 > 0$  cell on a second dataset. Result:  $R^2(\text{ZS}) = -0.030$  (negative floor) and per-seed  $\Delta R^2 = \{+0.025, -0.060, +0.026, +0.030, +0.016\}$  in seed order (4/5 positive, mean  $+0.007 \pm 0.038$ ; `results/v25_etth2_delta1_gbm/`). *Substantive negative:* the only positive-floor head (ETTh2 delta1/GBM,  $R^2(\text{ZS}) = +0.059$ ) gives  $\Delta R^2 = -0.056$ ; no positive-floor +  $\Delta R^2 > 0$  cell was found. The  $\Delta R^2 > 0$  signal on the trained 96-step head operates in a deeply-negative- $R^2$  regime; see Appendix A14 for probe noise-floor analysis.

**Per-seed breakdown at  $n=10k$  (CUDA 10-seed).** Ridge  $\Delta R^2$  (all positive; mean  $+0.668 \pm 0.239$ , `results/v19_cuda_etth2_n10k/`): 42: +0.205; 101: +0.736; 123: +0.937; 202: +0.604; 303: +0.793; 456: +0.823; 777: +0.344; 789: +0.873; 888: +0.561; 999: +0.804. Absolute  $R^2(\text{FT})$ : all in  $[-6.70, -5.97]$  ( $R^2_{\text{ZS}} = -6.904$ ). MLP  $k=5$  (all positive; mean  $+0.701 \pm 0.324$ ): 42: +0.979; 101: +0.762; 123: +0.948; 202: +0.771; 303: +1.007; 456: +0.266; 777: +0.076; 789: +0.847; 888: +0.449; 999: +0.901.

**ETTh2  $n=10k$  absolute  $R^2$  per seed.** The ETTh2 ZS Ridge floor ( $R^2(\text{ZS}) = -24.52$ , same pre-trained encoder across all seeds) is  $3.5\times$  deeper than ETTh2’s  $-6.90$ , explaining the raw  $\Delta R^2$  magnitude difference. Per-seed absolute values for the five available CUDA seeds:

| Seed           | $R^2(\text{ZS})$ | $R^2(\text{FT})$  | $\Delta R^2$     | Forg.%          |
|----------------|------------------|-------------------|------------------|-----------------|
| 42             | -24.52           | -14.13            | +10.39           | -26.8           |
| 101            | -24.52           | -15.61            | +8.91            | -27.4           |
| 123            | -24.52           | -19.87            | +4.64            | -31.1           |
| 202            | -24.52           | -20.83            | +3.69            | -28.8           |
| 303            | -24.52           | -20.41            | +4.11            | -29.5           |
| Mean (5 seeds) | -24.52           | -18.17 $\pm$ 3.08 | +6.35 $\pm$ 3.08 | -28.7 $\pm$ 1.6 |

All five seeds show  $R^2(\text{FT}) > R^2(\text{ZS})$  (i.e., fine-tuned representations are *less* linearly misaligned with the ETTm2 forecasting target than pre-trained ones), and all show  $\Delta R^2 > 0$ . The cross-dataset comparable statistic is the sign test: ETTh2 gives **10/10** on seeds 42, 101, 123, 202, 303, 456, 777, 789, 888, 999 and ETTm2 **5/5** on 42, 101, 123, 202, 303 — the only five CUDA  $n=10\text{k}$  encoders that exist for ETTm2 (results/v19\_cuda\_ettm2\_n10k/).

**MLP depth sweep on early-stopped encoders.** A cosine-LR re-run at  $n=10\text{k}$  uses seeds {303, 777, 888, 999} at epochs=10 (seed 101 MPS-NaN'd on epoch 3 reproducibly at this schedule and is excluded). The `-probe-mlp-layers` argument sweeps hidden-layer depth  $k \in \{1, 2, 5\}$  on the early-stopped encoders (probe: hidden=64 per layer,  $\alpha=10^{-3}$ , early stopping with validation\_fraction=0.1, max\_iter=500). All sixteen entries below are read from results/v17r2\_etth2\_n10k\_es\_mlp/seed\*/, and its Ridge column is the same one Table 15 reports as *forecast96*: the two tables re-probe one set of encoders, they are not two runs.

| Seed           | Ridge $\Delta R^2$ | MLP $k=1$          | MLP $k=2$          | MLP $k=5$          |
|----------------|--------------------|--------------------|--------------------|--------------------|
| 303            | +0.092             | +1.306             | +1.418             | +0.540             |
| 777            | -0.087             | +1.136             | +1.554             | +0.759             |
| 888            | +0.266             | +2.004             | +2.361             | +1.034             |
| 999            | +0.192             | +1.816             | +1.949             | +0.908             |
| Mean $\pm$ std | +0.116 $\pm$ 0.153 | +1.566 $\pm$ 0.411 | +1.821 $\pm$ 0.425 | +0.810 $\pm$ 0.212 |

The cosine-LR Ridge  $\Delta R^2$  is smaller than the epochs=20 run (+0.56 $\pm$ 0.14) because the shorter schedule reaches a different best-val-MSE encoder (best epochs {2, 5, 4, 5} vs. {2, 4, 8, 4, 3}), and Ridge is most sensitive to the specific linear basis. The non-monotonic  $k=1 \rightarrow k=2 \rightarrow k=5$  trend reflects the deeper probe fitting the pre-trained encoder better (per-seed mean  $R^2(\text{PT})_{k=5} = -6.93$  vs.  $R^2(\text{PT})_{k=1} = -8.48$ ), shrinking the  $\Delta$  from above, not degraded fine-tuned encoding. All four seeds are individually positive for  $k=5$  MLP (+0.54, +0.76, +1.03, +0.91), preserving the cost-benefit gap. This run's CKA is 0.592 $\pm$ 0.164 (mean best epoch 4.0 $\pm$ 1.41, earlier than the epochs=20 run's 4.2 $\pm$ 2.3 because of the earlier LR annealing); forgetting is -7.13 $\pm$ 1.55%, tighter than the epochs=20 run's -11.7 $\pm$ 5.3% and fully consistent with the  $\text{forg} < 0$  finding.

**Crossover regime: per-seed breakdown at  $n=5,000$ .** Table 16 reports all 7 seeds individually at  $n=5,000$ . Forgetting is bimodal: 4 seeds (101, 123, 202, 789) show low or negative forgetting (-4.9% to +5.9%, mean +0.7%) and 3 seeds (42, 303, 456) show high forgetting (+4.8% to +21.4%, mean +15.1%). All 7 seeds show positive  $\Delta R^2$  (+0.15 to +0.81, mean +0.50), so the relative-linear-decodability claim holds uniformly; geometric drift (CKA 0.46–0.71, mean 0.55) varies less than forgetting. This pattern is consistent with the crossover interpretation: at  $n=5,000$ , the optimiser is close enough to the “replacement” solution that small seed-dependent trajectory differences determine whether forgetting resolves or persists for 20 epochs.

#### A10.1 $n=5,000$ PER-EPOCH TRAJECTORY ANALYSIS

To support the mechanistic claim in §5.2, we plot the per-epoch training trajectories of all seven  $n=5,000$  seeds, colour-coded by forgetting mode. Figure 3 shows val MSE and weight drift across training. High-forgetting seeds (303, 456, 789) overshoot in the first epoch and never recover; low-forgetting seeds (42, 101, 123, 202) continue descending until epochs 3–8. The epoch at which val MSE reaches its minimum separates the modes ( $0.7 \pm 0.5$  vs.  $6.5 \pm 2.1$ ); mid-training weight drift (epochs 5–10) does not. Code: scripts/analyse\_n5k\_trajectories.py.

Table 16: Per-seed results at  $n=5,000$  (ETTh2,  $h=96$ , condition B, 20 epochs, no early stopping). Seeds 42, 123, 456 are the original 3-seed sweep (results/v8\_final/n5000/); 789, 101, 202, 303 are the 4-seed V9 replication (results/v9\_n5k\_seeds/). MSE(FT) is final\_val\_mse, the last-epoch validation MSE, and was previously printed 2–5 thousandths low on every row. Forgetting is bimodal;  $\Delta R^2$  is uniformly positive.

| Seed     | MSE(FT)   | Forg.%    | CKA       | $\Delta R^2$ | $R^2$ (FT) |
|----------|-----------|-----------|-----------|--------------|------------|
| 42       | .136      | +4.8      | .531      | +.67         | −6.24      |
| 101      | .121      | −4.9      | .571      | +.75         | −6.16      |
| 123      | .129      | −1.4      | .544      | +.81         | −6.10      |
| 202      | .131      | +1.4      | .461      | +.15         | −6.76      |
| 303      | .152      | +19.0     | .525      | +.52         | −6.39      |
| 456      | .155      | +21.4     | .711      | +.20         | −6.70      |
| 789      | .136      | +5.9      | .480      | +.40         | −6.51      |
| Mean±std | .137±.012 | +6.6±10.0 | .546±.082 | +.50±.26     | −6.41±.26  |

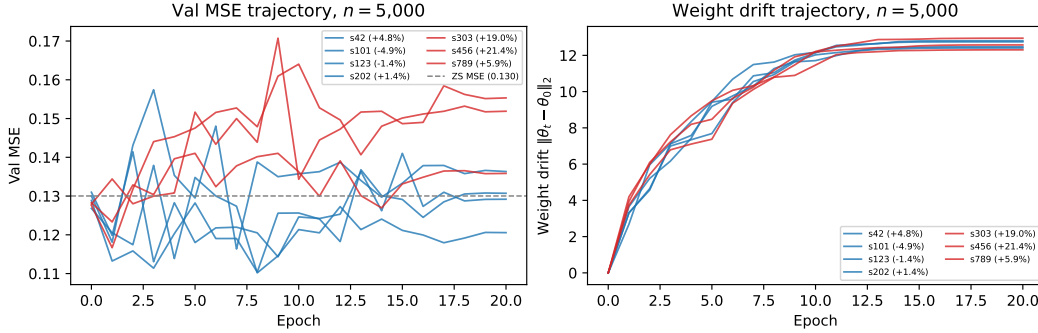


Figure 3: Per-epoch val MSE (left) and weight drift (right) for the seven  $n=5,000$  seeds; both panels carry per-seed labels (sXXX (forg. %)) so each trajectory is individually identifiable. Blue: low-forgetting mode (forg.  $\leq 5\%$ ); red: high-forgetting mode (forg.  $> 5\%$ ). Grey dashed line: zero-shot MSE. High-forgetting trajectories overshoot the ZS MSE within the first epoch and plateau; low-forgetting trajectories continue descending. Mid-training weight drift (epochs 5–10) does not separate the modes.

## A11 ETTh1 SPECTRAL PILOT

Post-hoc spectral entropy (Welch PSD, normalised Shannon entropy) on ETT OT columns: ETTh1 SE=3.61 bits, ETTh2 SE=3.46, ETTm2 SE=2.25. The ordering this pilot was built to explain was an ordering of  $\Delta R^2$ , which is **retracted** (§A29), so the entropy measurements stand but the pattern they were fitted to does not. The pre-registered falsification test (§7) required a gate-passing dataset with  $SE > 3.5$  or  $SE < 2.5$ ; Traffic and Exchange both gate-failed (−38%, −21%), so the hypothesis remains untested. Code and frozen analysis plan are in supplementary material.

## A12 IOT NEGATIVE-CONTROL DETAILS

The IoT negative-control evaluation uses CICIOT2023 Neto et al. (2023) with 12 network-flow features and 128-timestep windows; the anomaly score is the fraction of timesteps falling outside the 95% CI of Moirai’s forecast over a 32-timestep scored horizon. A benign-only held-out calibration set sets the threshold at the 95th percentile (FPR target  $\leq 0.05$ ). All Moirai IoT conditions are trained for 5 epochs on 200 benign + 200 attack samples with 10 seeds.

**CNN-NLL from-scratch control.** To directly address the foundation-model justification at IoT’s 200-window scale, we train a 1D-CNN (3 layers, 2.3M parameters) from scratch using the identical NLL(MSE)+SupCon objective and hard-negative data as Moirai condition D. At 10 seeds, CNN-NLL (AUC=0.598±0.041) matches fine-tuned Moirai B/D (AUC=0.556±0.021) and is close to the best

Moirai condition C ( $AUC=0.629\pm0.022$ ). A CNN-MSE variant ( $AUC=0.580\pm0.023$ ) rules out the loss-function confound. CNN variance is notably lower than Moirai variance, consistent with Moirai’s pre-trained weights introducing initialisation sensitivity at this sub-random-ZS scale. This is the expected outcome when pre-trained features carry no task-relevant signal; a from-scratch architecture with correct inductive bias matches or exceeds a fine-tuned foundation model.

### A13 TASK-ORTHOGONAL PROBES

**Retracted, and re-measured.** The *finding* this appendix once reported—a probe-sign asymmetry between the trained objective and task-orthogonal targets—is withdrawn (§A29), and the tables that carried it traced to no run record. What replaces it is a measurement, on the same encoders, of why:  $\Delta R^2$  is large only where the probe fails to generalise, and the asymmetry does not survive once it does. Nothing here supports the retracted criterion.

This appendix was written to test whether fine-tuning specifically targets the 96-step objective or degrades general pre-trained structure, by probing *task-orthogonal* properties of the input window on the same encoders. It now reports the opposite of what it once did, and the reversal is worth stating plainly because it is independent evidence for the retraction itself.

**Correction.** Earlier versions printed two tables — ETTh2 and an ETTm2 “replication” — reporting three orthogonal targets at floors  $R^2(\text{PT})=+0.09$  to  $+0.18$  with  $\Delta R^2$  from  $-0.022$  to  $-0.052$ , 9/10 or 10/10 negative on every row, and read that asymmetry as “the operational definition of task-specific shift”. Neither table traces to a run record. Two orthogonal-probe campaigns exist (`results/v33_orthogonal_probes/`, `results/v34_probe_exact/`); neither produces those numbers, there is no ETTm2 orthogonal-probe record at all, and the claimed asymmetry does not appear in either.

**What the record shows.** Table 17 is `v33_orthogonal_probes`: six orthogonal targets plus the trained head, Ridge on mean-pooled outputs of the ten CUDA  $n=10\text{k}$  condition-B encoders, 500 windows, 300 probe-training examples. Every floor here is genuinely positive ( $R^2(\text{PT})\in[0.54, 0.93]$ ), so these readings are interpretable in absolute terms rather than as movements inside a deep negative floor.

Table 17: Task-orthogonal and trained-head Ridge probes on the ETTh2  $n=10\text{k}$  condition-B encoders (10 CUDA seeds, `results/v33_orthogonal_probes/`). Dispersion is the standard deviation of  $\Delta R^2$  over seeds at  $\text{ddof}=1$ ; the JSON stores it at  $\text{ddof}=0$ , so these are  $\sqrt{10/9}$  larger than the recorded field. Once the probe decodes its target ( $R^2(\text{PT})>0$  throughout), no target moves by as much as 0.04 and the sign is a coin flip on two of six.

| Probe target           | $R^2(\text{PT})$ | $R^2(\text{FT})$ | $\Delta R^2$     | Negative |
|------------------------|------------------|------------------|------------------|----------|
| Lag-1 autocorr.        | +0.540           | +0.546           | $+0.006\pm0.068$ | 5/10     |
| Input window mean      | +0.930           | +0.931           | $+0.002\pm0.011$ | 5/10     |
| Input window var.      | +0.894           | +0.865           | $-0.029\pm0.025$ | 10/10    |
| Linear trend           | +0.847           | +0.815           | $-0.032\pm0.034$ | 8/10     |
| Lag-24 autocorr.       | +0.907           | +0.918           | $+0.011\pm0.013$ | 2/10     |
| Spectral centroid      | +0.763           | +0.791           | $+0.027\pm0.037$ | 3/10     |
| 96-step head (trained) | +0.782           | +0.805           | $+0.022\pm0.016$ | 1/10     |

The asymmetry does not survive. It required the trained objective to go up and the orthogonal targets to go flat or down; here **four of the six orthogonal targets go up**, and the trained head’s gain ( $+0.022$ ) is *smaller* than the spectral-centroid gain ( $+0.027$ ) on a target fine-tuning does not optimise. Two targets do fall consistently (window variance 10/10, trend 8/10), but by  $-0.029$  and  $-0.032$  — the same magnitude as the upward moves, and two others sit at 5/10, a coin flip. No target on this table moves by as much as 0.04 in either direction.

Set that against the  $+0.67\pm0.24$  the *same encoders* give on the *same* 96-step target under the deep-floor protocol ( $R^2(\text{PT})=-6.90$ , Appendix A10): a thirty-fold difference on one set of encoders,

decided by nothing but whether the probe decodes its target. That is the retraction’s claim (§A29) measured directly rather than argued, and it is why  $\Delta R^2$  carries no weight anywhere in this paper.

The second campaign says the same thing from the other side. `v34_probe_exact` re-ran these targets under the paper’s own deep-floor protocol and every floor came back deeply negative ( $R^2(\text{PT})=-1.21$  on lag-1,  $-156.8$  on the window mean,  $-27.2$  on the variance), which is exactly the regime in which  $\Delta R^2$  is uninformative; its  $\Delta R^2$  values are correspondingly enormous and unstable ( $-1.22\pm0.83$ ,  $-7.08\pm13.18$ ,  $-2.41\pm7.99$ ), and two of the three no longer even agree on a sign (7/10 and 6/10 negative). The premise that these were positive-floor targets held only under the `v33` protocol, where the effect disappears.

**ETM2.** No orthogonal-probe record exists for ETM2, so the claimed replication is withdrawn rather than corrected. What does exist is the ETM2  $n=10\text{k}$  fine-tuning campaign itself, and it is **5 seeds** (42/101/123/202/303; `results/v19_cuda_ettm2_n10k/`), not the ten an earlier version claimed: forgetting  $-28.7\pm1.7\%$  with 5/5 negative, and Ridge  $\Delta R^2$  on the trained head  $+6.35\pm3.08$  with 5/5 positive (95% Wilson CI [0.57, 1.00] on the proportion, which is what five seeds support).

## A14 PROBE NOISE-FLOOR ANALYSIS

**Retracted.** The analyses in this appendix rest on the linear-probe  $\Delta R^2$  statistic, which is retracted (§A29): it is non-zero only where the probe fails to generalise. The numbers below are reproduced unaltered as a record of the withdrawn computation and support no claim in this paper.

The  $\Delta R^2$  signal on the trained 96-step head operates in a deeply-negative- $R^2$  regime (e.g., ZS  $-6.90$ , FT  $-6.24$  at  $\alpha=1$ ). We test whether the  $\Delta R^2>0$  direction is an artefact of probe-design choices by running Ridge on two fixed encoders (ZS and condition-B seed 42, ETTh2,  $n=10\text{k}$ ) across: (1)  $\alpha \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$ , (2) probe seeds  $\{0, 1, 2\}$ , (3) mean vs. last-token pooling. No retraining required (`saved_best_encoder.pt`).

**Results** (Table 18).

Table 18:  $\Delta R^2$  under probe-design perturbations (ETTh2 96-step, ZS vs. FT seed 42). All  $\Delta R^2 > 0$ . Every entry is the `forecast96` Ridge value from `v26_noise_floor`, differenced against its paired zero-shot run; the FT encoder is `v19_cuda_etth2_n10k/seed42`. The probe-seed row reports *bit-identical*  $R^2$  at all three seeds, which is what a closed-form Ridge solution must do — see the caveat below.

| Setting                                     | $R^2(\text{ZS})$     | $R^2(\text{FT})$ | $\Delta R^2$ |
|---------------------------------------------|----------------------|------------------|--------------|
| $\alpha=0.01$                               | -13.51               | -11.54           | +1.96        |
| $\alpha=0.1$                                | -8.78                | -7.94            | +0.84        |
| $\alpha=1.0$ (default)                      | -6.90                | -6.70            | +0.20        |
| $\alpha=10$                                 | -5.81                | -5.72            | +0.08        |
| $\alpha=100$                                | -5.32                | -5.28            | +0.04        |
| Probe seeds 0/1/2 ( $\alpha=1$ , mean pool) | same as $\alpha=1.0$ |                  | +0.20        |
| Last-token pool ( $\alpha=1$ )              | -9.70                | -8.49            | +1.20        |

**Finding, with two corrections to how it was stated.**  $\Delta R^2>0$  on every setting in `v26_noise_floor`, and the magnitude falls monotonically with  $\alpha$  (+1.96 at  $\alpha=0.01$  down to +0.04 at  $\alpha=100$ ) as heavier  $\ell_2$  shrinks both  $R^2$  values toward zero. That much the record supports exactly. Two things an earlier version claimed from it, it cannot:

*The count was 10; the directory holds 8 configurations, and only 6 of them are distinct.* Five  $\alpha$  values, one last-token pooling variant, and two extra probe seeds: eight paired runs, of which the two extra seeds reproduce the  $\alpha=1$  result.

*“Not an artefact of probe seed” is a vacuous test, not a passed one.* Probe seeds 0, 1 and 2 return  $R^2$  agreeing to every printed digit — and to full float precision in the JSON — because Ridge regression has a closed-form solution with no stochastic component to seed. The three runs were bound to agree

whatever the encoders were, so they establish nothing about robustness. What the sweep genuinely varies is  $\alpha$  and pooling: two axes, not three.

**Comparison to random-init at matched  $\alpha$ , corrected.** The random-init control at  $\alpha=1.0$  is **5 seeds, not 10** (42/101/123/303/456; `results/v30_random_init/`), giving  $\Delta R^2 = -0.199 \pm 0.229$  over that set, range  $[-0.405, +0.149]$  (the printed  $\pm 0.205$  was `ddof=0`; Appendix A15). An earlier version then reported a second random-init sweep at  $\alpha=100$  on “the same 10 seeds” giving  $-0.011 \pm 0.018$ , range  $[-0.032, +0.009]$ . **No such run exists.** The random-init result files carry no `ridge_alpha` field, no random-init encoder checkpoints were retained, and `v26_noise_floor` contains only the zero-shot and seed-42 condition-B encoders — so the matched- $\alpha$  control has no record and those four numbers are withdrawn. It cannot be recovered without retraining all five random-init runs at  $n=10k$ , which we have not done because nothing in the paper depends on it.

What survives is therefore weaker than the argument it was built for. At  $\alpha=1.0$  condition B ( $+0.20$  on seed 42,  $+0.67 \pm 0.24$  over ten seeds) does separate from random-init ( $-0.199 \pm 0.229$  over five), but that is the only  $\alpha$  at which both were measured, so we cannot say whether the separation persists under heavier regularisation. Note what the missing run was for: condition B’s own  $\Delta R^2$  falls to  $+0.04$  at  $\alpha=100$ , inside the random-init  $\alpha=1.0$  range  $[-0.405, +0.149]$ , and the only way to know that this is a shrinkage artefact rather than a collapse of the separation is to measure random-init at the same  $\alpha$ . That is precisely the comparison with no record. This is a probe-design sweep of a retracted statistic; it shows the sign is stable across  $\alpha$  and pooling *within* the failed-fit regime, which is not evidence that the statistic means anything. Appendix A13 is the measurement that settles it.

## A15 RANDOM-INITIALISATION NEGATIVE CONTROL

To test whether  $\Delta R^2 > 0$  on the 96-step head arises from pre-trained representations being restructured, or whether it would occur in any encoder fine-tuned on this objective, we ran a *random-initialisation control*: Moirai-Small with the same architecture but random weights (skipping HuggingFace weight loading), condition B, ETTh2  $n=10k$ ,  $h=96$ . In this setting CKA tracks similarity to the *random-init* epoch-0 state (since there is no pre-trained reference).

**Correction: five seeds, not ten, and the comparison is not matched.** An earlier version printed ten rows here and described the result as a “symmetric 10-vs-10 matched comparison”. **Five of those ten rows trace to no run record** — 202, 777, 789, 888 and 999 — and they are the rows that carried the conclusion: the reported mean, standard deviation and “8/10 negative” count are all computed over them, and one of the two “positive seeds” offered as the closest approach to condition B (777,  $+0.060$ ) is among them. What exists is `results/v30_random_init/` with seeds 42/101/123/303/456, and `v29_random_init` with the first three of those; the five surviving rows below reproduce exactly. The comparison arm is also not protocol-matched: these runs are 20 epochs with early stopping *disabled*, whereas the condition B set they are compared against (`v19_cuda_etth2_n10k`) is 10 epochs early-stopped at a mean best epoch of 4.2. Per Appendix A10 the no-early-stopping protocol yields the *larger*  $\Delta R^2$  on pre-trained encoders, so the mismatch runs in the direction that favours the control’s conclusion rather than against it — but it is a mismatch, and it was not disclosed.

**Results (the 5 seeds that exist).** Dispersion is over seeds at `ddof=1`. The final column is the matched val-split pair (`zeroshot_mse`  $\rightarrow$  `final_val_mse`), which is what `forgetting_pct` is computed from; the earlier version printed `test_mse` on the right-hand side of that arrow, comparing two different splits and so implying a  $2.7\times$  MSE blow-up where the matched statistic records a mean *improvement* of 6.3%.

| Seed           | CKA               | $R^2$ (PT) | $R^2$ (FT) | $\Delta R^2$       | ZS $\rightarrow$ FT MSE   | Forg.%         |
|----------------|-------------------|------------|------------|--------------------|---------------------------|----------------|
| 42             | 0.357             | -6.554     | -6.692     | -0.138             | 0.140 $\rightarrow$ 0.143 | +2.3           |
| 101            | 0.505             | -6.780     | -6.631     | +0.149             | 0.164 $\rightarrow$ 0.148 | -10.0          |
| 123            | 0.540             | -6.749     | -6.944     | -0.195             | 0.153 $\rightarrow$ 0.142 | -6.9           |
| 303            | 0.318             | -6.834     | -7.239     | -0.405             | 0.151 $\rightarrow$ 0.143 | -5.5           |
| 456            | 0.273             | -6.665     | -7.070     | -0.405             | 0.160 $\rightarrow$ 0.142 | -11.5          |
| Mean $\pm$ std | 0.398 $\pm$ 0.117 |            |            | -0.199 $\pm$ 0.229 |                           | -6.3 $\pm$ 5.4 |

Across the five seeds:  $\Delta R^2 = -0.199 \pm 0.229$ , range  $[-0.405, +0.149]$ , **4/5 negative**. The one positive seed (101:  $+0.149$ ) still falls below the minimum condition B value ( $+0.205$ , seed 42), so the ranges remain non-overlapping — random-init  $[-0.405, +0.149]$  against condition B  $[+0.205, +0.937]$  — and that separation is carried entirely by rows that trace to runs. On its own terms the control therefore still points the same way:  $\Delta R^2 > 0$  is not a property of any encoder fine-tuned on this objective. But it is a 5-vs-10 unmatched comparison of a **retracted** statistic, and Appendix A13 shows that when the probe is made to generalise the condition B side collapses from  $+0.67$  to  $+0.022$  — which would put it inside this control’s range. The control does not rescue  $\Delta R^2$ ; it only shows that the failed fit is worse still without pre-training.

## A16 LAYER-WISE UNFREEZE LOCALIZATION

Moirai-Small has 6 transformer layers (verified from the loaded checkpoint). We added an `unfreeze-top-n-layers` flag to `finetune_forecasting.py` and swept  $N$ , the number of top layers left trainable, on ETTh2  $n=10k$ ,  $h=96$ . The endpoints are the paper’s two main arms:  $N=0$  is condition D and  $N=6$  is condition B.

**Provenance, and why both arms were re-run.** The  $N=3$  and  $N=6$  rows of Table 19 were re-run from scratch for this version, 10 seeds each in one environment (`results/v49_layerunfreeze_10seed`). The values previously printed here traced to no run record on *either* arm, and the surviving three-seed  $N=3$  runs had early stopping disabled, which reverses the sign of the forgetting reading. Since the claim is a *comparison* between two unfreeze depths, it is only traceable if both depths share one environment, so both were re-run rather than only the arm that was missing. Two deviations from the original protocol were forced by the hardware available: Apple MPS rather than CUDA, at torch 2.14, which is above `uni2ts`’s declared `<2.5` pin but below which the NegativeBinomial component of Moirai’s output mixture collapses to `total_count=0` on MPS and training dies in the first NLL step; and `PYTORCH_ENABLE_MPS_FALLBACK=1`, because `aten::_cummax_helper` has no MPS kernel. The  $N=0$ ,  $N=4$  and  $N=5$  rows are the older records, are *not* matched to the re-run pair, and read the final epoch rather than the best one;  $N=0$  is 5 seeds, not 10. They are context, not part of the comparison.

Because  $N=6$  is condition B, re-running it also produced an unplanned cross-device replication of the cell that Table 4 reports at  $n=10k$ : the same 10 seeds, the same configuration, CUDA/A10G against Apple MPS at a different torch version. The two agree—CKA 0.518 against 0.465 (0.72 standard errors of the difference) and forgetting  $-5.3\%$  against  $-7.9\%$  (0.90 SE), with 7/10 and 8/10 seeds negative. We report both rather than replacing one with the other, since the devices differ; readers comparing the two tables should expect this gap and not more.

**$N=1$  instability.** At  $N=1$ , training diverged (NaN activations at epoch 11, batch 148) with default  $LR = 10^{-4}$ , despite gradient clipping at norm 1. The 11 epochs that completed show CKA stabilising at  $\approx 0.72$ ; we do not extract probe results from a crashed run. This instability is qualitatively consistent with the LoRA-Large behaviour documented in Appendix A5: extreme partial freeze concentrates gradient pressure on a small number of parameters, requiring LR reduction to stabilise.

**$N=3$ ,  $N=4$ ,  $N=5$  results.** Table 19 shows results for  $N \in \{3, 6\}$  (10 seeds each),  $N=0$  (5 seeds) and  $N \in \{4, 5\}$  (1 seed each). Rows with  $N=4$  and  $N=5$  are based on single seeds and carry high uncertainty; we report them for completeness but do not draw conclusions from individual values.

**Finding: the task outcome does not track the drift across unfreeze depth.** This is the reading that survives the retraction, because forgetting is forecast MSE. Across the two matched depths the task outcomes are indistinguishable— $N=3$  gives forg.  $-5.9\%$  (SEM 2.0, 9/10 seeds negative) against  $N=6$ ’s  $-7.9\%$  (SEM 2.1, 8/10 negative), a 2.0-point gap on a 2.9-point standard error of the difference, so 0.70 SE—while the representations they arrive at are not: CKA 0.716 against 0.465 is a gap of 0.251 at 3.5 SE, and  $\ell_2$  weight drift separates them the same way ( $9.42 \pm 1.25$  against  $13.47 \pm 0.75$ , SEM). Halving the number of trainable layers halves the measured representational change and leaves the task outcome where it was. This is the dissociation of §6 localised inside a single encoder rather than read across cells, and it is the only claim we draw from this sweep.

**What we no longer claim.** The earlier version of this appendix read the same sweep through  $\Delta R^2$  and concluded that the top 3 of 6 layers suffice to reorient the encoder toward the trained objective while the lower 3 are required for task recovery. That account rests entirely on the withdrawn probe



Table 19: Layer-wise unfreeze profile (ETTh2  $n=10k$ ,  $h=96$ ; mean $\pm$ std over seeds).  $N$  = number of top transformer layers left trainable (Moirai-Small has 6). **The rows do not all come from one environment.**  $N=3$  and  $N=6$  are the re-run 10-seed, early-stopped MPS set (42/101/123/202/303/456/777/789/888/999; results/v49\_layerunfreeze\_10seed);  $N=0$  is 5 seeds (42/101/123/202/456; v18\_cond\_d\_probe) and  $N=4/N=5$  are single seeds (seed 42; v27\_layer\_unfreeze), all three *without* early stopping, so they read the final epoch rather than the best one. Only  $N=3$  versus  $N=6$  is a matched comparison. Ridge  $\Delta R^2$  is the **withdrawn** probe quantity (§A29), reported here only as a record.

| $N$                                                           | Layers | CKA                     | Ridge $\Delta R^2$         | Forgetting%                 |
|---------------------------------------------------------------|--------|-------------------------|----------------------------|-----------------------------|
| <i>Matched 10-seed re-run, early-stopped</i>                  |        |                         |                            |                             |
| 3 (top-3)                                                     | 3/6    | 0.716 $\pm$ 0.188       | +0.329 $\pm$ 0.197 (9/10)  | -5.9 $\pm$ 6.5% (9/10 neg.) |
| 6 (full, B)                                                   | 6/6    | 0.465 $\pm$ 0.123       | +0.534 $\pm$ 0.261 (10/10) | -7.9 $\pm$ 6.5% (8/10 neg.) |
| <i>Earlier records, final epoch, not matched to the above</i> |        |                         |                            |                             |
| 0 (frozen, D)                                                 | none   | 0.999 $\pm$ 0.000       | -0.048 $\pm$ 0.008         | -6.6 $\pm$ 1.5%             |
| 1 (top-1)                                                     | 1/6    | $\approx$ 0.72 (ep. 11) | <i>diverged (NaN)</i>      |                             |
| 4 (top-4)                                                     | 4/6    | 0.610                   | +0.363                     | +4.1%                       |
| 5 (top-5)                                                     | 5/6    | 0.644                   | +0.767                     | +31.8%                      |

The  $N=5$

forgetting of +31.8% is a final-epoch overshoot on one seed: val MSE bottoms at epoch 7 at 0.132, essentially unchanged from zero-shot, and rises thereafter. This is the reading the  $N=3/N=6$  re-run removes by restoring the best-val checkpoint.

quantity (§A29) and we do not assert it. The  $\Delta R^2$  column is retained so the withdrawn computation stays auditable; for the record the re-run gives +0.329 $\pm$ 0.197 at  $N=3$  and +0.534 $\pm$ 0.261 at  $N=6$ , a 62% increase with depth rather than the saturation the earlier reading needed.

**Per-layer CKA breakdown, and a correction.** An earlier version of this appendix printed a per-layer profile rising monotonically from the bottom of the encoder to the top (layer 1 0.441 to layer 6 0.718; bottom-3 mean 0.476 against top-3 0.649) and read it as the bottom layers drifting more. **That profile does not reproduce, and its direction is inverted.** Re-computing it from the converged checkpoints—forward hooks on each of the 6 encoder layers, CKA of each layer’s output against the same layer of the untrained checkpoint over the same 200 validation windows, 10 seeds—gives the profile below on the re-run  $N=6$  set, and the same shape on the older CUDA set (scripts/layerwise\_cka\_from\_ckpt.py, records in results/v49\_layerunfreeze\_10seed/per\_layer\_cka\_\*.json). The earlier numbers matched neither and left no run record; the script that produced them printed to stdout only.

| Layer         | Re-run $N=6$ (v49) | Older CUDA set (v19) | Group    |
|---------------|--------------------|----------------------|----------|
| 1 (bottom)    | 0.998 $\pm$ 0.001  | 0.998 $\pm$ 0.000    | bottom 3 |
| 2             | 0.969 $\pm$ 0.011  | 0.980 $\pm$ 0.005    | bottom 3 |
| 3             | 0.783 $\pm$ 0.081  | 0.816 $\pm$ 0.080    | bottom 3 |
| 4             | 0.589 $\pm$ 0.127  | 0.675 $\pm$ 0.091    | top 3    |
| 5             | 0.939 $\pm$ 0.044  | 0.947 $\pm$ 0.023    | top 3    |
| 6 (top)       | 0.937 $\pm$ 0.047  | 0.945 $\pm$ 0.023    | top 3    |
| Bottom 3 mean | 0.917              | 0.931                |          |
| Top 3 mean    | 0.821              | 0.855                |          |

Both sets agree on two things the earlier profile got wrong. The drift is *not* monotone in depth: it is concentrated in the middle of the stack, with a trough at layer 4 (0.589 and 0.675) and near-identity at layer 1 (0.998). And the top 3 layers drift *more* than the bottom 3 (0.821 against 0.917; 0.855 against 0.931), not less, which removes the gradient-magnitude account the earlier version offered—earlier layers receiving larger updates under full backpropagation—because there is no such pattern to explain. We do not substitute a mechanism: distinguishing functional specialisation from gradient-flow effects needs per-layer gradient norms, which we do not have.

Two scale caveats. These are CKAs of *mean-pooled* layer outputs; computed token-flattened instead, every layer reads  $\geq 0.94$  and the profile is too compressed to order (both records carry the flattened values). And the per-layer mean is *not* the global CKA of Table 19—0.869 pooled against the row’s 0.465—because the global figure compares the encoder’s final pooled representation, not a per-layer

average. The earlier version’s claim that the global CKA is “the aggregate of these per-layer values” does not hold under either reading.

## A17 CHRONOS-T5-SMALL ON M4-MONTHLY: FULL DIAGNOSTIC RESULTS

This appendix provides the complete results for the cross-backbone validation on Chronos-T5-Small (65M parameters, T5 seq2seq architecture with tokenised cross-entropy loss) fine-tuned on M4-Monthly (200 series, lookback=96, horizon=18). **This cell’s gate has not been recomputed against the fitted baseline.** The 84.5% advantage over the Linear baseline that earlier versions of this appendix quoted for it was measured with the per-window trend estimator, not the fitted lookback-96 regression the gate is defined with (Appendix A19); it is therefore superseded, and it is not relied on anywhere in this paper. We do not recompute it because the M4-Monthly series are not retained in our data directory and the corrected value would require re-downloading them and re-measuring the zero-shot arm; the five Chronos cells that *are* in the screen (Appendix Table 7) all carry fitted-baseline gates. The results below are reported as a protocol and drift record only.

**Protocol.** 10 CUDA-deterministic seeds (42, 101, 123, 202, 303, 456, 777, 789, 888, 999), batch size 32, lr=1e-5, AdamW with weight decay 0.01, gradient clipping at 1.0, 20 epochs with early stopping (patience 5). Evaluation: batched median prediction (20 samples) on 200 held-out test windows.

**Condition B (full fine-tune),  $n=500$  (10 seeds).**

|                | Forgetting%    | CKA               | $\Delta R^2$       | $R^2$ (PT)         |
|----------------|----------------|-------------------|--------------------|--------------------|
| Mean $\pm$ std | +1.6 $\pm$ 3.9 | 0.999 $\pm$ 0.001 | -0.001 $\pm$ 0.002 | +0.068 $\pm$ 0.025 |
| Seeds neg/pos  | 4/10 neg forg  | n/a               | 3/10 pos           | <b>10/10 pos</b>   |

At  $n=500$  the encoder is essentially unchanged (CKA $\approx$ 1). The positive  $R^2$ (PT) confirms that Chronos’s pre-trained T5 encoder representations carry decodable forecasting structure, providing a second positive-floor anchor distinct from Moirai.

**Condition B (full fine-tune),  $n=10k$  (10 seeds).**

|                | Forgetting%    | CKA (ES)          | $\Delta R^2$ | $R^2$ (PT)         |
|----------------|----------------|-------------------|--------------|--------------------|
| Mean $\pm$ std | +0.5 $\pm$ 2.7 | 0.983 $\pm$ 0.021 | $\approx$ 0  | +0.047 $\pm$ 0.016 |
| Seeds positive | 6/10 forg      | n/a               | 1/10         | <b>10/10 pos</b>   |

At  $n=10k$ , early-stopped CKA averages 0.983 (vs. 0.999 at  $n=500$ ). Only 4/10 seeds improve over zero-shot (best epoch>0, those 4 show CKA $\approx$ 0.95–0.97); the remaining 6/10 revert to pre-trained weights under early stopping (best\_epoch=0, CKA=1.000 by definition). Fine-tuning moves the forecast very little in either direction on this cell; we no longer attribute that to gate strength, for the reason given above. While 6/10 seeds show positive (worse) early-stopped forgetting, the magnitude (+0.5 $\pm$ 2.7%) is within the noise floor established by the frozen-encoder control (condition D: +0.6 $\pm$ 5.3%); a sign test is not statistically distinguishable from chance ( $p=0.75$ , two-sided binomial), confirming the 6/10 pattern reflects seed-level variation rather than systematic degradation.

**Final-epoch comparison (protocol robustness).** Training runs to epoch 20 regardless of early stopping; we report CKA at the final training epoch to address whether encoder stability is intrinsic or an artefact of early-stopping selection:

|                    | Final-epoch CKA   | Final-epoch Forg.% | $n$ |
|--------------------|-------------------|--------------------|-----|
| $n=500$ (10 seeds) | 0.968 $\pm$ 0.009 | +3.1 $\pm$ 5.8     | 500 |
| $n=10k$ (10 seeds) | 0.955 $\pm$ 0.006 | +3.4 $\pm$ 5.2     | 10k |

All 10 seeds at  $n=10k$  show final-epoch CKA>0.94 (min=0.945, max=0.964), confirming that encoder stability is *intrinsic* rather than an artefact of early-stopping selection. The mild CKA decline from 0.999 (early-stopped) to 0.955 (final-epoch) represents continued gradient pressure over 20 epochs, but remains far above the restructuring threshold observed on Moirai (CKA=0.52

on Small, 0.25 on Base). Final-epoch forgetting is  $+3.4 \pm 5.2\%$ , mild degradation vs. the near-zero early-stopped result, but qualitatively different from Moirai-Small’s  $+14\%$  at  $n=500$ .

**Mild CKA decline interpretation.** The  $0.999 \rightarrow 0.955$  CKA trajectory (early-stopped  $\rightarrow$  final-epoch at  $n=10k$ ) confirms a small but real gradient signal from the fine-tuning objective. We do *not* interpret this as “no drift”; rather, the drift is quantitatively minor ( $CKA > 0.94$ ) and functionally benign (final-epoch forgetting  $+3.4\%$  vs. Moirai-Small’s  $+14\%$  at  $n=500$ ). We previously read this off the gate; with this cell’s gate superseded and not recomputed, the observation stands on the CKA and forgetting measurements alone.

**Condition D (frozen encoder),  $n=10k$  (10 seeds).** Forgetting:  $+0.6 \pm 5.3\%$  (5/10 negative). Measured CKA is  $0.9993-0.9994$ , not 1: D holds the encoder’s weights fixed but leaves its input projection trainable, so the representation is not pinned by construction (Appendix A6). Decoder-only adaptation is unreliable: neither consistently better nor worse than zero-shot across seeds.

**Random-init control (6 seeds).**  $CKA=0.000$  (expected: random weights have no similarity to pre-trained).  $R^2(PT) < 0$  ( $-0.004 \pm 0.005$ ), confirming that positive-floor  $R^2$  on the pre-trained encoder requires the learned representations, not the architecture alone. Training loss remains high ( $> 8.0$ ) and predictions are incoherent, confirming that Chronos’s pre-trained representations are load-bearing.

**Comparison with Moirai (withdrawn).** An earlier version read this cell against Moirai/ETTh2 as a contrast between a moderate gate-pass and a strong one, and offered it as validating the value-gate as a regime indicator. That reading is withdrawn on both halves. The gate half does not survive the baseline correction: Moirai-Small/ETTh2 scores  $+0.400$  ( $h=96$ ) and  $+0.464$  ( $h=192$ ) against the fitted baseline, while the M4-Monthly gate was never recomputed against it, so the two sides of the contrast are not on a common scale. The probe half rests on the retracted  $\Delta R^2$  statistic (§A29). What remains is descriptive and gate-independent: Chronos’s encoder on M4-Monthly holds  $CKA \geq 0.95$  even at the final epoch with forgetting near zero under early stopping and  $+3.4\%$  at final epoch, whereas Moirai-Small on ETTh2 restructures to  $CKA \approx 0.52$  at  $n=10k$ . We report the pair as two observations, not as a regime law.

## A18 CORRECTIONS AND WITHDRAWALS: WHAT CHANGED FROM THE EARLIER VERSION

This work has been revised more than once, and every reading it has withdrawn is recorded here rather than in the body. The body keeps a single pointer to this appendix; nothing is removed from the record by the move, and the three detailed accounts that follow—the gate’s denominator (Appendix A19), ILI (Appendix A28) and the probe axis (§A29)—are unchanged.

**(a) The conclusion changed, not the emphasis.** An earlier version of this work reported eight degradation cells, all on ETTh1 or Weather at  $h \in \{96, 192\}$ . All eight fail clause (i) once the gate is computed against the linear regression the gate is *defined* as; none of the eight involved re-running a model (Appendix A19).

**(b) The non-Moirai gates.** An earlier version reported “eight of nine non-Moirai cells gate-pass” at  $h=24$ ; against the fitted baseline **none of the ten does**, and the earlier count was an artefact of the trend denominator. What the two extra backbones still supply is the dissociation, which does not depend on the gate.

**(c) The prospective arm’s scorecard.** The earlier version of that arm read TP 2, FP 6, precision 0.25, recall 1.00, and we defended the gate as a screen that never missed a degradation cell. With the corrected baseline there are no degradation cells to miss, so the screen has nothing to be right about.

**(d) ILI.** An earlier public version reported Moirai-Small on ILI as a 57% gate-pass; that number is withdrawn, and the cell gate-fails at  $R^2_{task}(PT) = -1.625$ . A separate script defect reported *zero* frozen-encoder improvement in all 10 seeds; corrected,  $\Delta_{enc} = -38.6 \pm 1.3$  (Appendix A28).

**(e) The linear-probe axis.** A  $\Delta R^2$  probe account of *what* restructured is retracted in full (§A29); the withdrawn numbers and the re-run that refutes them are kept in Appendices A10, A16 and A13. The layer-wise *task* readings survive, being forecast MSE, and are reported in the body.

## A19 THE GATE’S LINEAR BASELINE WAS MEASURED WRONGLY: A CORRECTION

**Two estimators, one name.** The value gate is  $R_{\text{task}}^2(\text{PT}) = 1 - \text{MSE}_{\text{ZS}} / \text{MSE}_{\text{Linear}}$  with threshold 0.20, and §2 defines  $\text{MSE}_{\text{Linear}}$  as a lookback-96 linear regression *fit on that same set*: a single ridge-OLS map from  $96 \times D$  inputs to  $h \times D$  outputs, estimated once on the cell’s training windows and applied unchanged out of sample. That estimator is `gate_linear_baseline.fit_linear_map` (applied by `apply_linear_map`). The gate was not computed with it. Through 26 Aug 2026 the denominator came from `gate_linear_baseline.linear_forecast`, which fits a slope and intercept to the last 96 steps *inside each evaluation window* and extrapolates them forward, using no training data at all. These are two different estimators sharing one label, not two implementations of one estimator, and the paper defines the first while reporting the second.

**The estimator the gate used is weaker than a constant.** Table 20 measures four denominators on the same 300 matched test windows on the train-normalised scale the stored zero-shot MSE is measured on. The load-bearing row is the third: on ETTh1  $h=96$ , simply predicting the training mean scores 0.9937 against the trend extrapolation’s 1.2207. A baseline that a constant predictor beats cannot support the inference the gate exists to license, namely that *there is a valuable pre-trained capability here worth preserving* — which is the sole premise the degradation reading rests on. The defined estimator is stronger by a factor of about three on ETTh1 and 9.4 on Weather. Two independent checks say the correction runs in the right direction: the fitted ETTh1  $h=96$  MSE of 0.4145 is in line with published multivariate Linear/DLinear results Zeng et al. (2023), and Moirai zero-shot losing to a supervised linear model on ETTh1 is a known result, whereas 1.2207 is not a physically plausible score for any linear baseline on this cell. Note also an asymmetry the definition imposes and which we retain: Moirai sees lookback +  $h$  steps of context while the linear baseline sees 96, so the corrected gate is still a lower bound on the linear model’s strength.

Table 20: Baseline strength: train-normalised MSE on 300 matched test windows. Row 1 is the denominator the gate used through 26 Aug 2026; row 4 is the denominator §2 defines. Row 1 is beaten by predicting the training mean (row 3) on *both* cells and is several times weaker than row 4. On the train-normalised scale the constant-mean predictor’s MSE is just the mean squared normalised target, so row 3 needs no fitting; a value near 1.0 is what an on-distribution test split should give.

| Denominator estimator                                           | ETTh1 $h=96$  | Weather $h=192$ |
|-----------------------------------------------------------------|---------------|-----------------|
| Per-window trend extrapolation ( <code>linear_forecast</code> ) | 1.2207        | 5.0707          |
| Persistence (last observed value held)                          | 1.1984        | 1.2227          |
| Constant training mean                                          | 0.9937        | 0.8619          |
| Fitted lookback-96 regression ( <code>fit_linear_map</code> )   | <b>0.4145</b> | <b>0.5403</b>   |

**What it costs, cell by cell.** Re-running all four arms with `gate_all_cells.py -baseline fitted` (now the default) moves most of the matrix:

- **Moirai: 5 of 21** cells clear 0.20 on the test split, against 20 of 21 before. Seventeen cells change status: 16 flip pass→fail, and one (Moirai-Large/ETTh2  $h=96$ ) flips fail→pass at +0.219.
- **Chronos: 0 of 5**, against 4 of 5 before: ETTh1 −0.123, ETTh2 −0.325, Weather −0.103, ETTm2 −1.281, Electricity +0.048.
- **TimesFM: 0 of 5**, against 5 of 5 before: ETTh1 −0.101, ETTh2 −0.287, Weather −0.077, ETTm2 −0.945, Electricity +0.132.
- **ILI (Moirai-Small): −1.625**. Appendix A28 previously reported this cell’s test gate as −0.033, which is its trend-baseline value, and before that a gate-pass of +57% was withdrawn there; it now carries −1.625. The reading of that appendix is unchanged and strengthened: ILI gate-fails, now by a wide margin.

Across arms, **5 of 32 cells pass on the test split and 7 of 31 on validation**. Every test-side survivor is Moirai on ETTh2; no other dataset and no other backbone retains a single gate-pass cell. Per-cell corrected Moirai values are in Table 3, and the survivors are given in full in Table 21. Three of the five survivors *improve* on the released checkpoint under full fine-tuning, by 31–42%, and the two whose

mean is harmful have only 4/10 and 5/10 seeds in that direction, so their means are outlier-driven rather than consistent (the Moirai-Small  $h=96$  cell also has a positive mean condition-D forgetting of +6.91 against 9/10 negative seeds, for the same reason).

**The  $h=96$  decomposition, since its mean is one seed.** Seed 42 of Moirai-Small/ETTh2  $h=96$  diverges to +98.9% under full fine-tuning *and* to +91.1% under the frozen encoder, which is why §5.1 calls it an optimisation failure rather than an encoder-drift effect; Appendix A6 adds the independent check that it cannot diverge at all under the strict freeze. Dropping it leaves  $\text{forg}_B = -0.20 \pm 4.15$  over 9 seeds (3 positive) and  $\text{forg}_D = -2.45 \pm 0.16$ , so the cell’s harmful mean does not survive the removal of a single run. **We nonetheless report the cell means including seed 42 everywhere in this paper**, and record the decomposition here rather than dropping the seed, because a rule that discards runs after seeing their outcome is not one we are willing to apply.

Table 21: The five cells that still clear  $R_{\text{task}}^2 \geq 0.20$  on held-out windows under the fitted baseline. Forg.% is percent change against zero-shot, so a *negative* entry means fine-tuning improved on the released checkpoint. The bracketed counts are the seeds whose sign matches the degradation direction: positive for B, negative for D. B–D positive means the frozen encoder was better.

| Cell                       | Gate   | B Forg.% (+) | D Forg.% (–)  | B–D val | B–D held-out |
|----------------------------|--------|--------------|---------------|---------|--------------|
| Small/ETTh2 $h=192, n=500$ | +0.464 | +1.99 (5/10) | –4.16 (10/10) | +17.4   | +6.1         |
| Small/ETTh2 $h=96, n=500$  | +0.400 | +9.71 (4/10) | +6.91 (9/10)  | +10.6   | +2.8         |
| Base/ETTh2 $h=96, n=1000$  | +0.361 | –31.44 (0/3) | –9.11 (3/3)   | +3.8    | –22.3        |
| Base/ETTh2 $h=192, n=1000$ | +0.275 | –36.35 (0/3) | –16.12 (3/3)  | –3.9    | –20.2        |
| Large/ETTh2 $h=96, n=500$  | +0.219 | –42.42 (0/3) | –21.62 (3/3)  | +7.2    | –20.8        |

**No cell now meets the degradation definition.** The definition is unchanged — gate  $\geq 0.20$ , *and* condition-B forgetting positive in every seed, *and* condition-D forgetting negative in every seed — and only the gate input moved. With the corrected denominator, `degradation_cells()` returns zero cells. There is one candidate, and the pre-existing unanimity rule excludes it:

```
excluded: Moirai-small/ETTh2 h192 n500 gate=+0.464
          forg_B=+1.99±3.32SEM (5/10 pos)
          forg_D=–4.16±0.20SEM (10/10 neg) B–D=+6.15
=> 1 excluded for seed disagreement
```

Half its seeds go the other way on condition B, so it fails the same all-seeds test that the earlier degradation cells were required to pass. We did not weaken or re-tune that rule to keep a cell; applying it as written leaves nothing—and Appendix A21 shows that weakening it, in seven directions and at both gate thresholds, does not recover one either.

**Prospective arm and threshold sweep.** With no positive instances, the pre-registered prospective test has nothing for the gate to find: on the eight pre-registered cells the gate scores **TP 0, FP 8, precision 0.00**, and recall is undefined rather than 1.00. The earlier reading of this arm (precision 0.25, TP 2, FP 6, recall 1.00) does not survive the correction. The threshold sweep of §7 (Table 8) reports precision 0.00 at *every* threshold in  $\{0.10, 0.20, 0.30, 0.40, 0.50, 0.60\}$ , with 8/8/6/5/3/2 cells flagged respectively: the failure is not a badly chosen operating point.

A further consequence is newly detectable. Within Moirai ( $n=22$  cells), the gate correlates with held-out B–D at Spearman  $\rho = -0.518$ : on the evidence in Table 5 the gate points *away* from the intervention outcome it was introduced to predict, in that higher gate goes with adapting the encoder having been the better choice. The interval around that point estimate excludes zero when cells are resampled,  $[-0.840, -0.040]$ , and includes zero when the six (backbone, dataset) clusters are,  $[-0.810, +0.500]$ ; we report the latter and read only the direction (Appendix A27). The CKA comparison is unchanged by all of this, being gate-independent:  $\rho = +0.168$ , clustered CI  $[-0.469, +0.717]$  within Moirai, and  $\rho = +0.567$ , clustered CI  $[+0.239, +0.828]$  pooled across backbones, which this paper’s own within-backbone rule says not to read.

**Provenance.** The substitution happened between 24 Jul and 26 Aug 2026 and is recorded nowhere: no commit message, no docstring, no code comment marks it, and the paper text was never updated to match. That is also why the paper carried both signs of the same quantity at once. The  $R_{\text{task}}^2$  values previously printed in Table 3 (–0.406 on ETTh1, –0.556 on ETTm2,

+0.279/ + 0.450 on ETTh2) were read from `results/gate_pass_etth1/metrics.json` and `results/gate_pass_ettm2/metrics.json`, which were produced by the older *fitted* LinearForecaster in `scripts/evaluate_moirai_zeroshot_forecasting.py`, while the gate-pass counts quoted around them came from the trend estimator. Nothing was suppressed to make the two agree; they simply were never compared.

The correction is auditable rather than a claim to be taken on trust. `gate_all_cells.py` takes `-baseline {fitted,trend}` with `fitted` as the default, threaded through all four arms; `linear_forecast` is retained and documented as `audit-trail-only`; and the trend-baseline numbers are preserved verbatim in `results/gate_test_side_trend.json` and `results/gate_val_side_trend.json`, so every superseded figure in this paper’s history can be recomputed and compared against its replacement.

One earlier fix should not be confused with this one. The gate’s evaluation windows are now built with `ext_lb = lookback + h`, matching the extended lookback the fine-tuning script evaluates with, rather than `lookback × 2`; the two coincide at  $h=96$  and diverge at  $h=192$ , where the older setting paired the stored zero-shot MSE with a linear baseline measured over a *different* set of target windows. That correction changed no cell’s gate status. This one changes 17 of the 21 Moirai cells and removes every degradation instance the paper had.

## A20 EIGHT DENOMINATORS, ONE NUMERATOR: IS THE SCREEN A PROPERTY OF THE ESTIMATOR?

The appendix above shows the gate’s verdict barely moving when the 0.20 threshold is swept but moving for 17 of 21 Moirai cells when the baseline estimator is swapped. So the sensitivity analysis this paper already had varied the parameter that does not matter, and the parameter that does had been varied exactly once—between a wrong estimator and a right one. This appendix varies it across a family, on all 32 screened cells (Table 22).

**What is held identical.**  $R_{\text{task}}^2 = 1 - \text{MSE}_{\text{ZS}} / \text{MSE}_{\text{base}}$ , and only the denominator depends on the estimator, so the numerator is read from `results/gate_test_side.json` and never recomputed: each column differs from the published one in exactly one respect. Before anything is reported, `scripts/gate_baseline_sensitivity.py` recomputes the *fitted* column through the alternative-baseline code path and checks it against both the published denominator *and* the published  $R_{\text{task}}^2$ ; all 32 cells agree to a relative tolerance of  $10^{-6}$ , and the script refuses to emit a table if any does not. The two checks are not redundant, and the ILI cell is why: its denominator matched while its  $R_{\text{task}}^2$  did not, because that cell aggregates as a mean over features of per-feature ratios where every other cell is a ratio of means (`gate_all_cells.ili_gate`, matching `finetune_ili`). A denominator check alone would have published one row of this table on a convention no other number in the paper uses.

**The estimators.** *Persistence* repeats the last observed value for all  $h$  steps and is the bottom of the ladder. *Seasonal naive* repeats the last seasonal period and fits nothing; periods are declared per dataset—24 steps for the hourly ETT, Weather and Electricity series, 96 for 15-minute ETTm2, 52 for weekly ILI—rather than inferred, because inferring one from an autocorrelation peak returns 1 on the flatter series and quietly substitutes a persistence baseline. *Per-feature AR* is the paper’s map with the cross-channel terms removed, so it isolates how much of the gate rests on multivariate information. *DLinear* is a single linear map on the trend/seasonal decomposition of the lookback, shared across channels. *MLP* is one two-layer network per channel (hidden 128) with the penalty selected per channel on a chronological holdout; the channels that hit the 200-iteration budget are recorded per cell. *Tuned ridge* is the paper’s own estimator with  $\lambda$  selected on a chronological 20% holdout of the training windows rather than fixed at  $10^{-4}$ ; the selected values span  $10^{-6}$  to  $10^4$ , and on all five ETTh2 cells the selection lands at  $10^3$  or  $10^4$ , so the fixed  $10^{-4}$  the paper uses is badly under-regularised there. *Boosted trees* is one `HistGradientBoostingRegressor` per feature over that feature’s lags plus the horizon step as an input feature; one model per (feature, step) would be 672 fits on a 7-feature  $h=96$  cell. Its rows are `windows × steps`, capped at 200,000 per feature, which on the Moirai cells uses 1,041–2,083 of the 10,165–48,777 available windows. Every cap and every selected penalty is recorded per cell in `results/gate_baselines.json` rather than only in the source.

**Four columns coincide on the single-channel arms, and that is arithmetic rather than a copy.** On the Chronos and TimesFM cells, which score one target channel, the fitted map, per-feature AR and DLinear are the *same estimator*: with  $F=1$  there are no cross-channel terms for AR to drop, and DLinear’s trend/seasonal split is an invertible linear reparameterisation of the lookback, so a shared linear head on it spans exactly the maps a lookback regression spans. Their columns therefore agree to three decimals by construction, and tuned ridge differs from them only through  $\lambda$ . A reader who sees four identical entries on those rows is seeing the ladder collapse where the data has one dimension, not a transcription error.

Table 22: **Gate  $R^2_{\text{task}}$  under eight denominators and a floor, on one numerator.** Bold clears 0.20. A dagger marks a denominator worse than the training-mean predictor on that cell: the ratio is still arithmetic, but it is not evidence of a pre-trained advantage, so daggered cells are excluded from the “and admissible” tally. The training-mean column is the floor, not a candidate estimator. The last row applies the three-clause degradation definition of §3 with clause (i) taken from each estimator and clauses (ii) and (iii) unchanged, over the 31 cells with a paired frozen-encoder run. Generated by scripts/gate\_baseline\_sensitivity.py.

| Cell                                  | Fitted ridge<br>(ours) | Training<br>mean | Persist-<br>ence    | Seasonal<br>naive   | Per-feature<br>AR | DLinear<br>(shared) | Tuned<br>ridge | MLP<br>(128)        | Boosted<br>trees    |
|---------------------------------------|------------------------|------------------|---------------------|---------------------|-------------------|---------------------|----------------|---------------------|---------------------|
| Moirai-S / ETTh2 ( $h=192$ )          | <b>+0.464</b>          | <b>+0.845</b>    | -0.063              | -0.267              | -0.283            | -0.738              | <b>+0.422</b>  | <b>+0.296</b>       | +0.934 <sup>†</sup> |
| Moirai-S / ETTh2 ( $h=96$ )           | <b>+0.400</b>          | <b>+0.813</b>    | -0.223              | -0.215              | -0.843            | -0.889              | <b>+0.374</b>  | <b>+0.406</b>       | <b>+0.806</b>       |
| Moirai-B / ETTh2 ( $h=96$ )           | <b>+0.361</b>          | <b>+0.801</b>    | -0.303              | -0.295              | -0.965            | -1.014              | <b>+0.333</b>  | <b>+0.367</b>       | <b>+0.793</b>       |
| Moirai-B / ETTh2 ( $h=192$ )          | <b>+0.275</b>          | <b>+0.791</b>    | -0.438              | -0.714              | -0.735            | -1.350              | <b>+0.218</b>  | +0.048              | +0.911 <sup>†</sup> |
| Moirai-L / ETTh2 ( $h=96$ )           | <b>+0.219</b>          | <b>+0.757</b>    | -0.592              | -0.582              | -1.400            | -1.460              | +0.185         | <b>+0.227</b>       | <b>+0.747</b>       |
| TimesFM-2.5 / Electricity ( $h=24$ )  | +0.132                 | +0.179           | +0.254 <sup>†</sup> | +0.301 <sup>†</sup> | +0.132            | +0.132              | +0.132         | +0.186 <sup>†</sup> | +0.146              |
| Moirai-S / ETTh2 ( $h=192$ )          | +0.108                 | <b>+0.830</b>    | +0.101              | +0.058              | <b>+0.350</b>     | +0.123              | +0.119         | <b>+0.545</b>       | <b>+0.826</b>       |
| Moirai-B / ETTh2 ( $h=192$ )          | +0.083                 | <b>+0.825</b>    | +0.076              | +0.032              | <b>+0.332</b>     | +0.099              | +0.095         | <b>+0.532</b>       | <b>+0.821</b>       |
| Chronos-T5-S / Electricity ( $h=24$ ) | +0.048                 | +0.100           | +0.182 <sup>†</sup> | +0.233 <sup>†</sup> | +0.048            | +0.048              | +0.048         | +0.107 <sup>†</sup> | +0.064              |
| Moirai-S / Electricity7 ( $h=192$ )   | -0.047                 | <b>+0.662</b>    | <b>+0.625</b>       | -0.035              | -0.516            | -0.440              | -0.071         | -0.367              | +0.893 <sup>†</sup> |
| Moirai-B / ETTh2 ( $h=96$ )           | -0.057                 | <b>+0.860</b>    | +0.165              | +0.047              | +0.014            | -0.193              | -0.067         | +0.193              | <b>+0.855</b>       |
| TimesFM-2.5 / Weather ( $h=24$ )      | -0.077                 | <b>+0.494</b>    | +0.535 <sup>†</sup> | <b>+0.209</b>       | -0.077            | -0.077              | -0.077         | -0.002              | +0.018              |
| Moirai-S / Electricity7 ( $h=96$ )    | -0.085                 | <b>+0.701</b>    | <b>+0.649</b>       | +0.093              | -0.404            | -0.400              | -0.102         | -0.337              | +0.856 <sup>†</sup> |
| TimesFM-2.5 / ETTh1 ( $h=24$ )        | -0.101                 | <b>+0.424</b>    | +0.124              | <b>+0.291</b>       | -0.101            | -0.101              | -0.091         | +0.005              | -0.005              |
| Chronos-T5-S / Weather ( $h=24$ )     | -0.103                 | <b>+0.481</b>    | +0.523 <sup>†</sup> | +0.190              | -0.103            | -0.103              | -0.103         | -0.027              | -0.006              |
| Moirai-S / ETTh2 ( $h=96$ )           | -0.110                 | <b>+0.853</b>    | +0.124              | -0.000              | -0.035            | -0.252              | -0.121         | +0.153              | <b>+0.848</b>       |
| Chronos-T5-S / ETTh1 ( $h=24$ )       | -0.123                 | <b>+0.412</b>    | +0.107              | <b>+0.278</b>       | -0.123            | -0.123              | -0.112         | -0.014              | -0.024              |
| Moirai-S / ETTh1 ( $h=96$ )           | -0.243                 | <b>+0.472</b>    | +0.570 <sup>†</sup> | <b>+0.216</b>       | -0.362            | -0.242              | -0.297         | -0.147              | +0.621 <sup>†</sup> |
| TimesFM-2.5 / ETTh2 ( $h=24$ )        | -0.287                 | <b>+0.475</b>    | +0.527 <sup>†</sup> | +0.021              | -0.287            | -0.287              | -0.238         | -0.108              | -0.131              |
| Chronos-T5-S / ETTh2 ( $h=24$ )       | -0.325                 | <b>+0.460</b>    | +0.513 <sup>†</sup> | -0.008              | -0.325            | -0.325              | -0.274         | -0.141              | -0.164              |
| Moirai-S / ETTh1 ( $h=192$ )          | -0.331                 | <b>+0.326</b>    | +0.406 <sup>†</sup> | +0.083              | -0.410            | -0.351              | -0.382         | -0.227              | +0.758 <sup>†</sup> |
| Moirai-L / ETTh1 ( $h=96$ )           | -0.454                 | <b>+0.382</b>    | +0.497 <sup>†</sup> | +0.083              | -0.594            | -0.453              | -0.517         | -0.342              | +0.556 <sup>†</sup> |
| Moirai-B / ETTh1 ( $h=96$ )           | -0.458                 | <b>+0.380</b>    | +0.496 <sup>†</sup> | +0.080              | -0.599            | -0.458              | -0.522         | -0.346              | +0.555 <sup>†</sup> |
| Moirai-B / ETTh1 ( $h=192$ )          | -0.501                 | <b>+0.239</b>    | +0.330 <sup>†</sup> | -0.034              | -0.590            | -0.524              | -0.559         | -0.383              | +0.727 <sup>†</sup> |
| Moirai-L / Weather ( $h=96$ )         | -0.621                 | <b>+0.204</b>    | +0.248 <sup>†</sup> | +0.143              | -0.531            | -0.437              | -0.611         | -0.611              | +0.055              |
| Moirai-S / Weather ( $h=192$ )        | -0.639                 | -0.037           | +0.276 <sup>†</sup> | +0.163 <sup>†</sup> | -0.775            | -0.670              | -0.639         | -0.648              | +0.600 <sup>†</sup> |
| Moirai-S / Weather ( $h=96$ )         | -0.867                 | +0.084           | +0.134 <sup>†</sup> | +0.014              | -0.763            | -0.655              | -0.855         | -0.855              | -0.088              |
| TimesFM-2.5 / ETTh2 ( $h=24$ )        | -0.945                 | <b>+0.576</b>    | +0.136              | +0.631 <sup>†</sup> | -0.945            | -0.945              | -0.901         | -0.861              | -0.531              |
| Moirai-B / Weather ( $h=96$ )         | -1.029                 | +0.004           | +0.059 <sup>†</sup> | -0.072              | -0.916            | -0.799              | -1.016         | -1.016              | -0.183              |
| Chronos-T5-S / ETTh2 ( $h=24$ )       | -1.281                 | <b>+0.503</b>    | -0.013              | +0.567 <sup>†</sup> | -1.281            | -1.281              | -1.229         | -1.182              | -0.795              |
| Moirai-B / Weather ( $h=192$ )        | -1.324                 | -0.470           | -0.027 <sup>†</sup> | -0.187 <sup>†</sup> | -1.517            | -1.368              | -1.323         | -1.337              | +0.433 <sup>†</sup> |
| Moirai-S / ILI ( $h=24$ )             | -1.625                 | -0.304           | +0.027 <sup>†</sup> | -0.786              | -1.625            | -1.625              | -1.638         | -1.301              | -1.563              |
| Clearing 0.20                         | 5/32                   | 25/32            | 14/32               | 8/32                | 2/32              | 0/32                | 4/32           | 6/32                | 18/32               |
| and admissible                        | 5/32                   | 25/32            | 2/32                | 4/32                | 2/32              | 0/32                | 4/32           | 6/32                | 7/32                |
| Below the floor                       | 0/32                   | 0/32             | 17/32               | 6/32                | 0/32              | 0/32                | 0/32           | 2/32                | 11/32               |
| Degradation cells                     | 0/31                   | 6/31             | 0/31                | 1/31                | 0/31              | 0/31                | 0/31           | 0/31                | 0/31                |

**A denominator has to clear a floor before the ratio means anything.** If  $\text{MSE}_{\text{base}}$  exceeds the training mean’s, the gate is dividing by the error of an estimator worse than answering “the average”, and a checkpoint can clear the threshold without beating anything—which is precisely what the trend extrapolation did. The fitted map is above the floor on **all 32 cells**, so the correction above did not trade one inadmissible denominator for another. Boosted trees fall below it on 11 cells and seasonal naive on 6. For the trees the mechanism is that they cannot extrapolate beyond their training range, and ETTh2’s test split is well outside it: the training-mean predictor itself scores 2.06 there at  $h=192$

against a normalised variance of 1, and the trees score 4.82. Persistence falls below the floor on 17 and the MLP on 2. Counting only admissible passes, the eight rungs give **5, 2, 4, 2, 0, 4, 6 and 7 of 32** in ladder order; the widest is boosted trees, which adds the four ETTh2 cells, and the narrowest is DLinear, which admits **none**.

**The negative headline strengthens. The identity of the five survivors does not.** No estimator admits many more cells than the fitted map and two admit far fewer, so “on most cells there is no demonstrated advantage” is not an artefact of one estimator family. But the passing *sets* are nearly disjoint. All five ETTh2 survivors fail against seasonal naive ( $-0.215$  to  $-0.714$ ), against per-feature AR ( $-0.283$  to  $-1.400$ ) and against DLinear ( $-0.738$  to  $-1.460$ ), which is the rung that beats the pre-trained model everywhere. Tuned ridge and the MLP each reproduce four of the five and not the same four: tuned ridge loses Moirai-Large/ETTh2  $h=96$  at  $+0.185$ , the MLP loses Moirai-Base/ETTh2  $h=192$  at  $+0.048$ . The denominators say why: on ETTh2  $h=192$  the univariate AR map scores 0.249 where the paper’s multivariate map scores 0.595, so the cross-channel terms *hurt* there, and “Moirai beats a target-data regression on ETTh2” is in part a statement about a regression with 672 free inputs. The four cells seasonal naive admits are a different set again: Chronos/ETTh1, TimesFM/ETTh1, TimesFM/Weather and Moirai-Small/ETTh1  $h=96$ . Where this paper says five cells clear the gate, it means five cells clear *this* gate.

**Does the zero survive? Under seven of the eight rungs, yes.** With clause (i) taken from each estimator, the definition returns **zero cells under the fitted map, persistence, per-feature AR, DLinear, tuned ridge, the MLP and boosted trees, and one under seasonal naive**: Moirai-Small/ETTh1  $h=96$ , which reads  $+0.216$  there against  $-0.243$  against the fitted map and then satisfies both remaining clauses unanimously— $\text{forg}_B=+18.5\%$  in 5/5 seeds,  $\text{forg}_D=-3.5\%$  in 5/5,  $B-D=+22.1$  at CKA 0.807. We report it rather than absorbing it: against one of the eight rungs this paper’s own definition has a positive instance.

Two things about that instance decide how much weight it carries. It clears 0.216 against a threshold of 0.20 and fails at 0.25. And it arises from a *weaker* admissible denominator, not a stronger one—on ETTh1  $h=96$  seasonal naive scores 0.657 where the fitted map scores 0.415. The direction of the sensitivity is therefore that loosening the baseline manufactures degradation cells, which is the same mechanism as the retracted eight rather than a stronger baseline revealing damage we missed. The floor column is that trend taken to its limit: under the training-mean denominator the count is **six**, and the six are Moirai-Small/ETTh1 at both horizons, Moirai-Base/ETTh1 at both, Moirai-Large/ETTh1  $h=96$  and Moirai-Large/Weather  $h=96$ —the cells the appendix above retracts. That is the retraction confirmed from the other side: the degradation instances the earlier version reported needed a denominator no stronger than a constant.

**The ladder read as a graded axis, and why the headline is a minimum.** A pass count per rung still treats the gate as binary. The graded reading assigns each cell one number,  $R_{\text{task}}^2(\text{best})$ , measured against the *strongest* admissible rival it faces. That is a **minimum** over rungs, not a maximum: a weaker denominator inflates  $R_{\text{task}}^2$ , so the strongest rival is the one that leaves the smallest score. Read that way, **no cell of the 32 clears 0.20 against every admissible rung**, and 15 clear it against at least one. The two counts move in opposite directions as the ladder grows—the first can only fall, the second can only rise—so only the first is a claim, and adding a ninth rung could not undo it. The five ETTh2 survivors have  $R_{\text{task}}^2(\text{best})$  between  $-0.738$  and  $-1.460$ , every one of them against DLinear. This is the quantity §6 uses as its value axis, and it is stored per cell in `results/gate_baselines.json` alongside which rung supplied it.

## A21 IS THE ZERO A PROPERTY OF THE RUNS OR OF THE DEFINITION?

The paper’s headline is that no cell meets its definition of damaging a demonstrated pre-trained capability, and the first fair objection is that the definition does the work. It requires condition-B forgetting positive in *every* seed and condition-D forgetting negative in *every* seed; unanimity over ten seeds is a demanding bar, and if a simple majority produced five degradation cells the negative result would be a property of the rule rather than of the runs. That objection is answerable from the stored per-seed sign counts with no new runs.

Table 23 walks a ladder of definitions from the weakest defensible one—the two means alone, with seed signs unchecked—up to ours, crossed with both gate thresholds so the sensitivity to *both* free



parameters is one table rather than two. Two exclusions apply identically at every rung, exactly as `degradation_cells()` applies them: cells with no zero-shot test reference cannot be scored at all (there are none), and the five Chronos cells are excluded because a head/decoder mismatch makes their per-condition forgetting non-comparable (limitation iv). Admitting them at the relaxed rungs would make loosening the rule appear to find degradation cells when what it found is a measurement already known to be invalid. Both exclusions are properties of the measurement rather than of the outcome, so applying them uniformly is not a selection on the result. That leaves 26 scorable cells, of which 6 clear a gate of 0.10 and 5 clear 0.20.

Table 23: Degradation cells admitted under eight definitions and two gate thresholds. Cells are named, not just counted; <sup>†</sup> marks a cell admitted only at the looser gate. The excluded confounded cells are Chronos-T5-Small on ETTh1, ETTh2, ETTm2, Weather and Electricity. Emitted by `scripts/degradation_sensitivity.py`.

| Definition of a degradation cell                             | Cells admitted   |                  | Which cells                                             |
|--------------------------------------------------------------|------------------|------------------|---------------------------------------------------------|
|                                                              | Gate $\geq 0.10$ | Gate $\geq 0.20$ |                                                         |
| Means only (seed signs unchecked)                            | 1                | 1                | S/ETTh2 <i>h</i> 192                                    |
| Means, $+$ $\geq$ half the seeds agree                       | 1                | 1                | S/ETTh2 <i>h</i> 192                                    |
| Means, $+$ $\geq 70\%$ of seeds agree                        | 0                | 0                | —                                                       |
| Means, $+$ $\geq 80\%$ of seeds agree                        | 0                | 0                | —                                                       |
| <b>Means, <math>+</math> every seed agrees (ours)</b>        | 0                | 0                | —                                                       |
| Means, $+$ one-sided $t$ on $\text{forg}_B$ at $\alpha=0.05$ | 0                | 0                | —                                                       |
| Seed counts only, $\geq$ half agree                          | 2                | 1                | S/ETTh2 <i>h</i> 192, S/ETTh2 <i>h</i> 192 <sup>†</sup> |
| Seed counts only, every seed agrees                          | 0                | 0                | —                                                       |
| <i>Gate-passing cells screened</i>                           | 6                | 5                |                                                         |

**Unanimity is not what produces the zero.** At the paper’s gate threshold the largest count anywhere on the ladder is **one**, and it is produced by the *weakest* rung, the two means with seed signs unchecked. Requiring 70% of seeds, 80%, unanimity, or a one-sided  $t$ -test on  $\text{forgetting}_B$  all give zero. Relaxing the gate to 0.10 raises the maximum to two, and only for one rung. There is no setting of these two parameters at which the paper’s conclusion becomes “several cells degrade”.

**The one cell that ever appears.** Moirai-Small on ETTh2 at  $h=192$  has  $\text{forgetting}_B +1.99$  with an SEM of 3.32 over ten seeds—a one-sided  $p$  of 0.28 against zero, and 5 of 10 seeds on each side. It is a coin flip, which is what the body already reports it as. The other four gate-passing cells are not near-misses in any direction: their  $\text{forgetting}_B$  is  $+9.71 \pm 10.58$ ,  $-31.44 \pm 3.65$ ,  $-36.35 \pm 2.80$  and  $-42.42 \pm 0.47$  SEM, so the three Base and Large cells are strongly *improved* by full fine-tuning rather than marginally harmed.

**Means and seed counts can disagree in direction, and we report both families rather than absorb the disagreement.** Moirai-Small/ETTh2 at  $h=96$  has a mean  $\text{forgetting}_D$  of  $+6.91$  while 9 of its 10 seeds are negative: a single outlier drags the mean across zero. A ladder that silently required the means clause at every rung would hide that, and “majority seed-consistency” is what the objection proposes as a *replacement* for unanimity, not as an extra hurdle on top of it. The last two rows therefore drop the means clause entirely, and they are the only rows that admit two cells—which is how Moirai-Small/ETTh2 at  $h=192$  becomes visible at all.

**Which definition the paper uses, and why.** We keep unanimity. The reason is not that it yields the fewest cells—at gate 0.20 four of the eight rungs also yield zero—but that a degradation claim is a claim that fine-tuning *reliably* destroys something, and a cell whose seeds disagree on the sign of the harm is not evidence of that. The cost of the choice is that it can discard a real but noisy effect, which is exactly what the  $h=192$  cell would be if its true  $\text{forgetting}_B$  were small and positive; that cell is therefore reported by name in the body rather than dropped silently.

## A22 PAIRED INFERENCE ON ENCODER ADAPTATION, AND WHAT THESE SEED COUNTS CAN RESOLVE

A unanimity count is not inference. This appendix reports  $\Delta_{\text{enc}}$  per cell as a paired estimate with an interval, an effect size and two  $p$ -values, and then states plainly what the matrix’s seed counts can and cannot establish.

**The exact test’s resolution is set by  $N$  alone.** A sign-flip permutation test on  $N$  paired differences has  $2^N$  assignments, so its smallest attainable two-sided  $p$  is  $2/2^N$ : 0.25 at  $N=3$ , 0.0625 at  $N=5$ , 0.002 at  $N=10$ . The paired- $N$  histogram across the 31 cells is {3:24, 5:4, 10:3}, so **only 3 of 31 cells can reach  $p<0.05$  without a distributional assumption**, and only one does (ILI,  $p=0.002$ ). Thirteen cells have a  $t$ -interval excluding zero where the exact test cannot reject at any effect size. We report both columns rather than the more favourable one: the  $t$ -test’s rejections are real inferences under its normality assumption, and the exact column is the price of not making that assumption.

**Why the reported interval is the  $t$ -interval, not the bootstrap.** At  $N=3$  every percentile-bootstrap resample mean lies inside  $[\min, \max]$  of the three differences, so a same-sign triple excludes zero *by construction*. Such an interval reports the sign of the data, not the uncertainty of the mean. The bootstrap is computed and stored in `results/paired_inference.json`; the table prints the paired  $t$ -interval beside the exact  $p$ .

**The aggregate straddles zero.** Averaging cell means and resampling (backbone, dataset) clusters, the 15 value-cells give  $\Delta_{\text{enc}}=-4.3$  with CI  $[-13.1, +4.5]$  and 7 of 15 positive; all 31 give  $+2.9$  with CI  $[-12.7, +15.1]$  and 16 of 31 positive. Neither excludes zero in either direction. This is the honest aggregate statement of the paper’s central contrast: across these benchmarks we cannot establish that allowing encoder adaptation systematically helps *or* hurts, which is a different and weaker claim than either regime law the literature offers.

Generated by `scripts/paired_inference.py`.

Table 24: **Paired seed-level inference on encoder adaptation.**  $\Delta_{\text{enc}} = L_{\text{full}} - L_{\text{frozen}}$  per seed, in percent of the cell’s zero-shot reference, so a positive value means allowing the encoder to adapt made the cell *worse*. B and D share a seed, an initialisation and a window sample, so this contrast is paired by construction.  $d$  is paired Cohen’s  $d$ ; dispersion is SEM, matching the body convention (§3) and not the population std the stored JSON fields use. Each row names its own  $n$ , which is not constant down the column (Table 5). **Two tests, because neither alone is adequate at these seed counts.**  $p_{\text{exact}}$  enumerates all  $2^N$  sign assignments, so it assumes nothing—but its resolution is fixed by  $N$ : the smallest two-sided value an  $N$ -seed cell can produce is  $2/2^N$ , which is 0.25 at  $N=3$  and 0.0625 at  $N=5$ . The 24 cells at  $N=3$  therefore *cannot* reach  $p < 0.05$  however large or consistent their effect, and a non-significant exact  $p$  on those cells is a statement about seed budget rather than about the effect. The CI and  $p_t$  are the paired  $t$  versions, which do resolve at  $N=3$  and pay for it with a normality assumption on three points that cannot be checked. Where the two agree the reading does not rest on that assumption; where they disagree, the evidence is assumption-dependent and is treated as such in the text. We do *not* report a percentile bootstrap interval here: every resample mean lies within the range of the observations, so at  $N=3$  with all seeds on one side of zero such an interval excludes zero by construction, restating the sign count it was meant to improve on. It is kept in `results/paired_inference.json` for the larger cells. In the two summary rows the last column is *not* an effect size: it gives the number of cells whose mean  $\Delta_{\text{enc}}$  is positive, out of the cells in that row. † marks cells clearing the gate threshold under at least one admissible baseline of the ladder (Appendix A20). ‡ marks the Chronos cells, whose per-condition forgetting is confounded by the head/decoder mismatch; their  $\Delta_{\text{enc}}$  is unaffected.

| Cell                                                                            | $N$ | $\Delta_{\text{enc}}$ | SEM  | 95% CI ( $t$ )  | $p_t$ | $p_{\text{exact}}$ | $d$    |
|---------------------------------------------------------------------------------|-----|-----------------------|------|-----------------|-------|--------------------|--------|
| Chronos / ETTh1 ( $h=24, n=8000$ ) <sup>†‡</sup>                                | 3   | −39.2                 | 13.9 | [−98.9, +20.5]  | 0.106 | 0.250              | −1.63  |
| Moirai-B / ETTh2 ( $h=192, n=1000$ ) <sup>†</sup>                               | 3   | −20.2                 | 1.1  | [−24.9, −15.5]  | 0.003 | 0.250              | −10.66 |
| Moirai-B / ETTh2 ( $h=96, n=1000$ ) <sup>†</sup>                                | 3   | −22.3                 | 2.9  | [−34.7, −10.0]  | 0.016 | 0.250              | −4.50  |
| Moirai-B / ETTm2 ( $h=192, n=1000$ ) <sup>†</sup>                               | 3   | +1.3                  | 3.4  | [−13.4, +16.0]  | 0.740 | 1.000              | +0.22  |
| Moirai-B / ETTm2 ( $h=96, n=1000$ ) <sup>†</sup>                                | 3   | +7.0                  | 2.4  | [−3.2, +17.3]   | 0.098 | 0.250              | +1.70  |
| Moirai-L / ETTh2 ( $h=96, n=500$ ) <sup>†</sup>                                 | 3   | −20.8                 | 1.4  | [−26.6, −15.0]  | 0.004 | 0.250              | −8.87  |
| Moirai-S / ETTh1 ( $h=96, n=1000$ ) <sup>†</sup>                                | 5   | +22.1                 | 2.3  | [+15.6, +28.6]  | 0.001 | 0.062              | +4.21  |
| Moirai-S / ETTh2 ( $h=192, n=500$ ) <sup>†</sup>                                | 10  | +6.1                  | 3.4  | [−1.5, +13.7]   | 0.101 | 0.100              | +0.58  |
| Moirai-S / ETTh2 ( $h=96, n=500$ ) <sup>†</sup>                                 | 10  | +2.8                  | 3.7  | [−5.6, +11.2]   | 0.472 | 0.557              | +0.24  |
| Moirai-S / ETTm2 ( $h=192, n=1000$ ) <sup>†</sup>                               | 5   | +4.7                  | 3.2  | [−4.2, +13.6]   | 0.217 | 0.312              | +0.65  |
| Moirai-S / ETTm2 ( $h=96, n=1000$ ) <sup>†</sup>                                | 5   | +2.6                  | 3.4  | [−6.9, +12.1]   | 0.495 | 0.562              | +0.34  |
| Moirai-S / Electricity7 ( $h=192, n=1000$ ) <sup>†</sup>                        | 3   | −2.6                  | 2.1  | [−11.7, +6.4]   | 0.338 | 0.500              | −0.72  |
| Moirai-S / Electricity7 ( $h=96, n=1000$ ) <sup>†</sup>                         | 3   | −3.2                  | 1.5  | [−9.9, +3.4]    | 0.170 | 0.250              | −1.22  |
| TimesFM / ETTh1 ( $h=24, n=1000$ ) <sup>†</sup>                                 | 3   | −1.4                  | 2.8  | [−13.5, +10.6]  | 0.658 | 0.750              | −0.30  |
| TimesFM / Weather ( $h=24, n=1000$ ) <sup>†</sup>                               | 3   | −1.6                  | 1.5  | [−8.1, +4.8]    | 0.387 | 0.500              | −0.63  |
| Chronos / Electricity ( $h=24, n=8000$ ) <sup>‡</sup>                           | 3   | −2.7                  | 1.9  | [−10.7, +5.4]   | 0.287 | 0.500              | −0.83  |
| Chronos / ETTh2 ( $h=24, n=8000$ ) <sup>‡</sup>                                 | 3   | −12.9                 | 0.6  | [−15.5, −10.4]  | 0.002 | 0.250              | −12.57 |
| Chronos / ETTm2 ( $h=24, n=8000$ ) <sup>‡</sup>                                 | 3   | −29.5                 | 6.4  | [−57.1, −1.9]   | 0.044 | 0.250              | −2.66  |
| Chronos / Weather ( $h=24, n=8000$ ) <sup>‡</sup>                               | 3   | −21.6                 | 3.2  | [−35.5, −7.6]   | 0.022 | 0.250              | −3.84  |
| Moirai-B / ETTh1 ( $h=192, n=1000$ )                                            | 3   | +47.0                 | 13.8 | [−12.4, +106.3] | 0.077 | 0.250              | +1.97  |
| Moirai-B / ETTh1 ( $h=96, n=1000$ )                                             | 3   | +37.5                 | 8.2  | [+2.1, +72.9]   | 0.045 | 0.250              | +2.63  |
| Moirai-B / Weather ( $h=192, n=1000$ )                                          | 3   | +12.5                 | 8.7  | [−25.1, +50.1]  | 0.288 | 0.500              | +0.83  |
| Moirai-B / Weather ( $h=96, n=1000$ )                                           | 3   | +9.4                  | 2.0  | [+1.0, +17.8]   | 0.040 | 0.250              | +2.78  |
| Moirai-L / ETTh1 ( $h=96, n=1000$ )                                             | 3   | +28.2                 | 4.9  | [+7.3, +49.1]   | 0.028 | 0.250              | +3.35  |
| Moirai-L / Weather ( $h=96, n=1000$ )                                           | 3   | +12.0                 | 7.5  | [−20.3, +44.3]  | 0.251 | 0.250              | +0.92  |
| Moirai-S / ETTh1 ( $h=192, n=1000$ )                                            | 5   | +13.5                 | 1.7  | [+8.8, +18.2]   | 0.001 | 0.062              | +3.53  |
| Moirai-S / ILI ( $h=24, n=452$ )                                                | 10  | −38.6                 | 1.3  | [−41.6, −35.6]  | 0.000 | 0.002              | −9.15  |
| Moirai-S / Weather ( $h=192, n=1000$ )                                          | 3   | +62.1                 | 13.6 | [+3.6, +120.6]  | 0.045 | 0.250              | +2.64  |
| Moirai-S / Weather ( $h=96, n=1000$ )                                           | 3   | +53.1                 | 10.2 | [+9.2, +97.0]   | 0.035 | 0.250              | +3.00  |
| TimesFM / ETTh2 ( $h=24, n=1000$ )                                              | 3   | −7.1                  | 3.1  | [−20.5, +6.2]   | 0.148 | 0.250              | −1.33  |
| TimesFM / ETTm2 ( $h=24, n=1000$ )                                              | 3   | −7.7                  | 2.6  | [−18.9, +3.5]   | 0.097 | 0.250              | −1.71  |
| <i>Mean of cell means, 95% CI from resampling (backbone, dataset) clusters:</i> |     |                       |      |                 |       |                    |        |
| 15 value-cells / 7 clusters                                                     |     | −4.3                  |      | [−13.1, +4.5]   |       |                    | 7/15   |
| all 31 cells / 15 clusters                                                      |     | +2.9                  |      | [−12.7, +15.1]  |       |                    | 16/31  |

## A23 THE GRADED VALUE AXIS: IS THE DISSOCIATION CONFINED TO CELLS WITH NOTHING TO LOSE?

The screen in the body is binary: a cell either clears  $R^2_{\text{task}}=0.20$  or it does not. That discards the only interesting covariate, and it leaves open the single strongest objection to this paper’s negative result—that drift fails to predict  $\Delta_{\text{enc}}$  only *because* 16 of our 31 cells have no measured advantage to lose, so we are reading noise where there is nothing to read. If that were right, the drift-to-outcome relation would strengthen with pre-trained value, and our finding would be a statement about weak benchmarks rather than about the diagnostic. This appendix estimates that dependence two ways and reports an interval rather than a verdict.

**The axis.** Value here is the graded score of Appendix A20:  $R^2_{\text{task}}$  against the *strongest admissible* rung of the eight-baseline ladder, which is a **minimum** over rungs, since  $R^2_{\text{task}}=1 - \text{MSE}_{\text{ZS}}/\text{MSE}_{\text{base}}$  and a weaker rival inflates it. It runs from  $-1.638$  (Moirai-Small/ILI) to  $+0.058$  (Moirai-Small/ETTh2  $h=96$ ) with mean  $-0.610$ , and its ceiling below 0.20 is the same fact as “no cell clears the gate against every admissible rung”. A cell with no admissible rung at all is *dropped*, never scored as zero value: a missing denominator and a denominator of zero are different facts, and the second would place the cell at the origin of this very axis.

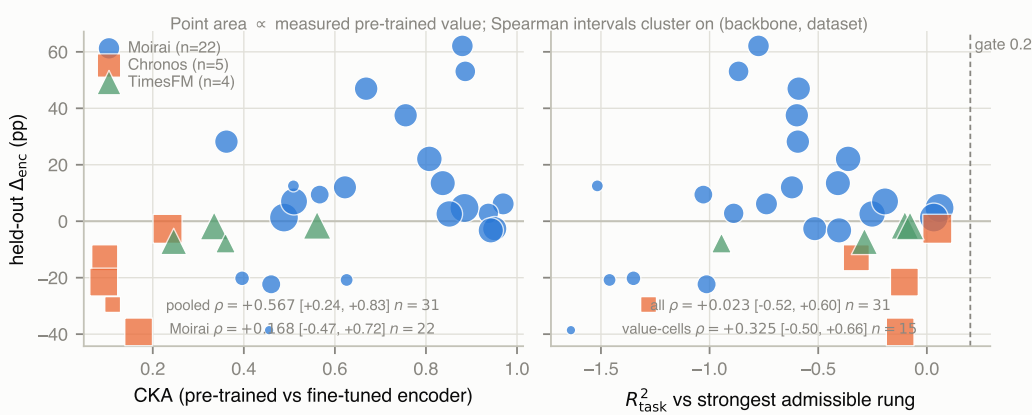


Figure 4: **Neither drift nor measured pre-trained value orders the encoder intervention.** Both panels share  $y = \text{held-out } \Delta_{\text{enc}}$  (positive = freezing the encoder did better). Point area is proportional to the graded value score in *both* panels, so the left panel answers the objection visually: if drift predicted the outcome wherever there was something to lose, the large points would trend and the small ones would scatter. Backbone is encoded by colour *and* marker because CKA is not comparable across architectures, which is why the pooled left-panel interval must not be read as a within-backbone one. Drawn by `paper_8/fig_value_axis.py` from `results/value_axis.json`; the script asserts its own counts against that file and fails rather than drawing a figure that disagrees with this text.

**The rank answer: the relation is not stronger where there is more to lose.** Restricting to the 15 value-cells gives  $\rho = +0.454$  with clustered CI [ $-0.490, +0.890$ ]; the 16 cells with no admissible value give  $\rho = +0.826$  with CI [ $-0.321, +0.913$ ] (Table 25). The restriction the objection asks for makes the relation *weaker*, not sharper, and neither interval excludes zero. We do not test the difference of those two  $\rho$ s: a difference of rank statistics on 15 and 16 cells from 7 and 9 clusters has no usable sampling distribution, and reporting one would be the kind of number this paper is arguing against.

**The slope answer, where the dependence is a single coefficient.** On the slope scale the interaction is estimable, and it is  $b_3 = -45.3$  pp per unit CKA per unit value with CI [ $-225.6, +55.0$ ], against a CKA slope at mean value of  $+31.2$  with CI [ $-34.1, +107.9$ ]. Two things to say about it and no more. The point estimate has the *opposite* sign to the objection—drift orders the outcome slightly *less* well where there is more value—and the interval spans zero so widely that only an interaction several times larger than the CKA slope itself would have been visible at 15 clusters. **So this is not a refutation of the objection; it is the statement that these cells cannot separate it from no**

Table 25: **The dependence, estimated two ways.** Rank rows are Spearman  $\rho$  with 10,000 bootstrap draws over (backbone, dataset) clusters, seed 0. The slope rows fit all cells at once with backbone identity absorbed as a fixed effect and both regressors centred, so  $b_1$  is the CKA slope *at mean value*;  $b_3$  is the quantity at issue. Emitted by `scripts/value_axis.py -latex`, which delegates the slope fit to `scripts/cka_fixed_effects.py` rather than refitting it.

| Relation                                                                                                                                                                               | $n$ | clusters | $\rho$ or $b$ | 95% CI over clusters          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|---------------|-------------------------------|
| CKA vs $\Delta_{\text{enc}}$ , the value-cells                                                                                                                                         | 15  | 7        | +0.454        | $[-0.490, +0.890]$ includes 0 |
| CKA vs $\Delta_{\text{enc}}$ , cells with no admissible value                                                                                                                          | 16  | 9        | +0.826        | $[-0.321, +0.913]$ includes 0 |
| CKA vs $\Delta_{\text{enc}}$ , within Moirai                                                                                                                                           | 22  | 6        | +0.168        | $[-0.469, +0.717]$ includes 0 |
| CKA vs $\Delta_{\text{enc}}$ , pooled (not a licensed reading)                                                                                                                         | 31  | 15       | +0.567        | $[+0.239, +0.828]$ excludes 0 |
| $R^2_{\text{task}}$ (best) vs $\Delta_{\text{enc}}$ , the value-cells                                                                                                                  | 15  | 7        | +0.325        | $[-0.497, +0.665]$ includes 0 |
| $R^2_{\text{task}}$ (best) vs $\Delta_{\text{enc}}$ , all cells                                                                                                                        | 31  | 15       | +0.023        | $[-0.516, +0.599]$ includes 0 |
| <i>Slope scale, all cells at once: <math>\Delta_{\text{enc}} = b_0 + b_1 \text{CKA} + \text{backbone FE} + b_2 v + b_3 (\text{CKA} \times v)</math>, <math>v = \text{value}</math></i> |     |          |               |                               |
| $b_1$ , CKA slope at mean value (pp per unit CKA)                                                                                                                                      | 31  | 15       | +31.2         | $[-34.1, +107.9]$ includes 0  |
| $b_3$ , CKA $\times$ value interaction                                                                                                                                                 | 31  | 15       | -45.3         | $[-225.6, +55.0]$ includes 0  |

**dependence at all.** That is a weaker claim than we would like to make, and it is the one the data support.

**And value does not order the outcome by itself.** The graded score against  $\Delta_{\text{enc}}$  gives  $\rho = +0.023$ , CI  $[-0.516, +0.599]$  across all 31 cells and  $\rho = +0.325$ , CI  $[-0.497, +0.665]$  across the 15 value-cells—the same non-result as CKA, on the covariate that was supposed to rescue it. The right panel of Figure 4 is that row: every point lies left of the 0.20 line, and the cells that improve most under encoder adaptation are not the ones with the least to lose.

## A24 CROSS-CELL STATISTICS, AND A POOLED ANALYSIS WE DO NOT LICENSE

Across the 31 intervention cells of Table 5 we ask whether either observational diagnostic—CKA, or  $\ell_2$  weight drift—orders the held-out intervention contrast B–D. Two predictors rather than one is deliberate: if *both* fail, the statement is about observational measures of fine-tuning in general and not about CKA specifically. The statistic is Spearman  $\rho$  with a 10,000-resample bootstrap CI (seed 0), reported at two resampling levels—over cells, and over the (backbone, dataset) clusters that Appendix A27 argues is the correct level—and emitted by `scripts/clustered_inference.py`. Pooled over all 31 cells:

| Predictor             | $n$ | Clusters | $\rho$ | 95% CI over cells             | 95% CI over clusters          |
|-----------------------|-----|----------|--------|-------------------------------|-------------------------------|
| CKA                   | 31  | 15       | +0.567 | $[+0.265, +0.759]$ excludes 0 | $[+0.239, +0.828]$ excludes 0 |
| $\ell_2$ weight drift | 25  | 9        | +0.118 | $[-0.300, +0.518]$ includes 0 | $[-0.489, +0.545]$ includes 0 |

The  $\ell_2$  row covers 25 of the 31 cells because the five Chronos cells and ILI retain no weight-drift measurement in their run records. No  $p$ -values are reported in this appendix or the next: they were previously computed from the cell-level resampling, which mis-states the dependence between cells, and this analysis is descriptive (Appendix A27).

**The pooled CKA row is not a licensed analysis, and it is the row that would help us.** The three backbones occupy almost disjoint CKA ranges—Moirai 0.36–0.97, Chronos 0.09–0.23, TimesFM 0.25–0.56—and differ systematically in outcome: every Chronos and TimesFM cell favours adaptation, and every large positive B–D is Moirai’s. A rank correlation pooled over 31 cells drawn from three such groups therefore measures which backbone a cell came from at least as much as how far its encoder moved: it is a textbook setting for a group-level artefact of the Simpson’s-paradox kind, and CKA magnitude is representation-dependent, which is why this paper compares CKA only within a backbone (§6). We stress the direction of the incentive. The pooled figure is the *favourable* one—it is the only reading in this appendix under which a diagnostic we introduced orders the intervention with an interval excluding zero—and the body nevertheless leads with the within-backbone figure.

We decline to make the positive claim the pooled number would support, because the restriction that forbids it was fixed before these cells were run rather than chosen after seeing them.

Table 26: Is the pooled CKA correlation a backbone artefact? The same statistic (Spearman  $\rho$  of CKA against held-out B–D, 10,000-resample bootstrap CI) computed pooled and then stratified by backbone, at both resampling levels. “Cl.” is the number of (backbone, dataset) clusters, which is the effective sample size for the clustered interval and is far below  $n$ . Values are emitted by `scripts/clustered_inference.py`. The pooled row is reported for audit, not read as evidence; the two small arms carry essentially no information at their cell counts, and for them the two levels coincide because every cell is already its own cluster.

| Grouping                  | $n$ | Cl. | CKA range | $\rho$ | 95% CI over cells | 95% CI over clusters             |
|---------------------------|-----|-----|-----------|--------|-------------------|----------------------------------|
| Pooled (against our rule) | 31  | 15  | 0.09–0.97 | +0.567 | [+0.265, +0.759]  | excl. 0 [−0.239, +0.828] excl. 0 |
| Moirai only               | 22  | 6   | 0.36–0.97 | +0.168 | [−0.328, +0.598]  | incl. 0 [−0.469, +0.717] incl. 0 |
| Chronos only              | 5   | 5   | 0.09–0.23 | +0.100 | [−1.000, +1.000]  | incl. 0 [−1.000, +1.000] incl. 0 |
| TimesFM only              | 4   | 4   | 0.25–0.56 | +0.400 | [−0.778, +1.000]  | incl. 0 [−0.778, +1.000] incl. 0 |

**Within Moirai the interval is wide, and we say so rather than round it to zero.** On the 22 Moirai cells  $\rho=+0.168$  with a cell-level CI [−0.328, +0.598], widening to [−0.469, +0.717] over the six clusters—a width of 0.93 on a scale whose whole range is 2, and 1.19 once the dependence is propagated, which is the honest bound this evidence supports. That interval comfortably admits a moderate positive relationship, and 22 cells cannot rule one out. The claim we license is the one the body makes: *CKA does not order our cells*. It is not that CKA is uninformative in general, and a reader who expects a real within-backbone relationship to exist will find nothing here that contradicts them; what they will not find is a cell ordering they could have acted on.

**Chronos and TimesFM cannot be asked at all.** With  $n=5$  and  $n=4$  the bootstrap resamples only a handful of distinct rank permutations and the interval spans nearly the entire admissible range, [−1.000, +1.000] and [−0.778, +1.000]. Both rows are tabulated for completeness and neither carries information. In particular the +0.400 on four TimesFM cells must not be read as agreement with the pooled sign: it is one of very few values the statistic can take on four cells, and its interval covers almost everything.

**$\ell_2$  weight drift does no better, and it is the second of the two diagnostics we defined.** Pooled—already the reading most likely to manufacture a correlation—weight drift gives  $\rho=+0.118$  with a CI that includes zero at both resampling levels, widening from [−0.300, +0.518] over cells to [−0.489, +0.545] over its nine clusters. Since §3 defines exactly these two representation diagnostics, their joint failure is the more useful statement: it is not that we picked the wrong summary of the encoder’s movement, but that neither summary we defined orders the outcome. The two are also not redundant and can move independently: a stack whose weight drift is *exactly* zero can read CKA well below one when the projections feeding it are trainable, which is what condition D does on TimesFM, where  $\ell_2$  drift over the transformer stack is 0.00 by construction while its measured CKA runs 0.38–0.89 (§3, §6; the analogous Moirai control is Appendix A6). A geometric battery—RBF CKA, SVCCA, orthogonal Procrustes, mean cosine and effective-rank change, on the 59 encoders that retained checkpoints—is Appendix A25. None of those measures orders the outcome either, and they do not order the *encoders* alike, so the failure is not a property of the summary we happened to pick.

**A backbone fixed-effects reading, and its limits.** The natural alternative to subsetting is to absorb backbone identity and let CKA enter as a continuous regressor, which answers the objection that we adopted the split that removes the significant relationship. We do fit it (Table 28, `scripts/cka_fixed_effects.py`): the within-backbone slope of held-out B–D on CKA is +39.1 pp per unit CKA with backbone fixed effects and +39.9 once  $\text{CKA} \times \text{backbone}$  interactions are added, against +47.7 with no backbone term at all. Both adjusted intervals include zero once clusters rather than cells are resampled. So the reduction from +47.7 to +39.1 shows that absorbing backbone identity does *not* by itself destroy the pooled slope—most of it survives adjustment—while the interval shows that 31 cells drawn from 15 clusters cannot pin the surviving slope down. What we do *not* do is fit variance components: with 22, 5 and 4 cells in the three groups there is no repeated-measures structure to estimate them from, so a mixed model over three groups of that size is

not identified and any interval it produced would be an artefact of the specification. Cells here are unequally powered and not independent, so every cross-cell number in this appendix is descriptive.

## A25 IS THE READING SPECIFIC TO CKA? SIX MEASURES ON 59 ENCODERS

The body reads drift through linear CKA, so a reader is entitled to ask whether “drift does not order the outcome” is a fact about representations or a fact about CKA. Appendix A24 gives the answer that covers the whole matrix:  $\ell_2$  weight drift, a geometrically unrelated quantity defined in §3 alongside CKA, fails on the same 31 cells. This appendix gives the second answer, on geometry, and states up front what it cannot do.

**What it cannot do.** Only the  $n=10k$  runs retained encoder checkpoints. The 22-cell Moirai matrix is  $n=500/1,000$  and kept none, so this battery *cannot* recompute the within-Moirai  $\rho$  under a different measure; that would require retraining the matrix, on different hardware, which would perturb every number in the paper. What it can do is test measure-specificity on the 59 encoders that do exist: Moirai-Small at  $h=96$ ,  $n=10k$ , on ETTh1, ETTh2 and ETTm2, under full fine-tuning and under two layer-unfreezing regimes (top 3 of 6, and all 6), in seven (dataset, run-set, regime) groups.

**Measured on the representations the paper’s CKA column describes.** Each run’s record stores `final_cka`, computed during training on the first 300 validation evaluation windows with the encoder output mean-pooled. The battery (`scripts/drift_metric_battery.py`) rebuilds exactly those windows—extended lookback 96+96, raw values, the model normalising internally—and recomputes linear CKA as a self-check: it agrees with the stored value to four decimal places on all 59 encoders, so every other column is known to describe the same representations rather than a similar-looking set. The checkpoint load asserts that `load_state_dict(strict=False)` reports no missing and no unexpected keys, because a namespace mismatch loads nothing silently and returns  $\text{CKA} \approx 1$ , which reads as “no drift”.

Table 27: **Six measures of representation change on 59 Moirai-Small encoders.** All computed on identical windows and identical pooling. “ $\rho$  with CKA” is the rank correlation of each measure with linear CKA over the 59 encoders; the last two columns correlate each measure with forgetting, pooled and then within each of the six groups of at least five encoders. Forgetting here is the *selection-split* figure—these runs stored no zero-shot test MSE, so a held-out contrast cannot be reconstructed from them, and §7 shows same-split scoring can invert the sign of the intervention. This arm is therefore a comparison of measures, not a held-out claim. Emitted by `scripts/drift_metric_battery.py`; the pooled column mixes three datasets and is descriptive for the reason Appendix A24 gives about pooling.

| Measure            | over 59 encoders |        |        | $\rho$ with CKA | $\rho$ with forgetting |                |
|--------------------|------------------|--------|--------|-----------------|------------------------|----------------|
|                    | min              | max    | mean   |                 | pooled                 | per group      |
| Linear CKA         | +0.194           | +0.915 | +0.588 | +1.000          | +0.350                 | [−0.19, +0.60] |
| Linear CKA (token) | +0.626           | +0.957 | +0.751 | +0.542          | −0.123                 | [−0.33, +0.55] |
| RBF CKA            | +0.624           | +0.913 | +0.734 | +0.412          | +0.117                 | [−0.36, +0.90] |
| SVCCA              | +0.726           | +0.939 | +0.776 | +0.553          | −0.038                 | [−0.70, +0.35] |
| Procrustes sim.    | +0.715           | +0.947 | +0.818 | +0.675          | +0.127                 | [−0.52, +0.80] |
| Mean cosine        | +0.660           | +0.990 | +0.840 | +0.073          | −0.287                 | [−0.67, +0.60] |
| Eff. rank ratio    | +0.969           | +1.250 | +1.119 | −0.251          | +0.245                 | [+0.00, +0.65] |

**The more invariant the measure, the less drift there is to see.** Linear CKA spans 0.194 to 0.915 across these encoders. SVCCA—invariant to any invertible linear map, so it asks whether the same information survives in a possibly rotated and rescaled basis—spans only 0.726 to 0.939, and orthogonal-Procrustes similarity 0.715 to 0.947. The cell that CKA scores at 0.194 is at 0.73 on SVCCA. This is the concrete form of the objection limitation (7) concedes: a large part of what CKA reports as drift is a change of basis under which the subspace, and with it the linearly decodable content, is largely preserved. It is also why we do not claim CKA is invalid—it measures what it measures—only that a paper cannot infer damage from it.

**The measures do not order the encoders alike, so “drift” is not one quantity.** Rank correlations among them run from +0.073 (linear CKA against mean cosine, the least invariant measure in the

battery) through +0.412 (linear CKA against RBF CKA) to +0.845 (RBF CKA against Procrustes). Choosing a similarity measure chooses an ordering of these encoders. We had expected the opposite—a family of near-clones—and report the disagreement because it makes the next paragraph the load-bearing one.

**No measure orders the outcome either.** Against selection-split forgetting, the pooled rank correlations run from +0.350 (linear CKA) to −0.287 (mean cosine), and the pooled figures are the flattering ones: within groups, where dataset identity is held fixed, every similarity measure changes sign across the six groups (linear CKA [−0.19, +0.60], RBF CKA [−0.36, +0.90], SVCCA [−0.70, +0.35], Procrustes [−0.52, +0.80], mean cosine [−0.67, +0.60]). Six measures spanning no invariance to full linear-map invariance therefore reproduce the body’s reading rather than overturning it: the disagreement among them does not resolve into any one of them ordering the outcome. The single exception is not a similarity measure at all. The effective-rank ratio—how many directions the representation uses after fine-tuning against before—is above 1 on 55 of the 59 encoders (0.969–1.250, so fine-tuning generally *expands* the spectrum) and is the one column with no negative group value ([+0.00, +0.65]). At six groups, one backbone and one sample size we record that as a lead for future work, not a diagnostic; it is exactly the kind of pattern this paper’s own prospective test (§7) was built to keep us honest about.

## A26 WHAT PREDICTS THE INTERVENTION, AND WHY NONE OF IT IS A SHORTCUT

If some quantity measurable from one run ordered B–D, a practitioner could skip the frozen control. We tested every candidate we had, within Moirai only, because it is the only arm with enough cells to ask (22). The statistic is again Spearman  $\rho$  against held-out B–D with a 10,000-resample bootstrap CI at both resampling levels, from `scripts/clustered_inference.py`:

| Predictor (Moirai only)            | $n$ | Cl. | $\rho$ | 95% CI over cells           | 95% CI over clusters        |
|------------------------------------|-----|-----|--------|-----------------------------|-----------------------------|
| $\text{forg}_B$ (condition B only) | 22  | 6   | +0.894 | [+0.667, +0.981] excludes 0 | [+0.594, +0.965] excludes 0 |
| $\text{forg}_D$                    | 22  | 6   | +0.073 | [−0.324, +0.429] includes 0 | [−0.343, +0.412] includes 0 |
| Gate $R_{\text{task}}^2$ (PT)      | 22  | 6   | −0.518 | [−0.840, −0.040] excludes 0 | [−0.810, +0.500] includes 0 |
| CKA                                | 22  | 6   | +0.168 | [−0.328, +0.598] includes 0 | [−0.469, +0.717] includes 0 |
| $\ell_2$ weight drift              | 21  | 5   | +0.318 | [−0.122, +0.673] includes 0 | [−0.329, +0.861] includes 0 |

Weight drift covers 21 of the 22 cells because ILI’s run records retain no drift measurement. The gate row is the same figure reported in Appendix A19 as a consequence of the baseline correction; it appears here beside its competitors. **These 22 cells come from six clusters**, and the gate row is the one place in the paper where that distinction changes a verdict: see Appendix A27.

**The  $\text{forg}_B$  result is an identity, not a discovery.**  $B-D \equiv \text{forg}_B - \text{forg}_D$  holds *exactly*, because both terms are percentage changes against the same zero-shot reference: over all cells the maximum deviation is  $\max |(\text{forg}_B - \text{forg}_D) - (B-D)| = 0.000$ . The dispersion is then overwhelmingly on the B side, SD 26.2 against  $\text{forg}_D$ ’s 8.1, so  $B-D$  is close to a rescaled copy of  $\text{forg}_B$  and  $\rho = +0.894$  is close to arithmetic rather than to a finding. It confers no ability to skip condition D:  $\text{forg}_B$  exists only once condition B has been run, and at that point running D is one further run on the same cell. This is exactly the kind of correlation that looks like a shortcut and is not, and the near-zero  $\text{forg}_D$  row is the reason—the rule works only because the frozen arm happens to be nearly constant on this backbone, which is an empirical property of these cells and not a theorem.

**The gate’s continuous score points the wrong way, but only the sign is supported.** On the same 22 cells the gate correlates *negatively* with held-out B–D,  $\rho = -0.518$ . Resampling cells gives CI [−0.840, −0.040], an interval that excludes zero; resampling the six clusters gives [−0.810, +0.500], which does not. **We report the clustered interval and withdraw the “excludes zero” reading of this correlation** (Appendix A27); it is the one claim in the paper whose verdict the correct resampling level overturns, and the correction runs against us. The mechanism is legible in Table 5: the cells with the largest measured pre-trained advantage are the Moirai/ETTh2 cells, and three of those five are the cells where full fine-tuning beats the frozen encoder by the most ( $B-D$  −20.2 to −22.3 pp), while the cells with the most negative gates—Weather and ETTh1—are where freezing wins by tens of points, up to +62.1 pp. A practitioner following “preserve where the gate fires” would therefore freeze



precisely where adapting the encoder pays best. We report this at the strength it has: a within-Moirai point estimate on 22 retrospective cells from six series, with a clustered interval that spans zero. **The negative sign is a description of these cells, not an interval-supported claim, and the magnitude is not estimated to any useful precision.** The mechanism above is legible cell by cell in Table 5 and does not depend on the correlation surviving; that is why we still report the direction.

**The other two arms cannot adjudicate.** On TimesFM ( $n=4$ ) the gate gives  $\rho=+0.800$ , CI  $[+0.333, +1.000]$ ,  $\text{forg}_B$  gives  $+1.000$  and CKA  $+0.400$ , CI  $[-0.778, +1.000]$ ; on Chronos ( $n=5$ ) the gate gives  $\rho=+0.500$ , CI  $[-0.667, +1.000]$ , and  $\text{forg}_B$  gives  $-0.100$ . These are reported for completeness and are uninformative at four and five cells: a bootstrap over four cells resamples only a few distinct rank patterns, so even the TimesFM gate interval, which nominally excludes zero, is an artefact of that granularity and we read nothing from it. Chronos’s  $\text{forg}_B$  carries a further defect and is flagged `forg_confounded` in the code: a head/decoder mismatch means its B-side number is not a clean per-condition forgetting measurement, so it is retained only to show where the  $\text{forg}_B$  rule breaks and is never quoted as a forgetting value.

**Nothing here is a replacement diagnostic.** The honest summary of this appendix and of Appendix A24 together is that no observational quantity we measured orders the intervention in a usable direction: CKA and  $\ell_2$  drift do not order it, the gate orders it *backwards*, and the one predictor that tracks it does so by identity and only after the run that would have told you the answer anyway. On the evidence in Table 5, the only reliable way to know whether encoder adaptation helped a given cell is to run both conditions on that cell.

## A27 CELLS ARE NOT INDEPENDENT: CLUSTERED INTERVALS FOR EVERY CROSS-CELL STATISTIC

Every interval in Appendices A24 and A26 was previously computed by resampling *cells*, which treats each cell as an independent observation. They are not. A cell is a (backbone, size, dataset, horizon) combination, so the 31 cells draw on 3 backbones and 7 series: the Moirai-Small, -Base and -Large runs on ETTh2 at  $h=96$  and  $h=192$  share the series, the chronological split, the normalisation constants and—at fixed size—the pretrained checkpoint. Resampling cells therefore counts shared facts repeatedly and produces intervals that are too narrow, and a  $p$ -value computed from that scheme is not conventional inferential evidence. This is a fair objection to how the paper reported these numbers and it is answerable without new runs.

**The resampling unit.** We take the cluster to be the (backbone, dataset) pair: the level at which the pretraining corpus, the series, the split and the checkpoint family are shared. Clusters are drawn with replacement and all of a drawn cluster’s cells enter together (`scripts/cluster_keys.py`, one definition shared by `cell_matrix.py` and `cka_fixed_effects.py` so the two cannot drift apart). A draw spanning fewer than three distinct predictor values admits no rank correlation and is discarded, which is why the draws kept are reported alongside every interval; on the 31-cell and 22-cell analyses the discard rate is under 0.1%, and it is material only on the four- and five-cell arms, which carry no information anyway. The effective sample size falls sharply: 31 cells are 15 clusters, the 22 Moirai cells are 6, and the 25 cells with a weight-drift record are 9.

**What changes, in full.** Of the eleven cross-cell correlations reported in this paper, **exactly one changes verdict** when the interval is taken over clusters instead of cells, and it changes against us. Both levels appear side by side in Table 26 and in the two tables of Appendices A24 and A26; the summary is:

- **Unchanged, and these carry the body’s claims.** CKA does not order B–D within Moirai at either level ( $\rho=+0.168$ ; cells  $[-0.328, +0.598]$ , clusters  $[-0.469, +0.717]$ ).  $\ell_2$  weight drift does not order it at either level, pooled ( $\rho=+0.118$ ) or within Moirai ( $\rho=+0.318$ ).  $\text{forg}_B$  still orders it with an interval far from zero ( $\rho=+0.894$ ; clusters  $[+0.594, +0.965]$ )—which is expected, since that relationship is close to an identity (Appendix A26). The unlicensed pooled CKA correlation still excludes zero, and we still decline to read it.
- **Changed: the gate’s anti-predictive interval.**  $\rho=-0.518$  between the gate and held-out B–D moves from  $[-0.840, -0.040]$  over cells to  $[-0.810, +0.500]$  over the six Moirai clusters. **We withdraw the “excludes zero” reading** and report the direction as a description

of these 22 cells. The mechanism—the highest-gate cells are exactly the ETTh2 cells where adaptation pays best—remains visible cell by cell in Table 5 and does not rest on the interval.

**Backbone fixed effects.** The cluster bootstrap also lets us report the adjustment the previous version of this appendix declined to fit (Table 28).

Table 28: Within-backbone slope of held-out B–D on CKA, in percentage points per unit of CKA, with backbone identity absorbed as a fixed effect rather than by subsetting. Intervals are 10,000 cluster-bootstrap draws over the 15 (backbone, dataset) clusters of 31 cells; draws in which the fixed effects are unidentified or the design is rank-deficient are discarded, and the kept count is reported. Emitted by `scripts/clustered_inference.py` from `scripts/cka_fixed_effects.py`.

| Specification                    | $b_1$ (pp per unit CKA) | 95% CI over clusters         | Draws kept |
|----------------------------------|-------------------------|------------------------------|------------|
| No backbone term                 | +47.71                  | —                            | —          |
| CKA + backbone FE                | +39.08                  | [−15.37, +98.50] includes 0  | 9895       |
| CKA + backbone FE + CKA×backbone | +39.91                  | [−18.95, +109.71] includes 0 | 8522       |

Most of the unadjusted slope survives adjustment—+47.7 falls only to +39.1—so it is not the case that absorbing backbone identity destroys the pooled relationship, and we do not claim it. What the intervals say is that 15 clusters cannot pin the surviving slope down. **A wide interval is not evidence of absence.** At this cluster count an interval including zero means the design cannot resolve the sign, which is weaker than “CKA carries no information” and is the only thing we assert.

**Two things this appendix does not do.** It does not fit variance components: with 22, 5 and 4 cells in the three backbone groups a mixed model is not identified, and any interval it produced would be a property of the specification. And it does not recompute  $p$ -values at the cluster level. A  $p$ -value would need a null distribution for a rank correlation among dependent observations at an effective sample size near 6, and the paper describes this analysis as descriptive; printing  $p=0.0136$  beside that sentence was a contradiction, so the  $p$ -columns are removed rather than re-derived.

## A28 ILI: GATE CHECK AND WITHDRAWN FROZEN CONTROL

The cell is Moirai-Small on national influenza-like illness: 966 weekly observations, 7 features,  $h=24$ , lookback=104, split 579/193/194 chronologically.

**ILI gate-fails on held-out windows, by a wide margin.** Measured against the fitted linear baseline the gate is defined with (Appendix A19), and computed from the runs this paper reports (10 seeds), the test-segment gate is  $-1.625$ ; that is the value carried in Table 5. **Two earlier figures for this cell are withdrawn:** a gate-pass of +57%, and the trend-baseline pair +0.139 on validation and  $-0.033$  on test, both of which came from the per-window trend estimator rather than the defined one and are superseded. The per-feature decomposition of that trend value—five of seven features positive but only two clearing 0.20, and NUM. OF PROVIDERS at  $-0.86$  because a near-straight test series is fitted almost exactly by extrapolating its own trend—explained the superseded number and is dropped with it. Against the fitted baseline the validation split gives  $-10.206$ , but that mean is not the quantity to read: it is dominated by NUM. OF PROVIDERS, whose near-straight validation segment the ridge map fits to MSE 0.034 and which therefore alone contributes  $-44.0$ . The median per-feature reading is  $-3.40$  on validation and  $-1.51$  on test, and **0 of 7 features clear 0.20 on either split**, so the verdict is robust even where the mean is not. ILI is therefore not a degradation cell and no claim in §6 rests on its gate; what changes is that it can no longer be cited as a moderate-gate cell, which makes the falsified prediction below correspondingly weaker evidence. A heavily restructuring encoder on a cell with *no* demonstrated zero-shot advantage is much less surprising than one on a moderate-gate cell.

**A withdrawn frozen control.** On ILI the encoder restructures heavily (CKA  $0.456 \pm 0.008$ ) and full fine-tuning improves by  $-80.0 \pm 0.6\%$  on validation and  $-50.77 \pm 0.49\%$  held out (10/10 negative on both, 0/10 positive). We previously reported B–D= $-80.7$  pp here, that is, that encoder adaptation supplied *all* of the gain. **That is withdrawn.** The frozen control it was measured against had never been trained: the script assigned the zero-shot MSE as the fine-tuned MSE and wrote a forgetting of exactly 0.0, so the “0.0% in all 10 seeds” was one constant, not ten measurements. Re-run so that condition D actually trains (encoder frozen, the 3.2M parameters outside it updated), the gap

is  $-62.6 \pm 1.4$  on validation and  $-38.6 \pm 1.3$  held out (10/10 negative on both): the frozen encoder recovers a large share of the gain rather than none of it, improving by  $-12.18 \pm 1.28\%$  held out on its own (10/10 negative). D reaches CKA 0.76–0.90, not 1.000, because the projections feeding the encoder remain trainable (Appendix A6). Condition B is essentially unaffected: it gives  $-80.0 \pm 0.6$  against the previously published  $-80.7 \pm 0.3$ , and CKA  $0.456 \pm 0.008$  against the published 0.478. The forgetting figures agree within about one standard error; the CKA means differ by 2.8 standard errors, though 0.478 sits inside the per-seed range we measure (0.409–0.505), so we describe the two B arms as consistent in outcome rather than identical. Either way the entire B–D discrepancy was the untrained control, not condition B. Table 29 gives the per-seed values behind both conditions; the point of printing all ten is that the withdrawn reading was a constant where these are ten distinct measurements.

Table 29: ILI per-seed results (Moirai-Small,  $h=24$ ,  $\text{lookback}=104$ ), both conditions, regenerated for this revision. The  $\Delta R^2$  column of the earlier version of this table is dropped with the retracted probe axis; condition D is shown because it previously did not exist as a measurement.

| Seed | B (full fine-tuning) |                         | D (frozen encoder) |                         |
|------|----------------------|-------------------------|--------------------|-------------------------|
|      | CKA                  | Forg.% (val / held-out) | CKA                | Forg.% (val / held-out) |
| 42   | 0.458                | $-78.6 / -51.5$         | 0.804              | $-16.2 / -14.6$         |
| 101  | 0.465                | $-81.4 / -50.1$         | 0.898              | $-13.5 / -6.7$          |
| 123  | 0.476                | $-80.0 / -48.9$         | 0.756              | $-27.6 / -19.5$         |
| 202  | 0.457                | $-81.4 / -52.1$         | 0.791              | $-19.7 / -13.6$         |
| 303  | 0.409                | $-78.1 / -49.0$         | 0.823              | $-18.3 / -11.5$         |
| 456  | 0.463                | $-75.5 / -50.0$         | 0.851              | $-15.4 / -13.2$         |
| 777  | 0.437                | $-81.6 / -52.5$         | 0.822              | $-18.0 / -12.4$         |
| 789  | 0.449                | $-80.3 / -49.5$         | 0.865              | $-12.7 / -4.9$          |
| 888  | 0.505                | $-82.0 / -50.8$         | 0.848              | $-16.1 / -12.8$         |
| 999  | 0.437                | $-80.8 / -53.4$         | 0.807              | $-16.1 / -12.7$         |
| Mean | 0.456                | $-80.0 / -50.8$         | 0.827              | $-17.4 / -12.2$         |

**Interpretation.** The pre-registered prediction had two components, and neither now supports the hypothesis. The CKA component (0.90–0.95, intermediate behaviour) is **falsified**: CKA is in the restructuring regime (0.456), not intermediate. The probe-asymmetry component ( $\geq 7/10$  seeds  $\Delta R^2 > 0$ ) was reported as confirmed, but rests on the statistic we retract (§A29) and is therefore **withdrawn** rather than confirmed. The prediction as a whole is falsified, not partially so. What it establishes is narrow: gate strength does not linearly predict drift magnitude. We can no longer read a restructuring-to-stability transition off this cell at all, because on held-out windows ILI does not clear the gate. ILI’s low CKA reflects the small dataset (966 timesteps,  $\approx 450$  training windows per feature) which permits aggressive restructuring within 20 epochs.

## A29 RETRACTION: THE PROBE-SIGN ASYMMETRY AXIS

An earlier version of this work reported a second diagnostic axis alongside CKA drift: a *probe-sign asymmetry*, in which fine-tuning raised linear decodability of the trained objective ( $\Delta R^2 > 0$ ) while leaving task-orthogonal probes flat or lower. We reported it as replicating across every gate-passing cell. **We retract it.** It does not survive a protocol with every choice fixed in advance, and nothing in this paper’s remaining claims depends on it.

**What fails.** The statistic reproduces numerically: the headline  $\Delta R^2 = +0.67 \pm 0.24$  on Moirai-Small/ETTh2 at  $n=10k$  is recoverable to four significant figures, but only in a regime where the probe does not generalise. Under a temporal split the pre-trained probe scores  $R^2(\text{PT})$  between  $-2.7$  and  $-164.1$ : it is not decoding the target at all, so  $\Delta R^2$  measures relative movement between two failed fits. Re-running the identical probes, encoders and windows under a randomised split, where all seven probes generalise ( $R^2(\text{PT}) \in [0.678, 0.994]$ ), no  $|\Delta R^2|$  exceeds 0.010.

| Split                             | $R^2(\text{PT})$ range | $\max  \Delta R^2 $ |
|-----------------------------------|------------------------|---------------------|
| Temporal (as originally reported) | $-164.1$ to $-2.7$     | 21.4                |
| Randomised, same windows          | $+0.678$ to $+0.994$   | <b>0.010</b>        |

The effect is therefore an artifact of probe failure, not evidence about the encoder. It also fails our own published criterion for discarding a probe: we stated that a probe with negative  $R^2$  should not be read as evidence, and then read one.

**What this costs, and what it does not.** Every claim resting on probe asymmetry is withdrawn: the “dissociation” bullet in §1, criterion (i) of §2, and the per-cell  $\Delta R^2$  readings in §6 and the appendix tables, which we retain only as a record of the retracted computation. Table 4 carried three such columns. They no longer appear there, and we transcribe them here instead, so that the withdrawal is legible against what was printed rather than quietly absorbed into a regenerated table. Unlike every other table in this paper these values are **not** re-derivable by `scripts/rederive_all.sh`: the probe records behind them are scattered across result directories at one to three seeds per  $n$ , and we are not going to write an emitter whose output is numbers we are telling the reader not to use. They are a transcript, they carry no dispersion where  $k=1$ , and they support no claim.

*Retracted. Transcribed from the withdrawn version of Table 4; not re-derived.*

| Samples | $R^2$ (FT)      | $\Delta R^2$ (Ridge) | $\Delta R^2$ (MLP) |
|---------|-----------------|----------------------|--------------------|
| 500     | −6.79 ( $k=2$ ) | + .12 ( $k=2$ )      | − .26 ( $k=1$ )    |
| 1,000   | −6.77 ( $k=1$ ) | + .13 ( $k=1$ )      | + .30 ( $k=1$ )    |
| 2,000   | −6.51 ( $k=3$ ) | + .39 $\pm$ .35      | + .73 $\pm$ .75    |
| 5,000   | −6.41 ( $k=7$ ) | + .50 $\pm$ .26      | n/a                |
| 10,000  | n/a             | + .67 $\pm$ .24      | n/a                |

The zero-shot floor is  $R_{ZS}^2 = -6.90$ , so  $R^2$ (FT) is negative in every row: the probe never decodes the target, which is precisely the regime this retraction is about, and the  $\Delta R^2$  column is then movement between two failed fits. The MSE, forgetting and CKA columns that remain in Table 4 are unaffected and are the only ones to read.

The paper’s other findings share no machinery with the probes: the value gate is a downstream MSE comparison, CKA drift is computed from encoder representations directly, and the frozen-encoder comparison is read in forecast MSE. The value gate was nonetheless found to have a defect of its own, independently of this retraction and by a different route—its linear denominator was not the estimator the gate is defined as—and that correction is documented separately in Appendix A19. The two are unrelated: neither the probe re-analysis nor the gate re-computation used any result from the other.

**Why we report this.** A negative result about one’s own tooling is cheap to bury and expensive to discover downstream. The re-analysis code and both split protocols are released with the paper.